```
  LOC       OBJECT CODE      ADDR1     ADDR2    STMT

                                                   2 ********************************************************************
                                                   3 *
                                                   4 *        Zvector E6 instruction tests for VRI-j encoded:
                                                   5 *
                                                   6 *        E64A VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                   7 *
                                                   8 *        and partial testing of
                                                   9 *
                                                  10 *        E64E VCVBQ    - VECTOR CONVERT TO BINARY (128)
                                                  11 *
                                                  12 *        during cross check tests for VCVBQ
                                                  13 *
                                                  14 *        James Wekel March 2026
                                                  15 ********************************************************************

                                                  17 ********************************************************************
                                                  18 *
                                                  19 *        basic instruction tests
                                                  20 *
                                                  21 ********************************************************************
                                                  22 *  This program tests proper functioning of the z/arch E6 VRI-j vector
                                                  23 *  convert to decimal (128), VCVDQ, instruction.
                                                  24 *  Exceptions are not tested.
                                                  25 *
                                                  26 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                  27 *  obvious coding errors.  None of the tests are thorough.  They are
                                                  28 *  NOT designed to test all aspects of any of the instructions.
                                                  29 *
                                                  30 ********************************************************************
                                                  31 *
                                                  32 *   *Testcase zvector-e6-24-VCVDQ
                                                  33 *   *
                                                  34 *   *        Zvector E6 instruction tests for VRI-j encoded:
                                                  35 *   *
                                                  36 *   *        E64A VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                  37 *   *
                                                  38 *   *        # ---------------------------------------------------------
                                                  39 *   *        #  This tests only the basic function of the instruction.
                                                  40 *   *        #  Exceptions are NOT tested.
                                                  41 *   *        # ---------------------------------------------------------
                                                  42 *   *
                                                  43 *   mainsize    2
                                                  44 *   numcpu      1
                                                  45 *   sysclear
                                                  46 *   archlvl     z/Arch
                                                  47 *
                                                  48 *   loadcore    "$(testpath)/zvector-e6-24-VCVDQ.core" 0x0
                                                  49 *
                                                  50 *   diag8cmd    enable    # (needed for messages to Hercules console)
                                                  51 *   runtest 5
                                                  52 *   diag8cmd    disable   # (reset back to default)
                                                  53 *
                                                  54 *   *Done
                                                  55 *
                                                  56 ********************************************************************
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                    58 ***********************************************************************
                                                    59 *        FCHECK Macro - Is a Facility Bit set?
                                                    60 *
                                                    61 *        If the facility bit is NOT set, an message is issued and
                                                    62 *        the test is skipped.
                                                    63 *
                                                    64 *        Fcheck uses R0, R1 and R2
                                                    65 *
                                                    66 * eg.    FCHECK 134,'vector-packed-decimal'
                                                    67 ***********************************************************************
                                                    68          MACRO
                                                    69          FCHECK &BITNO,&NOTSETMSG
                                                    70 .*                         &BITNO : facility bit number to check
                                                    71 .*                         &NOTSETMSG : 'facility name'
                                                    72          LCLA  &FBBYTE           Facility bit in Byte
                                                    73          LCLA  &FBBIT            Facility bit within Byte
                                                    74
                                                    75          LCLA  &L(8)
                                                    76 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                    77
                                                    78 &FBBYTE  SETA  &BITNO/8
                                                    79 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                    80 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                    81
                                                    82          B     X&SYSNDX
                                                    83 *                                        Fcheck data area
                                                    84 *                                        skip messgae
                                                    85 SKT&SYSNDX DC  C'     Skipping tests: '
                                                    86          DC    C&NOTSETMSG
                                                    87          DC    C' facility (bit &BITNO) is not installed.'
                                                    88 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                    89 *                                        facility bits
                                                    90          DS    FD                        gap
                                                    91 FB&SYSNDX DS   4FD
                                                    92          DS    FD                        gap
                                                    93 *
                                                    94 X&SYSNDX EQU *
                                                    95          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                    96          STFLE FB&SYSNDX                 get facility bits
                                                    97
                                                    98          XGR   R0,R0
                                                    99          IC    R0,FB&SYSNDX+&FBBYTE      get fbit byte
                                                   100          N     R0,=F'&FBBIT'             is bit set?
                                                   101          BNZ   XC&SYSNDX
                                                   102 *
                                                   103 * facility bit not set, issue message and exit
                                                   104 *
                                                   105          LA    R0,SKL&SYSNDX             message length
                                                   106          LA    R1,SKT&SYSNDX             message address
                                                   107          BAL   R2,MSG
                                                   108
                                                   109          B     EOJ
                                                   110 XC&SYSNDX EQU *
                                                   111          MEND
```

```
  LOC       OBJECT CODE      ADDR1     ADDR2    STMT

                                                 113 ***********************************************************************
                                                 114 *     Instruction Macros
                                                 115 ***********************************************************************
                                                 116 ***********************************************************************
                                                 117 * (pending VCVBQ - VECTOR CONVERT TO BINARY (128)
                                                 118 *  inclusion in SATK ASAM)
                                                 119 *
                                                 120 *     VCVBQ Macro to help build VCVBQ instruction
                                                 121 *        VCVBQ   m3
                                                 122 *
                                                 123 *        Note: v1 and v2 are fixed vector registers: v1 = 3; v2 = 2
                                                 124 *              m3 can be specified in hex eg. x'0'
                                                 125 ***********************************************************************
                                                 126          MACRO
                                                 127          VCVBQ &M3
                                                 128 .*                                      &M3  - m3 for VCVBQ instruction
                                                 129          LCLA  &V2M3
                                                 130 &M3ZZ    SETA  +(&M3*16)
                                                 131                                                                SPACE 1
                                                 132          DS    0H                      E64E VCVBQ
                                                 133          DC    X'E6'                   - VECTOR CONVERT TO BINARY (128)
                                                 134          DC    X'32'                    v3, v2
                                                 135          DC    X'00'                   reserved
                                                 136          DC    HL1'&M3ZZ'               m3, reserved
                                                 137          DC    X'00'                   reserved, RXB
                                                 138          DC    X'4E'
                                                 139                                                                SPACE 1
                                                 140          MEND

                                                 142 ***********************************************************************
                                                 143 * (pending VCVDQ - VECTOR CONVERT TO DECIMAL (128)
                                                 144 *  inclusion in SATK ASAM)
                                                 145 *
                                                 146 *     VCVDQ Macro to help build VCVBQ instruction
                                                 147 *        VCVDQ &I3,&M4
                                                 148 *
                                                 149 *        Note: v1 and v2 are fixed vector registers: v1=2, v2=2
                                                 150 *              i3 must be specified in hex eg. x'0'
                                                 151 *              m4 must be specified in hex eg. x'0'
                                                 152 ***********************************************************************
                                                 153          MACRO
                                                 154          VCVDQ &I3,&M4
                                                 155 .*                                      &I3  - i3 for VCVDQ instruction
                                                 156 .*                                      &M4  - M4 for VCVDQ instruction
                                                 157          LCLA  &V2I3M4
                                                 158 &M4I3    SETA  +(&M4*4096)+(&I3*16)
                                                 159                                                                SPACE 1
                                                 160          DS    0H                      E64A VCVDQ
                                                 161          DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
                                                 162          DC    X'22'                    v1, v2
                                                 163          DC    X'00'                   reserved
                                                 164          DC    HL2'&M4I3'               m4, i3, rXB
                                                 165          DC    X'4A'
                                                 166                                                                SPACE 1
                                                 167          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   169 ***********************************************************************
                                                   170 *        Low core PSWs
                                                   171 ***********************************************************************

                              00000000  000048BB   173 ZVE6TST  START 0
00000000                      00000000             174          USING ZVE6TST,R0             Low core addressability
                                                   175
                              00000140  00000000   176 SVOLDPSW EQU   ZVE6TST+X'140'         z/Arch Supervisor call old PSW

00000000                      00000000  000001A0   178          ORG   ZVE6TST+X'1A0'         z/Architecure RESTART PSW
000001A0  00000001 80000000                        179          DC    X'0000000180000000'
000001A8  00000000 00000200                        180          DC    AD(BEGIN)

000001B0                      000001B0  000001D0   182          ORG   ZVE6TST+X'1D0'         z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                        183          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                        184          DC    AD(X'DEAD')

000001E0                      000001E0  00000200   186          ORG   ZVE6TST+X'200'         Start of actual test program...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   188
                                                   189 **********************************************************************
                                                   190 *               The actual "ZVE6TST" program itself...
                                                   191 **********************************************************************
                                                   192 *
                                                   193 *  Architecture Mode: z/Arch
                                                   194 *  Register Usage:
                                                   195 *
                                                   196 *   R0       (work)
                                                   197 *   R1-4     (work)
                                                   198 *   R5       Testing control table - current test base
                                                   199 *   R6-R7    (work)
                                                   200 *   R8       First base register
                                                   201 *   R9       Second base register
                                                   202 *   R10      Third base register
                                                   203 *   R11      E6TEST call return
                                                   204 *   R12      E6TESTS register
                                                   205 *   R13      (work)
                                                   206 *   R14      Subroutine call
                                                   207 *   R15      Secondary Subroutine call or work
                                                   208 *
                                                   209 **********************************************************************

00000200                       00000200            211          USING  BEGIN,R8                 FIRST Base Register
00000200                       00001200            212          USING  BEGIN+4096,R9            SECOND Base Register
00000200                       00002200            213          USING  BEGIN+8192,R10           THIRD Base Register
                                                   214
00000200  0580                                     215 BEGIN    BALR  R8,0                      Initalize FIRST base register
00000202  0680                                     216          BCTR  R8,0                      Initalize FIRST base register
00000204  0680                                     217          BCTR  R8,0                      Initalize FIRST base register
                                                   218
00000206  4190 8800                      00000800  219          LA    R9,2048(,R8)              Initalize SECOND base register
0000020A  4190 9800                      00000800  220          LA    R9,2048(,R9)              Initalize SECOND base register
                                                   221
0000020E  41A0 9800                      00000800  222          LA    R10,2048(,R9)             Initalize THIRD base register
00000212  41A0 A800                      00000800  223          LA    R10,2048(,R10)            Initalize THIRD base register
                                                   224
00000216  B600 8474                      00000674  225          STCTL R0,R0,CTLR0               Store CR0 to enable AFP
0000021A  9604 8475                      00000675  226          OI    CTLR0+1,X'04'             Turn on AFP bit
0000021E  9602 8475                      00000675  227          OI    CTLR0+1,X'02'             Turn on Vector bit
00000222  B700 8474                      00000674  228          LCTL  R0,R0,CTLR0               Reload updated CR0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                    230 ********************************************************************
                                                    231 * Is Vector packed-decimal enhancement 3 facility installed  (bit 199)
                                                    232 ********************************************************************
                                                    233
                                                    234          FCHECK 199,'vector packed-decimal-enhancement 3'
00000226  47F0 80B8                     000002B8    235+         B     X0001
                                                    236+*                                  Fcheck data area
                                                    237+*                                  skip messgae
0000022A  40404040 4040E292                         238+SKT0001  DC    C'      Skipping tests: '
00000240  A58583A3 96994097                         239+         DC    C'vector packed-decimal-enhancement 3'
00000263  40868183 899389A3                         240+         DC    C' facility (bit 199) is not installed.'
                              0000005E  00000001    241+SKL0001  EQU   *-SKT0001
                                                    242+*                                  facility bits
00000288  00000000 00000000                         243+         DS    FD                  gap
00000290  00000000 00000000                         244+FB0001   DS    4FD
000002B0  00000000 00000000                         245+         DS    FD                  gap
                                                    246+*
                              000002B8  00000001    247+X0001    EQU   *
000002B8  4100 0004                     00000004    248+         LA    R0,((X0001-FB0001)/8)-1
000002BC  B2B0 8090                     00000290    249+         STFLE FB0001              get facility bits
000002C0  B982 0000                                 250+         XGR   R0,R0
000002C4  4300 80A8                     000002A8    251+         IC    R0,FB0001+24        get fbit byte
000002C8  5400 8498                     00000698    252+         N     R0,=F'1'            is bit set?
000002CC  4770 80E0                     000002E0    253+         BNZ   XC0001
                                                    254+*
                                                    255+* facility bit not set, issue message and exit
                                                    256+*
000002D0  4100 005E                     0000005E    257+         LA    R0,SKL0001          message length
000002D4  4110 802A                     0000022A    258+         LA    R1,SKT0001          message address
000002D8  4520 8390                     00000590    259+         BAL   R2,MSG
000002DC  47F0 8458                     00000658    260+         B     EOJ
                              000002E0  00000001    261+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   263 ********************************************************************
                                                   264 *              Do tests in the E6TESTS table
                                                   265 ********************************************************************
                                                   266
000002E0  58C0 849C                     0000069C   267          L     R12,=A(E6TESTS)         get table of test addresses
                                                   268
                              000002E4  00000001   269 NEXTE6   EQU   *
000002E4  5850 C000                     00000000   270          L     R5,0(0,R12)             get test address
000002E8  1255                                     271          LTR   R5,R5                      have a test?
000002EA  4780 8348                     00000548   272          BZ    ENDTEST                       done?
                                                   273
000002EE                      00000000             274          USING E6TEST,R5
                                                   275
000002EE  4800 5004                     00000004   276          LH    R0,TNUM                 save current test number
000002F2  5000 8E04                     00001004   277          ST    R0,TESTING              for easy reference
                                                   278
000002F6  58B0 5000                     00000000   279          L     R11,TSUB                get address of test routine
000002FA  05BB                                     280          BALR  R11,R11                 do test
                                                   281
                                                   282 *  check CC
000002FC  45F0 811C                     0000031C   283          BAL   R15,CCCHECK
                                                   284
                                                   285 * do cross check
00000300  45F0 81C0                     000003C0   286          BAL   R15,XCHECK
                                                   287
                                                   288 *  validate results
                                                   289
00000304  E310 501C 0014                0000001C   290          LGF   R1,READDR               expected result address
0000030A  D50F 5028 1000      00000028  00000000   291          CLC   V1OUTPUT,0(R1)
00000310  4770 82D0                     000004D0   292          BNE   FAILMSG                 no, issue failed message
                                                   293
                              00000314  00000001   294 DONEXT   EQU   *
00000314  41C0 C004                     00000004   295          LA    R12,4(0,R12)            next test address
00000318  47F0 80E4                     000002E4   296          B     NEXTE6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   298 *--------------------------------------------------------------------
                                                   299 * CC check
                                                   300 *--------------------------------------------------------------------
                             0000031C  00000001    301 CCCHECK  EQU   *
0000031C  D200 5071 84A8     00000071  000006A8    302          MVC   CCFAILED,=CL1'N'       initializae cc failed
                                                   303 *
                                                   304 * extract CC from PSW
                                                   305 *
00000322  5810 5068                    00000068    306          L     R1,CCPSW
00000326  8810 000C                    0000000C    307          SRL   R1,12
0000032A  5410 84A0                    000006A0    308          N     R1,=XL4'3'
0000032E  4210 5070                    00000070    309          STC   R1,CCFOUND             save cc
                                                   310 *
                                                   311 * check CC was expected
                                                   312 *
00000332  D500 500A 5070     0000000A  00000070    313          CLC   CC,CCFOUND
00000338  4770 813E                    0000033E    314          BNE   CCMSG
                                                   315
0000033C  07FF                                     316          BR    R15                    return from CCcheck
                                                   317
                                                   318 *********************************************************************
                                                   319 * cc was not as expected
                                                   320 *********************************************************************
                             0000033E  00000001    321 CCMSG    EQU   *
0000033E  D200 5071 84A9     00000071  000006A9    322          MVC   CCFAILED,=CL1'Y'        set cc failed
                                                   323 *
                                                   324 * FILL IN MESSAGE
                                                   325 *
00000344  4820 5004                    00000004    326          LH    R2,TNUM                 get test number and convert
00000348  4E20 8F36                    00001136    327          CVD   R2,DECNUM
0000034C  D211 8F20 8F0A     00001120  0000110A    328          MVC   PRT3,EDIT
00000352  DE11 8F20 8F36     00001120  00001136    329          ED    PRT3,DECNUM
00000358  D202 8E67 8F2D     00001067  0000112D    330          MVC   CCPRTNUM(3),PRT3+13     fill in message with test #
                                                   331
0000035E  D207 8E90 5010     00001090  00000010    332          MVC   CCPRTNAME,OPNAME        fill in message with instruction
                                                   333
00000364  B982 0022                                334          XGR   R2,R2                   get CC as U8
00000368  4320 500A                    0000000A    335          IC    R2,CC
0000036C  4E20 8F36                    00001136    336          CVD   R2,DECNUM               and convert
00000370  D211 8F20 8F0A     00001120  0000110A    337          MVC   PRT3,EDIT
00000376  DE11 8F20 8F36     00001120  00001136    338          ED    PRT3,DECNUM
0000037C  D200 8EA6 8F2F     000010A6  0000112F    339          MVC   CCPRTEXP(1),PRT3+15     fill in message with CC field
                                                   340
00000382  B982 0022                                341          XGR   R2,R2                   get CCFOUND as U8
00000386  4320 5070                    00000070    342          IC    R2,CCFOUND
0000038A  4E20 8F36                    00001136    343          CVD   R2,DECNUM               and convert
0000038E  D211 8F20 8F0A     00001120  0000110A    344          MVC   PRT3,EDIT
00000394  DE11 8F20 8F36     00001120  00001136    345          ED    PRT3,DECNUM
0000039A  D200 8EB6 8F2F     000010B6  0000112F    346          MVC   CCPRTGOT(1),PRT3+15    fill in message with ccfound
                                                   347
000003A0  4100 005E                    0000005E    348          LA    R0,CCPRTLNG            message length
000003A4  4110 8E5A                    0000105A    349          LA    R1,CCPRTLINE           messagfe address
000003A8  50F0 81B8                    000003B8    350          ST    R15,CCR15              save r15
000003AC  45F0 8356                    00000556    351          BAL   R15,RPTERROR
                                                   352
000003B0  58F0 81B8                    000003B8    353          L     R15,CCR15
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000003B4  07FF                                     354          BR    R15
                                                   355 *
000003B8  00000000 00000000                        356 CCR15    DS    FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   358 *--------------------------------------------------------------------
                                                   359 * cross check that the result can be converted back to the source
                                                   360 * Note: skip xcheck - if skip requested (S or Y)
                                                   361 *                  - if instruction is not VCVDQ
                                                   362 *                  - if CC check failed
                                                   363 *                  - if CC=3 (overflowed)
                                                   364 *--------------------------------------------------------------------
                                000003C0  00000001  365 XCHECK   EQU   *
000003C0  D207 5040 8480      00000040  00000680  366          MVC   SKIPXC,=CL8'SKIP XC '
000003C6  D500 5007 84AA      00000007  000006AA  367          CLC   XCSKIP,=CL1'S'         skip xcheck requested
000003CC  478F 0000                     00000000  368          BE    0(R15)                    skip == S, so exit
000003D0  D500 5007 84A9      00000007  000006A9  369          CLC   XCSKIP,=CL1'Y'         skip xcheck requested
000003D6  478F 0000                     00000000  370          BE    0(R15)                    skip == Y, so exit
                                                   371 *
                                                   372 *        cross check depends on the instruction
                                                   373 *
000003DA  D207 5040 8488      00000040  00000688  374          MVC   SKIPXC,=CL8'        '
000003E0  D507 5010 8490      00000010  00000690  375          CLC   OPNAME,=CL8'VCVDQ   '
000003E6  477F 0000                     00000000  376          BNE    0(R15)                  so exit
                                                   377 *
                                                   378 *        cross check depends on CC=3 overflow check
                                                   379 *
000003EA  D500 5070 84AB      00000070  000006AB  380          CLC   CCFOUND,=XL1'3'
000003F0  478F 0000                     00000000  381          BE    0(R15)                   so exit
                                                   382 *
                                                   383 *        cross check depends on CC check
                                                   384 *
000003F4  D500 5071 84A9      00000071  000006A9  385          CLC   CCFAILED,=CL1'Y'
000003FA  478F 0000                     00000000  386          BE    0(R15)                   so exit
                                                   387 *
                                                   388 * do cross check: VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                   389 *
                                                   390          VCVBQ x'00'

000003FE                                           392+         DS    0H                       E64E VCVBQ
000003FE  E6                                       393+         DC    X'E6'                    - VECTOR CONVERT TO BINARY (128)
000003FF  32                                       394+         DC    X'32'                     v3, v2
00000400  00                                       395+         DC    X'00'                    reserved
00000401  00                                       396+         DC    HL1'0'                    m3, reserved
00000402  00                                       397+         DC    X'00'                    reserved, RXB
00000403  4E                                       398+         DC    X'4E'

00000404  E730 5050 000E                00000050  400          VST   V3,XCOUTPUT
                                                   401
0000040A  E310 500C 0014                0000000C  402          LGF   R1,V2ADDR             expected source address
00000410  D50F 5050 1000                00000000  403          CLC   XCOUTPUT(16),0(R1)
00000416  4770 821C                     0000041C  404          BNE   XCFAILMSG
0000041A  07FF                                     405          BR    R15                   return from xcheck
                                                   406
                                                   407
                                                   408 * xcheck failed message
0000041C                                           409 XCFAILMSG DS   0H
0000041C  4820 5004                     00000004  410          LH    R2,TNUM               get test number and convert
00000420  4E20 8F36                     00001136  411          CVD   R2,DECNUM
00000424  D211 8F20 8F0A      00001120  0000110A  412          MVC   PRT3,EDIT
0000042A  DE11 8F20 8F36      00001120  00001136  413          ED    PRT3,DECNUM
```

```
  LOC        OBJECT CODE        ADDR1     ADDR2     STMT

00000430  D202 8EC5 8F2D      000010C5  0000112D   414          MVC   XCPTNUM(3),PRT3+13   fill in message with test #
                                                   415
00000436  D207 8EEE 5010      000010EE  00000010   416          MVC   XCPNAME,OPNAME       fill in message with instruction
                                                   417
0000043C  B982 0022                                418          XGR   R2,R2
00000440  4320 5008                     00000008   419          IC    R2,I3                get i3 and convert
00000444  4E20 8F36                     00001136   420          CVD   R2,DECNUM
00000448  D211 8F20 8F0A      00001120  0000110A   421          MVC   PRT3,EDIT
0000044E  DE11 8F20 8F36      00001120  00001136   422          ED    PRT3,DECNUM
00000454  D202 8EFF 8F2D      000010FF  0000112D   423          MVC   XCPI3(3),PRT3+13     fill in message with i3 field
                                                   424 *
                                                   425
0000045A  B982 0022                                426          XGR   R2,R2
0000045E  4320 5009                     00000009   427          IC    R2,M4                get m4 and convert
00000462  4E20 8F36                     00001136   428          CVD   R2,DECNUM
00000466  D211 8F20 8F0A      00001120  0000110A   429          MVC   PRT3,EDIT
0000046C  DE11 8F20 8F36      00001120  00001136   430          ED    PRT3,DECNUM
00000472  D201 8F07 8F2E      00001107  0000112E   431          MVC   XCPM4(2),PRT3+14     fill in message with m4 field
                                                   432 *
00000478  50F0 82C8                     000004C8   433          ST    R15,XCR15            save r15
0000047C  4100 0052                     00000052   434          LA    R0,XCPLNG            message length
00000480  4110 8EB8                     000010B8   435          LA    R1,XCPLINE           messagfe address
00000484  45F0 8356                     00000556   436          BAL   R15,RPTERROR
00000488  58F0 82C8                     000004C8   437          L     R15,XCR15
                                                   438
0000048C  5800 8498                     00000698   439          L     R0,=F'1'             set failed test indicator
00000490  5000 8E00                     00001000   440          ST    R0,FAILED
00000494  07FF                                     441          BR    R15                  return from xcheck
                                                   442
00000498                                           443          DS    0FD
00000498  00000000 00000000                        444 XCRESULT DS    XL16
000004A8  00000000 00000000                        445 XCV1     DS    XL16
000004B8  00000000 00000000                        446 XCV2     DS    XL16
000004C8  00000000 00000000                        447 XCR15    DS    FD
                                                   448
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   450 ***********************************************************************
                                                   451 * result not as expected:
                                                   452 *        issue message with test number, instruction under test
                                                   453 *              and instruction m4
                                                   454 ***********************************************************************
                                000004D0  00000001 455 FAILMSG  EQU   *
000004D0  4820 5004                       00000004 456          LH    R2,TNUM               get test number and convert
000004D4  4E20 8F36                       00001136 457          CVD   R2,DECNUM
000004D8  D211 8F20 8F0A      00001120  0000110A 458          MVC   PRT3,EDIT
000004DE  DE11 8F20 8F36      00001120  00001136 459          ED    PRT3,DECNUM
000004E4  D202 8E15 8F2D      00001015  0000112D 460          MVC   PRTNUM(3),PRT3+13     fill in message with test #
                                                   461
000004EA  D207 8E3E 5010      0000103E  00000010 462          MVC   PRTNAME,OPNAME        fill in message with instruction
                                                   463 *
000004F0  B982 0022                                464          XGR   R2,R2
000004F4  4320 5008                       00000008 465          IC    R2,I3                 get I3 and convert
000004F8  4E20 8F36                       00001136 466          CVD   R2,DECNUM
000004FC  D211 8F20 8F0A      00001120  0000110A 467          MVC   PRT3,EDIT
00000502  DE11 8F20 8F36      00001120  00001136 468          ED    PRT3,DECNUM
00000508  D202 8E4F 8F2D      0000104F  0000112D 469          MVC   PRTI3(3),PRT3+13      fill in message with i3 field
                                                   470 *
0000050E  B982 0022                                471          XGR   R2,R2
00000512  4320 5009                       00000009 472          IC    R2,M4                 get m4 and convert
00000516  4E20 8F36                       00001136 473          CVD   R2,DECNUM
0000051A  D211 8F20 8F0A      00001120  0000110A 474          MVC   PRT3,EDIT
00000520  DE11 8F20 8F36      00001120  00001136 475          ED    PRT3,DECNUM
00000526  D201 8E57 8F2E      00001057  0000112E 476          MVC   PRTM4(2),PRT3+14      fill in message with m4 field
                                                   477 *
0000052C  4100 0052                       00000052 478          LA    R0,PRTLNG             message length
00000530  4110 8E08                       00001008 479          LA    R1,PRTLINE            messagfe address
00000534  45F0 8356                       00000556 480          BAL   R15,RPTERROR

                                                   482 ***********************************************************************
                                                   483 * continue after a failed test
                                                   484 ***********************************************************************
                                00000538  00000001 485 FAILCONT EQU   *
00000538  5800 8498                       00000698 486          L     R0,=F'1'           set failed test indicator
0000053C  5000 8E00                       00001000 487          ST    R0,FAILED
                                                   488
00000540  41C0 C004                       00000004 489          LA    R12,4(0,R12)       next test address
00000544  47F0 80E4                       000002E4 490          B     NEXTE6

                                                   492 ***********************************************************************
                                                   493 * end of testing; set ending psw
                                                   494 ***********************************************************************
                                00000548  00000001 495 ENDTEST  EQU   *
00000548  5810 8E00                       00001000 496          L     R1,FAILED          did a test fail?
0000054C  1211                                     497          LTR   R1,R1
0000054E  4780 8458                       00000658 498          BZ    EOJ                   No, exit
00000552  47F0 8470                       00000670 499          B     FAILTEST              Yes, exit with BAD PSW
                                                   500
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   502 ***********************************************************************
                                                   503 *        RPTERROR          Report instruction test in error
                                                   504 *                             R0 = MESSGAE LENGTH
                                                   505 *                             R1 = ADDRESS OF MESSAGE
                                                   506 ***********************************************************************

00000556  50F0 8374                     00000574   508 RPTERROR ST    R15,RPTSAVE          Save return address
0000055A  5050 8378                     00000578   509          ST    R5,RPTSVR5           Save R5
                                                   510 *
                                                   511 *        Use Hercules Diagnose for Message to console
                                                   512 *
0000055E  9002 8380                     00000580   513          STM   R0,R2,RPTDWSAV       save regs used by MSG
00000562  4520 8390                     00000590   514          BAL   R2,MSG               call Hercules console MSG display
00000566  9802 8380                     00000580   515          LM    R0,R2,RPTDWSAV       restore regs


0000056A  5850 8378                     00000578   517          L     R5,RPTSVR5           Restore R5
0000056E  58F0 8374                     00000574   518          L     R15,RPTSAVE          Restore return address
00000572  07FF                                     519          BR    R15                  Return to caller

00000574  00000000                                 521 RPTSAVE  DC    F'0'                 R15 save area
00000578  00000000                                 522 RPTSVR5  DC    F'0'                 R5 save area

00000580  00000000 00000000                        524 RPTDWSAV DC    2D'0'                R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   526 **********************************************************************
                                                   527 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                   528 *              R2 = return address
                                                   529 **********************************************************************

00000590  4900 84A4                     000006A4   531 MSG      CH    R0,=H'0'                 Do we even HAVE a message?
00000594  07D2                                     532          BNHR  R2                       No, ignore

00000596  9002 83CC                     000005CC   534          STM   R0,R2,MSGSAVE            Save registers

0000059A  4900 84A6                     000006A6   536          CH    R0,=AL2(L'MSGMSG)        Message length within limits?
0000059E  47D0 83A6                     000005A6   537          BNH   MSGOK                    Yes, continue
000005A2  4100 005F                     0000005F   538          LA    R0,L'MSGMSG              No, set to maximum

000005A6  1820                                     540 MSGOK    LR    R2,R0                    Copy length to work register
000005A8  0620                                     541          BCTR  R2,0                     Minus-1 for execute
000005AA  4420 83D8                     000005D8   542          EX    R2,MSGMVC                Copy message to O/P buffer

000005AE  4120 200A                     0000000A   544          LA    R2,1+L'MSGCMD(,R2)       Calculate true command length
000005B2  4110 83DE                     000005DE   545          LA    R1,MSGCMD                Point to true command

000005B6  83120008                                 547          DC    X'83',X'12',X'0008'      Issue Hercules Diagnose X'008'
000005BA  4780 83C6                     000005C6   548          BZ    MSGRET                   Return if successful
                                                   549
000005BE  1222                                     550          LTR   R2,R2                    Is Diag8 Ry (R2) 0?
000005C0  4780 83C6                     000005C6   551          BZ    MSGRET                     an error occurred but coninue
                                                   552
000005C4  0000                                     553          DC    H'0'                     CRASH for debugging purposes

000005C6  9802 83CC                     000005CC   555 MSGRET   LM    R0,R2,MSGSAVE            Restore registers
000005CA  07F2                                     556          BR    R2                       Return to caller




000005CC  00000000 00000000                        558 MSGSAVE  DC    3F'0'                    Registers save area
000005D8  D200 83E7 1000       000005E7  00000000  559 MSGMVC   MVC   MSGMSG(0),0(R1)          Executed instruction

000005DE  D4E2C7D5 D6C8405C                        561 MSGCMD   DC    C'MSGNOH * '             *** HERCULES MESSAGE COMMAND ***
000005E7  40404040 40404040                        562 MSGMSG   DC    CL95' '                  The message text to be displayed
                                                   563
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   565 ***********************************************************************
                                                   566 *        Normal completion or Abnormal termination PSWs
                                                   567 ***********************************************************************

00000648  00020001 80000000                        569 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

00000658  B2B2 8448                      00000648  571 EOJ      LPSWE EOJPSW                 Normal completion

00000660  00020001 80000000                        573 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

00000670  B2B2 8460                      00000660  575 FAILTEST LPSWE FAILPSW                Abnormal termination

                                                   577 ***********************************************************************
                                                   578 *        Working Storage
                                                   579 ***********************************************************************

00000674  00000000                                 581 CTLR0    DS    F                 CR0
00000678  00000000                                 582          DS    F
0000067C  00000000                                 583 FPCINIT  DC    XL4'00000000'     FPC before test

                                                   585
00000680                                           586          LTORG ,                 Literals pool
00000680  E2D2C9D7 40E7C340                        587                =CL8'SKIP XC '
00000688  40404040 40404040                        588                =CL8'        '
00000690  E5C3E5C4 D8404040                        589                =CL8'VCVDQ   '
00000698  00000001                                 590                =F'1'
0000069C  0000478C                                 591                =A(E6TESTS)
000006A0  00000003                                 592                =XL4'3'
000006A4  0000                                     593                =H'0'
000006A6  005F                                     594                =AL2(L'MSGMSG)
000006A8  D5                                       595                =CL1'N'
000006A9  E8                                       596                =CL1'Y'
000006AA  E2                                       597                =CL1'S'
000006AB  03                                       598                =XL1'3'
                                                   599
                                                   600 *        some constants
                                                   601
                               00000400  00000001  602 K        EQU   1024              One KB
                               00001000  00000001  603 PAGE     EQU   (4*K)             Size of one page
                               00010000  00000001  604 K64      EQU   (64*K)            64 KB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                             00100000  00000001   605 MB       EQU   (K*K)             1 MB
                                                  606
                                                  607
                             AABBCCDD  00000001   608 REG2PATT EQU   X'AABBCCDD'    Polluted Register pattern
                             000000DD  00000001   609 REG2LOW  EQU         X'DD'    (last byte above)
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

                                                  611 *=====================================================================
                                                  612 *
                                                  613 *  NOTE: start data on an address that is easy to display
                                                  614 *        within Hercules
                                                  615 *
                                                  616 *=====================================================================
                                                  617
000006AC                    000006AC  00001000   618          ORG   ZVE6TST+X'1000'
00001000  00000000                               619 FAILED   DC    F'0'                        some test failed?
00001004  00000000                               620 TESTING  DC    F'0'                        current test #

                                                  622 ********************************************************************
                                                  623 *        TEST failed : result messgae
                                                  624 ********************************************************************
                                                  625 *
                                                  626 *        failed message and associated editting
                                                  627 *
00001008  40404040 4040E385                      628 PRTLINE  DC    C'      Test # '
00001015  A7A7A7                                 629 PRTNUM   DC    C'xxx'
00001018  40868189 9385847A                      630          DC    c' failed: wrong result for instruction '
0000103E  A7A7A7A7 A7A7A7A7                      631 PRTNAME  DC    CL8'xxxxxxxx'
00001046  40A689A3 884089F3                      632          DC    C' with i3='
0000104F  A7A7A7                                 633 PRTI3    DC    C'xxx'
00001052  6B                                     634          DC    C','
00001053  4094F47E                               635          DC    C' m4='
00001057  A7A7                                   636 PRTM4    DC    C'xx'
00001059  4B                                     637          DC    C'.'
                            00000052  00000001   638 PRTLNG   EQU   *-PRTLINE

                                                  640 ********************************************************************
                                                  641 *        TEST failed : CC message
                                                  642 ********************************************************************
                                                  643 *
                                                  644 *        failed message and associated editting
                                                  645 *
0000105A  40404040 4040E385                      646 CCPRTLINE DC   C'      Test # '
00001067  A7A7A7                                 647 CCPRTNUM DC    C'xxx'
0000106A  40868189 9385847A                      648          DC    C' failed: wrong CC     for instruction '
00001090  A7A7A7A7 A7A7A7A7                      649 CCPRTNAME DC    CL8'xxxxxxxx'
00001098  4085A797 8583A385                      650          DC    C' expected: cc='
000010A6  A7                                     651 CCPRTEXP DC    C'x'
000010A7  6B                                     652          DC    C','
000010A8  40998583 8589A585                      653          DC    C' received: cc='
000010B6  A7                                     654 CCPRTGOT DC    C'x'
000010B7  4B                                     655          DC    C'.'
                            0000005E  00000001   656 CCPRTLNG   EQU   *-CCPRTLINE

                                                  658 ********************************************************************
                                                  659 *        TEST failed : XCHECK
                                                  660 ********************************************************************
                                                  661 *
                                                  662 *        XCHECK failed message
                                                  663 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000010B8  40404040 4040E385                         664 XCPLINE  DC    C'       Test # '
000010C5  A7A7A7                                    665 XCPTNUM  DC    C'xxx'
000010C8  40868189 9385847A                         666          DC    c' failed: cross check  for instruction '
000010EE  A7A7A7A7 A7A7A7A7                         667 XCPNAME  DC    CL8'xxxxxxxx'
000010F6  40A689A3 884089F3                         668          DC    C' with i3='
000010FF  A7A7A7                                    669 XCPI3    DC    C'xxx'
00001102  6B                                        670          DC    C','
00001103  4094F47E                                  671          DC    C' m4='
00001107  A7A7                                      672 XCPM4    DC    C'xx'
00001109  4B                                        673          DC    C'.'
                             00000052  00000001     674 XCPLNG   EQU   *-XCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   676 ***********************************************************************
                                                   677 *        TEST failed : message working storge
                                                   678 ***********************************************************************
0000110A  40212020 20202020                        679 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                   680
0000111C  7E7E7E6E                                 681          DC    C'===>'
00001120  40404040 40404040                        682 PRT3     DC    CL18' '
00001132  4C7E7E7E                                 683          DC    C'<==='
00001136  00000000 00000000                        684 DECNUM   DS    CL16


                                                   686 *
                                                   687 *        Vector instruction results, pollution and input
                                                   688 *
00001146  00000000 00000000                        689          DS    XL16                                        gap
00001156  FFFFFFFF FFFFFFFF                        690 V1FUDGE  DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'    V1 FUDGE
00001166  12345678 90123456                        691 V1INPUT  DC    XL16'1234567890123456789012345678901D'    V1 input
00001176  00000000 00000000                        692          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   694 ***********************************************************************
                                                   695 *        E6TEST DSECT
                                                   696 ***********************************************************************

                                                   698 E6TEST   DSECT ,
00000000  00000000                                 699 TSUB     DC    A(0)                 pointer  to test
00000004  0000                                     700 TNUM     DC    H'00'                Test Number
00000006  00                                       701          DC    X'00'
00000007  40                                       702 XCSKIP   DC    CL1' '               Y = skip cross check
00000008  00                                       703 I3       DC    HL1'00'              I3 used
00000009  00                                       704 M4       DC    HL1'00'              M4 used
0000000A  00                                       705 CC       DC    HL1'00'              cc
0000000B  00                                       706 CCMASK   DC    HL1'00'              not expected CC mask
                                                   707
0000000C  00000000                                 708 V2ADDR   DC    A(0)                 address of v2: 16-byte packed decimal
00000010  40404040 40404040                        709 OPNAME   DC    CL8' '               E6 name
00000018  00000000                                 710 RELEN    DC    A(0)                 result length
0000001C  00000000                                 711 READDR   DC    A(0)                 expected result address
00000020  00000000 00000000                        712          DS    FD                      gap
00000028  00000000 00000000                        713 V1OUTPUT DS    XL16                 V1 Output
00000038  00000000 00000000                        714          DS    FD                      gap
00000040  40404040 40404040                        715 SKIPXC   DC    CL8' '               was cross check skipped?
00000048  00000000 00000000                        716          DS    FD                      gap
00000050  00000000 00000000                        717 XCOUTPUT DS    XL16                 Cross check Output
00000060  00000000 00000000                        718          DS    FD                      gap
                                                   719
00000068  00000000 00000000                        720 CCPSW    DS    2F             extract PSW after test (has CC)
00000070  00                                       721 CCFOUND  DS    X              extracted cc
00000071  00                                       722 CCFAILED DS    X              CC  check failed
                                                   723
00000074  00000000                                 724 WK1      DS    F
00000078                                           725          DS    0F
                                                   726 **
                                                   727 *        test routine will be here (from VRI-j macro)

                            00000000  000048BB     729 ZVE6TST  CSECT ,
00001188                                           730          DS    0F

                                                   732
                                                   733 *
                                                   734 * macro to generate individual test
                                                   735 *
                                                   736          MACRO
                                                   737          VRI_J &INST,&I3,&M4,&CC,&SKIP
                                                   738 .*                                  &INST  - VRI-j instruction under test
                                                   739 .*                                  &i3    - i3 field
                                                   740 .*                                  &m4    - m4 field
                                                   741 .*                                  &CC    - expected CC
                                                   742 .*                                  &SKIP  - S = skip cross check
                                                   743 .*
                                                   744          LCLA  &XCC(4)  &CC has mask values for FAILED condition codes
                                                   745 &XCC(1)  SETA  7                    CC != 0
                                                   746 &XCC(2)  SETA  11                   CC != 1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   747 &XCC(3)  SETA  13                 CC != 2
                                                   748 &XCC(4)  SETA  14                 CC != 3
                                                   749
                                                   750          GBLA  &TNUM
                                                   751 &TNUM    SETA  &TNUM+1
                                                   752
                                                   753          DS    0FD
                                                   754          USING *,R5               base for test data and test routine
                                                   755
                                                   756 T&TNUM   DC    A(X&TNUM)          address of test routine
                                                   757          DC    H'&TNUM'           test number
                                                   758          DC    X'00'
                                                   759
                                                   760          DC    CL1'&SKIP'         S or Y = skip cross check
                                                   761          DC    &i3                i3
                                                   762          DC    &M4                m4
                                                   763          DC    HL1'&CC'           cc
                                                   764          DC    HL1'&XCC(&CC+1)'   cc failed mask
                                                   765
                                                   766 V2_&TNUM DC    A(RE&TNUM+16)      address of v2: 16-byte packed decimal
                                                   767          DC    CL8'&INST'         instruction name
                                                   768          DC    A(16)              result length
                                                   769          DC    A(RE&TNUM)         address of expected resul
                                                   770          DS    FD                 gap
                                                   771 V1O&TNUM DS    XL16               V1 output
                                                   772          DS    FD                    gap
                                                   773          DC    CL8' '             was cross check skipped?
                                                   774          DS    FD                    gap
                                                   775 XCO&TNUM DS    XL16               Cross check Output
                                                   776          DS    FD                    gap
                                                   777          DS    2F           extract PSW after test (has CC)
                                                   778          DS    X            extracted cc
                                                   779          DS    X            CC Check failed
                                                   780 .*
                                                   781 *
                                                   782 X&TNUM   DS    0F
                                                   783          VL    V2,V1FUDGE
                                                   784
                                                   785          LGF   R2,V2_&TNUM         get v2 address
                                                   786          VL    V2,0(R2)
                                                   787
                                                   788          &INST &I3,&M4             test instruction (dest is source)
                                                   789
                                                   790          EPSW  R2,R0              exptract psw
                                                   791          ST    R2,CCPSW               to save CC
                                                   792
                                                   793          VST   V2,V1O&TNUM        save instruction result
                                                   794          BR    R11                return
                                                   795
                                                   796 RE&TNUM  DS    0F                 expected 16 byte result
                                                   797          DROP  R5
                                                   798
                                                   799          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   801 *
                                                   802 * macro to generate table of pointers to individual tests
                                                   803 *
                                                   804          MACRO
                                                   805          PTTABLE
                                                   806          GBLA  &TNUM
                                                   807          LCLA  &CUR
                                                   808 &CUR     SETA  1
                                                   809 .*
                                                   810 TTABLE   DS    0F
                                                   811 .LOOP    ANOP
                                                   812 .*
                                                   813          DC    A(T&CUR)          TEST &CUR
                                                   814 .*
                                                   815 &CUR     SETA  &CUR+1
                                                   816          AIF   (&CUR LE &TNUM).LOOP
                                                   817 *
                                                   818          DC    A(0)              END OF TABLE
                                                   819          DC    A(0)
                                                   820 .*
                                                   821          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   823 ***********************************************************************
                                                   824 *        E6 VRI-j tests
                                                   825 ***********************************************************************
                                                   826          PRINT DATA
                                                   827 *
                                                   828 *        E64A VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                   829 *
                                                   830 *----------------------------------------------------------------------
                                                   831 *   VRI-j   instruction, i3, m4, CC, SKIP     (Skip cross check if "S")
                                                   832 *   followed by
                                                   833 *              v1 - 16 byte expected result - packed-decimal
                                                   834 *              v2 - 16 byte source vector   - 128-bit binary
                                                   835 *----------------------------------------------------------------------
                                                   836 *  VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                   837 *----------------------------------------------------------------------
                                                   838 *        i3:   x'9f"   (IOM=1, RDC=31)
                                                   839 *        m4:   x'1'    (LB=0, P1=0, CS=1)
                                                   840 *
                                                   841          VRI_J VCVDQ,X'9F',X'1',0,N
00001188                                           842+         DS    0FD
00001188                       00001188            843+         USING *,R5                base for test data and test routine
00001188  000011FC                                 844+T1       DC    A(X1)               address of test routine
0000118C  0001                                     845+         DC    H'1'                test number
0000118E  00                                       846+         DC    X'00'
0000118F  D5                                       847+         DC    CL1'N'              S or Y = skip cross check
00001190  9F                                       848+         DC    X'9F'               i3
00001191  01                                       849+         DC    X'1'                m4
00001192  00                                       850+         DC    HL1'0'              cc
00001193  07                                       851+         DC    HL1'7'              cc failed mask
00001194  00001234                                 852+V2_1     DC    A(RE1+16)           address of v2: 16-byte packed decimal
00001198  E5C3E5C4 D8404040                        853+         DC    CL8'VCVDQ'          instruction name
000011A0  00000010                                 854+         DC    A(16)               result length
000011A4  00001224                                 855+         DC    A(RE1)              address of expected resul
000011A8  00000000 00000000                        856+         DS    FD                  gap
000011B0  00000000 00000000                        857+V1O1     DS    XL16                V1 output
000011B8  00000000 00000000
000011C0  00000000 00000000                        858+         DS    FD                     gap
000011C8  40404040 40404040                        859+         DC    CL8' '              was cross check skipped?
000011D0  00000000 00000000                        860+         DS    FD                     gap
000011D8  00000000 00000000                        861+XCO1     DS    XL16                Cross check Output
000011E0  00000000 00000000
000011E8  00000000 00000000                        862+         DS    FD                     gap
000011F0  00000000 00000000                        863+         DS    2F            extract PSW after test (has CC)
000011F8  00                                       864+         DS    X             extracted cc
000011F9  00                                       865+         DS    X             CC Check failed
                                                   866+*
000011FC                                           867+X1       DS    0F
000011FC  E720 8F56 0006             00001156      868+         VL    V2,V1FUDGE
00001202  E320 500C 0014             00001194      869+         LGF   R2,V2_1              get v2 address
00001208  E722 0000 0006             00000000      870+         VL    V2,0(R2)

0000120E                                           873+         DS    0H                        E64A VCVDQ
0000120E  E6                                       874+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000120F  22                                       875+         DC    X'22'                      v1, v2
00001210  00                                       876+         DC    X'00'                     reserved
00001211  19F0                                     877+         DC    HL2'6640'                  m4, i3, rXB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001213  4A                                       878+         DC    X'4A'

00001214  B98D 0020                                880+         EPSW  R2,R0              exptract psw
00001218  5020 5068                      00000068  881+         ST    R2,CCPSW               to save CC
0000121C  E720 5028 000E                 000011B0  882+         VST   V2,V1O1            save instruction result
00001222  07FB                                     883+         BR    R11                return
00001224                                           884+RE1      DS    0F                 expected 16 byte result
00001224                                           885+         DROP  R5
00001224  00000000 00000000                        886          DC    XL16'0000000000000000 000000000000000C'      v1
0000122C  00000000 0000000C
00001234  00000000 00000000                        887          DC    XL16'0000000000000000 0000000000000000'      v2
0000123C  00000000 00000000
                                                   888
                                                   889          VRI_J VCVDQ,X'9F',X'1',0,N
00001248                                           890+         DS    0FD
00001248                         00001248          891+         USING *,R5               base for test data and test routine
00001248  000012BC                                 892+T2       DC    A(X2)              address of test routine
0000124C  0002                                     893+         DC    H'2'               test number
0000124E  00                                       894+         DC    X'00'
0000124F  D5                                       895+         DC    CL1'N'             S or Y = skip cross check
00001250  9F                                       896+         DC    X'9F'              i3
00001251  01                                       897+         DC    X'1'               m4
00001252  00                                       898+         DC    HL1'0'             cc
00001253  07                                       899+         DC    HL1'7'             cc failed mask
00001254  000012F4                                 900+V2_2     DC    A(RE2+16)          address of v2: 16-byte packed decimal
00001258  E5C3E5C4 D8404040                        901+         DC    CL8'VCVDQ'         instruction name
00001260  00000010                                 902+         DC    A(16)              result length
00001264  000012E4                                 903+         DC    A(RE2)             address of expected resul
00001268  00000000 00000000                        904+         DS    FD                 gap
00001270  00000000 00000000                        905+V1O2     DS    XL16               V1 output
00001278  00000000 00000000
00001280  00000000 00000000                        906+         DS    FD                    gap
00001288  40404040 40404040                        907+         DC    CL8' '             was cross check skipped?
00001290  00000000 00000000                        908+         DS    FD                    gap
00001298  00000000 00000000                        909+XCO2     DS    XL16               Cross check Output
000012A0  00000000 00000000
000012A8  00000000 00000000                        910+         DS    FD                    gap
000012B0  00000000 00000000                        911+         DS    2F           extract PSW after test (has CC)
000012B8  00                                       912+         DS    X            extracted cc
000012B9  00                                       913+         DS    X            CC Check failed
                                                   914+*
000012BC                                           915+X2       DS    0F
000012BC  E720 8F56 0006                 00001156  916+         VL    V2,V1FUDGE
000012C2  E320 500C 0014                 00001254  917+         LGF   R2,V2_2             get v2 address
000012C8  E722 0000 0006                 00000000  918+         VL    V2,0(R2)

000012CE                                           921+         DS    0H                       E64A VCVDQ
000012CE  E6                                       922+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
000012CF  22                                       923+         DC    X'22'                     v1, v2
000012D0  00                                       924+         DC    X'00'                    reserved
000012D1  19F0                                     925+         DC    HL2'6640'                 m4, i3, rXB
000012D3  4A                                       926+         DC    X'4A'

000012D4  B98D 0020                                928+         EPSW  R2,R0              exptract psw
000012D8  5020 5068                      00000068  929+         ST    R2,CCPSW               to save CC
000012DC  E720 5028 000E                 00001270  930+         VST   V2,V1O2            save instruction result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000012E2  07FB                                     931+          BR     R11                 return
000012E4                                           932+RE2       DS     0F                  expected 16 byte result
000012E4                                           933+          DROP   R5
000012E4  00000000 00000000                        934           DC     XL16'0000000000000000 000000000000001C'      v1
000012EC  00000000 0000001C
000012F4  00000000 00000000                        935           DC     XL16'0000000000000000 0000000000000001'      v2
000012FC  00000000 00000001
                                                   936
                                                   937           VRI_J VCVDQ,X'9F',X'1',0,N
00001308                                           938+          DS     0FD
00001308                     00001308              939+          USING *,R5                 base for test data and test routine
00001308  0000137C                                 940+T3        DC     A(X3)               address of test routine
0000130C  0003                                     941+          DC     H'3'                test number
0000130E  00                                       942+          DC     X'00'
0000130F  D5                                       943+          DC     CL1'N'              S or Y = skip cross check
00001310  9F                                       944+          DC     X'9F'               i3
00001311  01                                       945+          DC     X'1'                m4
00001312  00                                       946+          DC     HL1'0'              cc
00001313  07                                       947+          DC     HL1'7'              cc failed mask
00001314  000013B4                                 948+V2_3      DC     A(RE3+16)           address of v2: 16-byte packed decimal
00001318  E5C3E5C4 D8404040                        949+          DC     CL8'VCVDQ'          instruction name
00001320  00000010                                 950+          DC     A(16)               result length
00001324  000013A4                                 951+          DC     A(RE3)              address of expected resul
00001328  00000000 00000000                        952+          DS     FD                  gap
00001330  00000000 00000000                        953+V1O3      DS     XL16                V1 output
00001338  00000000 00000000
00001340  00000000 00000000                        954+          DS     FD                     gap
00001348  40404040 40404040                        955+          DC     CL8' '              was cross check skipped?
00001350  00000000 00000000                        956+          DS     FD                     gap
00001358  00000000 00000000                        957+XCO3      DS     XL16                Cross check Output
00001360  00000000 00000000
00001368  00000000 00000000                        958+          DS     FD                     gap
00001370  00000000 00000000                        959+          DS     2F           extract PSW after test (has CC)
00001378  00                                       960+          DS     X            extracted cc
00001379  00                                       961+          DS     X            CC Check failed
                                                   962+*
0000137C                                           963+X3        DS     0F
0000137C  E720 8F56 0006               00001156    964+          VL     V2,V1FUDGE
00001382  E320 500C 0014               00001314    965+          LGF    R2,V2_3              get v2 address
00001388  E722 0000 0006               00000000    966+          VL     V2,0(R2)

0000138E                                           969+          DS     0H                        E64A VCVDQ
0000138E  E6                                       970+          DC     X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000138F  22                                       971+          DC     X'22'                      v1, v2
00001390  00                                       972+          DC     X'00'                     reserved
00001391  19F0                                     973+          DC     HL2'6640'                  m4, i3, rXB
00001393  4A                                       974+          DC     X'4A'

00001394  B98D 0020                                976+          EPSW   R2,R0               exptract psw
00001398  5020 5068                    00000068    977+          ST     R2,CCPSW                to save CC
0000139C  E720 5028 000E               00001330    978+          VST    V2,V1O3             save instruction result
000013A2  07FB                                     979+          BR     R11                 return
000013A4                                           980+RE3       DS     0F                  expected 16 byte result
000013A4                                           981+          DROP   R5
000013A4  00000000 00000000                        982           DC     XL16'0000000000000000 000000000000001D'      v1
000013AC  00000000 0000001D
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000013B4  FFFFFFFF FFFFFFFF                        983          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
000013BC  FFFFFFFF FFFFFFFF
                                                   984
                                                   985          VRI_J VCVDQ,X'9F',X'1',0,N
000013C8                                           986+         DS    0FD
000013C8                       000013C8            987+         USING *,R5                 base for test data and test routine
000013C8  0000143C                                 988+T4       DC    A(X4)                address of test routine
000013CC  0004                                     989+         DC    H'4'                 test number
000013CE  00                                       990+         DC    X'00'
000013CF  D5                                       991+         DC    CL1'N'               S or Y = skip cross check
000013D0  9F                                       992+         DC    X'9F'                i3
000013D1  01                                       993+         DC    X'1'                 m4
000013D2  00                                       994+         DC    HL1'0'               cc
000013D3  07                                       995+         DC    HL1'7'               cc failed mask
000013D4  00001474                                 996+V2_4     DC    A(RE4+16)            address of v2: 16-byte packed decimal
000013D8  E5C3E5C4 D8404040                        997+         DC    CL8'VCVDQ'           instruction name
000013E0  00000010                                 998+         DC    A(16)                result length
000013E4  00001464                                 999+         DC    A(RE4)               address of expected resul
000013E8  00000000 00000000                       1000+         DS    FD                   gap
000013F0  00000000 00000000                       1001+V1O4     DS    XL16                 V1 output
000013F8  00000000 00000000
00001400  00000000 00000000                       1002+         DS    FD                      gap
00001408  40404040 40404040                       1003+         DC    CL8' '               was cross check skipped?
00001410  00000000 00000000                       1004+         DS    FD                      gap
00001418  00000000 00000000                       1005+XCO4     DS    XL16                 Cross check Output
00001420  00000000 00000000
00001428  00000000 00000000                       1006+         DS    FD                      gap
00001430  00000000 00000000                       1007+         DS    2F             extract PSW after test (has CC)
00001438  00                                      1008+         DS    X              extracted cc
00001439  00                                      1009+         DS    X              CC Check failed
                                                  1010+*
0000143C                                          1011+X4       DS    0F
0000143C  E720 8F56 0006             00001156     1012+         VL    V2,V1FUDGE
00001442  E320 500C 0014             000013D4     1013+         LGF   R2,V2_4                get v2 address
00001448  E722 0000 0006             00000000     1014+         VL    V2,0(R2)

0000144E                                          1017+         DS    0H                           E64A VCVDQ
0000144E  E6                                      1018+         DC    X'E6'                         VECTOR CONVERT TO DECIMAL (128)
0000144F  22                                      1019+         DC    X'22'                         v1, v2
00001450  00                                      1020+         DC    X'00'                        reserved
00001451  19F0                                    1021+         DC    HL2'6640'                     m4, i3, rXB
00001453  4A                                      1022+         DC    X'4A'

00001454  B98D 0020                               1024+         EPSW  R2,R0                exptract psw
00001458  5020 5068                  00000068     1025+         ST    R2,CCPSW                 to save CC
0000145C  E720 5028 000E             000013F0     1026+         VST   V2,V1O4              save instruction result
00001462  07FB                                    1027+         BR    R11                  return
00001464                                          1028+RE4      DS    0F                   expected 16 byte result
00001464                                          1029+         DROP  R5
00001464  00000000 00009223                       1030          DC    XL16'0000000000009223 372036854775807C'      v1
0000146C  37203685 4775807C
00001474  00000000 00000000                       1031          DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
0000147C  7FFFFFFF FFFFFFFF
                                                  1032
                                                  1033          VRI_J VCVDQ,X'9F',X'1',0,N
00001488                                          1034+         DS    0FD
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001488                     00001488              1035+          USING *,R5               base for test data and test routine
00001488  000014FC                                 1036+T5        DC    A(X5)              address of test routine
0000148C  0005                                     1037+          DC    H'5'               test number
0000148E  00                                       1038+          DC    X'00'
0000148F  D5                                       1039+          DC    CL1'N'             S or Y = skip cross check
00001490  9F                                       1040+          DC    X'9F'              i3
00001491  01                                       1041+          DC    X'1'               m4
00001492  00                                       1042+          DC    HL1'0'             cc
00001493  07                                       1043+          DC    HL1'7'             cc failed mask
00001494  00001534                                 1044+V2_5      DC    A(RE5+16)          address of v2: 16-byte packed decimal
00001498  E5C3E5C4 D8404040                        1045+          DC    CL8'VCVDQ'         instruction name
000014A0  00000010                                 1046+          DC    A(16)              result length
000014A4  00001524                                 1047+          DC    A(RE5)             address of expected resul
000014A8  00000000 00000000                        1048+          DS    FD                 gap
000014B0  00000000 00000000                        1049+V1O5      DS    XL16               V1 output
000014B8  00000000 00000000
000014C0  00000000 00000000                        1050+          DS    FD                    gap
000014C8  40404040 40404040                        1051+          DC    CL8' '             was cross check skipped?
000014D0  00000000 00000000                        1052+          DS    FD                    gap
000014D8  00000000 00000000                        1053+XCO5      DS    XL16               Cross check Output
000014E0  00000000 00000000
000014E8  00000000 00000000                        1054+          DS    FD                    gap
000014F0  00000000 00000000                        1055+          DS    2F           extract PSW after test (has CC)
000014F8  00                                       1056+          DS    X            extracted cc
000014F9  00                                       1057+          DS    X            CC Check failed
                                                   1058+*
000014FC                                           1059+X5        DS    0F
000014FC  E720 8F56 0006               00001156    1060+          VL    V2,V1FUDGE
00001502  E320 500C 0014               00001494    1061+          LGF   R2,V2_5             get v2 address
00001508  E722 0000 0006               00000000    1062+          VL    V2,0(R2)

0000150E                                           1065+          DS    0H                       E64A VCVDQ
0000150E  E6                                       1066+          DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000150F  22                                       1067+          DC    X'22'                     v1, v2
00001510  00                                       1068+          DC    X'00'                    reserved
00001511  19F0                                     1069+          DC    HL2'6640'                 m4, i3, rXB
00001513  4A                                       1070+          DC    X'4A'

00001514  B98D 0020                                1072+          EPSW  R2,R0              exptract psw
00001518  5020 5068                    00000068    1073+          ST    R2,CCPSW               to save CC
0000151C  E720 5028 000E               000014B0    1074+          VST   V2,V1O5            save instruction result
00001522  07FB                                     1075+          BR    R11                return
00001524                                           1076+RE5       DS    0F                 expected 16 byte result
00001524                                           1077+          DROP  R5
00001524  00000000 00009223                        1078           DC    XL16'0000000000009223 372036854775807D'      v1
0000152C  37203685 4775807D
00001534  FFFFFFFF FFFFFFFF                        1079           DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
0000153C  80000000 00000001
                                                   1080
                                                   1081           VRI_J VCVDQ,X'9F',X'1',0,N
00001548                                           1082+          DS    0FD
00001548                     00001548              1083+          USING *,R5               base for test data and test routine
00001548  000015BC                                 1084+T6        DC    A(X6)              address of test routine
0000154C  0006                                     1085+          DC    H'6'               test number
0000154E  00                                       1086+          DC    X'00'
0000154F  D5                                       1087+          DC    CL1'N'             S or Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001550  9F                                      1088+          DC    X'9F'             i3
00001551  01                                      1089+          DC    X'1'              m4
00001552  00                                      1090+          DC    HL1'0'            cc
00001553  07                                      1091+          DC    HL1'7'            cc failed mask
00001554  000015F4                                1092+V2_6      DC    A(RE6+16)         address of v2: 16-byte packed decimal
00001558  E5C3E5C4 D8404040                       1093+          DC    CL8'VCVDQ'        instruction name
00001560  00000010                                1094+          DC    A(16)             result length
00001564  000015E4                                1095+          DC    A(RE6)            address of expected resul
00001568  00000000 00000000                       1096+          DS    FD                gap
00001570  00000000 00000000                       1097+V1O6      DS    XL16              V1 output
00001578  00000000 00000000
00001580  00000000 00000000                       1098+          DS    FD                   gap
00001588  40404040 40404040                       1099+          DC    CL8' '            was cross check skipped?
00001590  00000000 00000000                       1100+          DS    FD                   gap
00001598  00000000 00000000                       1101+XCO6      DS    XL16              Cross check Output
000015A0  00000000 00000000
000015A8  00000000 00000000                       1102+          DS    FD                   gap
000015B0  00000000 00000000                       1103+          DS    2F          extract PSW after test (has CC)
000015B8  00                                      1104+          DS    X           extracted cc
000015B9  00                                      1105+          DS    X           CC Check failed
                                                  1106+*
000015BC                                          1107+X6        DS    0F
000015BC  E720 8F56 0006             00001156     1108+          VL    V2,V1FUDGE
000015C2  E320 500C 0014             00001554     1109+          LGF   R2,V2_6            get v2 address
000015C8  E722 0000 0006             00000000     1110+          VL    V2,0(R2)

000015CE                                          1113+          DS    0H                      E64A VCVDQ
000015CE  E6                                      1114+          DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
000015CF  22                                      1115+          DC    X'22'                    v1, v2
000015D0  00                                      1116+          DC    X'00'                   reserved
000015D1  19F0                                    1117+          DC    HL2'6640'                m4, i3, rXB
000015D3  4A                                      1118+          DC    X'4A'

000015D4  B98D 0020                               1120+          EPSW  R2,R0             exptract psw
000015D8  5020 5068                  00000068     1121+          ST    R2,CCPSW              to save CC
000015DC  E720 5028 000E             00001570     1122+          VST   V2,V1O6           save instruction result
000015E2  07FB                                    1123+          BR    R11               return
000015E4                                          1124+RE6       DS    0F                expected 16 byte result
000015E4                                          1125+          DROP  R5
000015E4  99999999 99999999                       1126           DC    XL16'9999999999999999 999999999999999C'      v1
000015EC  99999999 9999999C
000015F4  0000007E 37BE2022                       1127           DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
000015FC  C0914B26 7FFFFFFF
                                                  1128
                                                  1129           VRI_J VCVDQ,X'9F',X'1',0,N
00001608                                          1130+          DS    0FD
00001608                          00001608        1131+          USING *,R5              base for test data and test routine
00001608  0000167C                                1132+T7        DC    A(X7)             address of test routine
0000160C  0007                                    1133+          DC    H'7'              test number
0000160E  00                                      1134+          DC    X'00'
0000160F  D5                                      1135+          DC    CL1'N'            S or Y = skip cross check
00001610  9F                                      1136+          DC    X'9F'             i3
00001611  01                                      1137+          DC    X'1'              m4
00001612  00                                      1138+          DC    HL1'0'            cc
00001613  07                                      1139+          DC    HL1'7'            cc failed mask
00001614  000016B4                                1140+V2_7      DC    A(RE7+16)         address of v2: 16-byte packed decimal
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001618  E5C3E5C4 D8404040                        1141+          DC    CL8'VCVDQ'         instruction name
00001620  00000010                                 1142+          DC    A(16)              result length
00001624  000016A4                                 1143+          DC    A(RE7)             address of expected resul
00001628  00000000 00000000                        1144+          DS    FD                 gap
00001630  00000000 00000000                        1145+V1O7      DS    XL16               V1 output
00001638  00000000 00000000
00001640  00000000 00000000                        1146+          DS    FD                    gap
00001648  40404040 40404040                        1147+          DC    CL8' '             was cross check skipped?
00001650  00000000 00000000                        1148+          DS    FD                    gap
00001658  00000000 00000000                        1149+XCO7      DS    XL16               Cross check Output
00001660  00000000 00000000
00001668  00000000 00000000                        1150+          DS    FD                    gap
00001670  00000000 00000000                        1151+          DS    2F           extract PSW after test (has CC)
00001678  00                                       1152+          DS    X            extracted cc
00001679  00                                       1153+          DS    X            CC Check failed
                                                   1154+*
0000167C                                           1155+X7        DS    0F
0000167C  E720 8F56 0006              00001156     1156+          VL    V2,V1FUDGE
00001682  E320 500C 0014              00001614     1157+          LGF   R2,V2_7             get v2 address
00001688  E722 0000 0006              00000000     1158+          VL    V2,0(R2)

0000168E                                           1161+          DS    0H                       E64A VCVDQ
0000168E  E6                                       1162+          DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000168F  22                                       1163+          DC    X'22'                     v1, v2
00001690  00                                       1164+          DC    X'00'                    reserved
00001691  19F0                                     1165+          DC    HL2'6640'                 m4, i3, rXB
00001693  4A                                       1166+          DC    X'4A'

00001694  B98D 0020                                1168+          EPSW  R2,R0              exptract psw
00001698  5020 5068                   00000068     1169+          ST    R2,CCPSW               to save CC
0000169C  E720 5028 000E              00001630     1170+          VST   V2,V1O7            save instruction result
000016A2  07FB                                     1171+          BR    R11                return
000016A4                                           1172+RE7       DS    0F                 expected 16 byte result
000016A4                                           1173+          DROP  R5
000016A4  99999999 99999999                        1174           DC    XL16'9999999999999999 999999999999999D'      v1
000016AC  99999999 9999999D
000016B4  FFFFFF81 C841DFDD                        1175           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
000016BC  3F6EB4D9 80000001
                                                   1176 *
                                                   1177 * overflow
                                                   1178 *
                                                   1179           VRI_J VCVDQ,X'9F',X'1',3,N
000016C8                                           1180+          DS    0FD
000016C8                     000016C8              1181+          USING *,R5               base for test data and test routine
000016C8  0000173C                                 1182+T8        DC    A(X8)              address of test routine
000016CC  0008                                     1183+          DC    H'8'               test number
000016CE  00                                       1184+          DC    X'00'
000016CF  D5                                       1185+          DC    CL1'N'             S or Y = skip cross check
000016D0  9F                                       1186+          DC    X'9F'              i3
000016D1  01                                       1187+          DC    X'1'               m4
000016D2  03                                       1188+          DC    HL1'3'             cc
000016D3  0E                                       1189+          DC    HL1'14'            cc failed mask
000016D4  00001774                                 1190+V2_8      DC    A(RE8+16)          address of v2: 16-byte packed decimal
000016D8  E5C3E5C4 D8404040                        1191+          DC    CL8'VCVDQ'         instruction name
000016E0  00000010                                 1192+          DC    A(16)              result length
000016E4  00001764                                 1193+          DC    A(RE8)             address of expected resul
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000016E8  00000000 00000000                        1194+         DS    FD                  gap
000016F0  00000000 00000000                        1195+V108     DS    XL16                V1 output
000016F8  00000000 00000000
00001700  00000000 00000000                        1196+         DS    FD                     gap
00001708  40404040 40404040                        1197+         DC    CL8' '              was cross check skipped?
00001710  00000000 00000000                        1198+         DS    FD                     gap
00001718  00000000 00000000                        1199+XC08     DS    XL16                Cross check Output
00001720  00000000 00000000
00001728  00000000 00000000                        1200+         DS    FD                     gap
00001730  00000000 00000000                        1201+         DS    2F            extract PSW after test (has CC)
00001738  00                                       1202+         DS    X             extracted cc
00001739  00                                       1203+         DS    X             CC Check failed
                                                   1204+*
0000173C                                           1205+X8       DS    0F
0000173C  E720 8F56 0006                00001156   1206+         VL    V2,V1FUDGE
00001742  E320 500C 0014                000016D4   1207+         LGF   R2,V2_8              get v2 address
00001748  E722 0000 0006                00000000   1208+         VL    V2,0(R2)

0000174E                                           1211+         DS    0H                        E64A VCVDQ
0000174E  E6                                       1212+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000174F  22                                       1213+         DC    X'22'                      v1, v2
00001750  00                                       1214+         DC    X'00'                     reserved
00001751  19F0                                     1215+         DC    HL2'6640'                  m4, i3, rXB
00001753  4A                                       1216+         DC    X'4A'

00001754  B98D 0020                                1218+         EPSW  R2,R0               exptract psw
00001758  5020 5068                     00000068   1219+         ST    R2,CCPSW                to save CC
0000175C  E720 5028 000E                000016F0   1220+         VST   V2,V108             save instruction result
00001762  07FB                                     1221+         BR    R11                 return
00001764                                           1222+RE8       DS    0F                  expected 16 byte result
00001764                                           1223+         DROP  R5
00001764  00000000 00000000                        1224          DC    XL16'0000000000000000 000000000000000C'      v1
0000176C  00000000 0000000C
00001774  0000007E 37BE2022                        1225          DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
0000177C  C0914B26 80000000
                                                   1226
                                                   1227          VRI_J VCVDQ,X'9F',X'1',3,N
00001788                                           1228+         DS    0FD
00001788                       00001788            1229+         USING *,R5                base for test data and test routine
00001788  000017FC                                 1230+T9       DC    A(X9)               address of test routine
0000178C  0009                                     1231+         DC    H'9'                test number
0000178E  00                                       1232+         DC    X'00'
0000178F  D5                                       1233+         DC    CL1'N'              S or Y = skip cross check
00001790  9F                                       1234+         DC    X'9F'               i3
00001791  01                                       1235+         DC    X'1'                m4
00001792  03                                       1236+         DC    HL1'3'              cc
00001793  0E                                       1237+         DC    HL1'14'             cc failed mask
00001794  00001834                                 1238+V2_9     DC    A(RE9+16)           address of v2: 16-byte packed decimal
00001798  E5C3E5C4 D8404040                        1239+         DC    CL8'VCVDQ'          instruction name
000017A0  00000010                                 1240+         DC    A(16)               result length
000017A4  00001824                                 1241+         DC    A(RE9)              address of expected resul
000017A8  00000000 00000000                        1242+         DS    FD                  gap
000017B0  00000000 00000000                        1243+V109     DS    XL16                V1 output
000017B8  00000000 00000000
000017C0  00000000 00000000                        1244+         DS    FD                     gap
000017C8  40404040 40404040                        1245+         DC    CL8' '              was cross check skipped?
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000017D0  00000000 00000000                       1246+         DS    FD                   gap
000017D8  00000000 00000000                       1247+XC09     DS    XL16              Cross check Output
000017E0  00000000 00000000
000017E8  00000000 00000000                       1248+         DS    FD                   gap
000017F0  00000000 00000000                       1249+         DS    2F          extract PSW after test (has CC)
000017F8  00                                      1250+         DS    X           extracted cc
000017F9  00                                      1251+         DS    X           CC Check failed
                                                  1252+*
000017FC                                          1253+X9       DS    0F
000017FC  E720 8F56 0006            00001156      1254+         VL    V2,V1FUDGE
00001802  E320 500C 0014            00001794      1255+         LGF   R2,V2_9              get v2 address
00001808  E722 0000 0006            00000000      1256+         VL    V2,0(R2)

0000180E                                          1259+         DS    0H                        E64A VCVDQ
0000180E  E6                                      1260+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000180F  22                                      1261+         DC    X'22'                      v1, v2
00001810  00                                      1262+         DC    X'00'                     reserved
00001811  19F0                                    1263+         DC    HL2'6640'                  m4, i3, rXB
00001813  4A                                      1264+         DC    X'4A'

00001814  B98D 0020                               1266+         EPSW  R2,R0             exptract psw
00001818  5020 5068                 00000068      1267+         ST    R2,CCPSW              to save CC
0000181C  E720 5028 000E            000017B0      1268+         VST   V2,V1O9           save instruction result
00001822  07FB                                    1269+         BR    R11               return
00001824                                          1270+RE9      DS    0F                expected 16 byte result
00001824                                          1271+         DROP  R5
00001824  00000000 00000000                       1272          DC    XL16'0000000000000000 000000000000000D'      v1
0000182C  00000000 0000000D
00001834  FFFFFF81 C841DFDD                       1273          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
0000183C  3F6EB4D9 80000000
                                                  1274
                                                  1275 *-----------------------------------------------------------------------
                                                  1276 *        i3:   x'9f"   (IOM=1, RDC=31)
                                                  1277 *        m4:   x'3'    (LB=0, P1=1, CS=1)
                                                  1278 *
                                                  1279          VRI_J VCVDQ,X'9F',X'3',0,N
00001848                                          1280+         DS    0FD
00001848                          00001848        1281+         USING *,R5              base for test data and test routine
00001848  000018BC                                1282+T10      DC    A(X10)            address of test routine
0000184C  000A                                    1283+         DC    H'10'             test number
0000184E  00                                      1284+         DC    X'00'
0000184F  D5                                      1285+         DC    CL1'N'            S or Y = skip cross check
00001850  9F                                      1286+         DC    X'9F'             i3
00001851  03                                      1287+         DC    X'3'              m4
00001852  00                                      1288+         DC    HL1'0'            cc
00001853  07                                      1289+         DC    HL1'7'            cc failed mask
00001854  000018F4                                1290+V2_10    DC    A(RE10+16)        address of v2: 16-byte packed decimal
00001858  E5C3E5C4 D8404040                       1291+         DC    CL8'VCVDQ'        instruction name
00001860  00000010                                1292+         DC    A(16)             result length
00001864  000018E4                                1293+         DC    A(RE10)           address of expected resul
00001868  00000000 00000000                       1294+         DS    FD                gap
00001870  00000000 00000000                       1295+V1O10    DS    XL16              V1 output
00001878  00000000 00000000
00001880  00000000 00000000                       1296+         DS    FD                   gap
00001888  40404040 40404040                       1297+         DC    CL8' '            was cross check skipped?
00001890  00000000 00000000                       1298+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001898  00000000 00000000                     1299+XCO010    DS    XL16                 Cross check Output
000018A0  00000000 00000000
000018A8  00000000 00000000                     1300+          DS    FD                      gap
000018B0  00000000 00000000                     1301+          DS    2F             extract PSW after test (has CC)
000018B8  00                                    1302+          DS    X              extracted cc
000018B9  00                                    1303+          DS    X              CC Check failed
                                                1304+*
000018BC                                        1305+X10       DS    0F
000018BC  E720 8F56 0006              00001156  1306+          VL    V2,V1FUDGE
000018C2  E320 500C 0014              00001854  1307+          LGF   R2,V2_10              get v2 address
000018C8  E722 0000 0006              00000000  1308+          VL    V2,0(R2)

000018CE                                        1311+          DS    0H                          E64A VCVDQ
000018CE  E6                                    1312+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
000018CF  22                                    1313+          DC    X'22'                        v1, v2
000018D0  00                                    1314+          DC    X'00'                       reserved
000018D1  39F0                                  1315+          DC    HL2'14832'                   m4, i3, rXB
000018D3  4A                                    1316+          DC    X'4A'

000018D4  B98D 0020                             1318+          EPSW  R2,R0                exptract psw
000018D8  5020 5068                   00000068  1319+          ST    R2,CCPSW                 to save CC
000018DC  E720 5028 000E              00001870  1320+          VST   V2,V1O10             save instruction result
000018E2  07FB                                  1321+          BR    R11                  return
000018E4                                        1322+RE10      DS    0F                   expected 16 byte result
000018E4                                        1323+          DROP  R5
000018E4  00000000 00000000                     1324           DC    XL16'0000000000000000 000000000000000F'      v1
000018EC  00000000 0000000F
000018F4  00000000 00000000                     1325           DC    XL16'0000000000000000 0000000000000000'      v2
000018FC  00000000 00000000
                                                1326
                                                1327           VRI_J VCVDQ,X'9F',X'3',0,N
00001908                                        1328+          DS    0FD
00001908                     00001908           1329+          USING *,R5                 base for test data and test routine
00001908  0000197C                              1330+T11       DC    A(X11)               address of test routine
0000190C  000B                                  1331+          DC    H'11'                test number
0000190E  00                                    1332+          DC    X'00'
0000190F  D5                                    1333+          DC    CL1'N'               S or Y = skip cross check
00001910  9F                                    1334+          DC    X'9F'                i3
00001911  03                                    1335+          DC    X'3'                 m4
00001912  00                                    1336+          DC    HL1'0'               cc
00001913  07                                    1337+          DC    HL1'7'               cc failed mask
00001914  000019B4                              1338+V2_11     DC    A(RE11+16)           address of v2: 16-byte packed decimal
00001918  E5C3E5C4 D8404040                     1339+          DC    CL8'VCVDQ'           instruction name
00001920  00000010                              1340+          DC    A(16)                result length
00001924  000019A4                              1341+          DC    A(RE11)              address of expected resul
00001928  00000000 00000000                     1342+          DS    FD                   gap
00001930  00000000 00000000                     1343+V1O11     DS    XL16                 V1 output
00001938  00000000 00000000
00001940  00000000 00000000                     1344+          DS    FD                      gap
00001948  40404040 40404040                     1345+          DC    CL8' '               was cross check skipped?
00001950  00000000 00000000                     1346+          DS    FD                      gap
00001958  00000000 00000000                     1347+XCO011    DS    XL16                 Cross check Output
00001960  00000000 00000000
00001968  00000000 00000000                     1348+          DS    FD                      gap
00001970  00000000 00000000                     1349+          DS    2F             extract PSW after test (has CC)
00001978  00                                    1350+          DS    X              extracted cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001979  00                                      1351+         DS    X           CC Check failed
                                                  1352+*
0000197C                                          1353+X11      DS    0F
0000197C  E720 8F56 0006                00001156  1354+         VL    V2,V1FUDGE
00001982  E320 500C 0014                00001914  1355+         LGF   R2,V2_11            get v2 address
00001988  E722 0000 0006                00000000  1356+         VL    V2,0(R2)

0000198E                                          1359+         DS    0H                      E64A VCVDQ
0000198E  E6                                      1360+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000198F  22                                      1361+         DC    X'22'                    v1, v2
00001990  00                                      1362+         DC    X'00'                   reserved
00001991  39F0                                    1363+         DC    HL2'14832'               m4, i3, rXB
00001993  4A                                      1364+         DC    X'4A'

00001994  B98D 0020                               1366+         EPSW  R2,R0              exptract psw
00001998  5020 5068                     00000068  1367+         ST    R2,CCPSW               to save CC
0000199C  E720 5028 000E                00001930  1368+         VST   V2,V1O11           save instruction result
000019A2  07FB                                    1369+         BR    R11                return
000019A4                                          1370+RE11     DS    0F                 expected 16 byte result
000019A4                                          1371+         DROP  R5
000019A4  00000000 00000000                       1372          DC    XL16'0000000000000000 000000000000001F'      v1
000019AC  00000000 0000001F
000019B4  00000000 00000000                       1373          DC    XL16'0000000000000000 0000000000000001'      v2
000019BC  00000000 00000001
                                                  1374
                                                  1375          VRI_J VCVDQ,X'9F',X'3',0,S
000019C8                                          1376+         DS    0FD
000019C8                       000019C8           1377+         USING *,R5               base for test data and test routine
000019C8  00001A3C                                1378+T12      DC    A(X12)             address of test routine
000019CC  000C                                    1379+         DC    H'12'              test number
000019CE  00                                      1380+         DC    X'00'
000019CF  E2                                      1381+         DC    CL1'S'             S or Y = skip cross check
000019D0  9F                                      1382+         DC    X'9F'              i3
000019D1  03                                      1383+         DC    X'3'               m4
000019D2  00                                      1384+         DC    HL1'0'             cc
000019D3  07                                      1385+         DC    HL1'7'             cc failed mask
000019D4  00001A74                                1386+V2_12    DC    A(RE12+16)         address of v2: 16-byte packed decimal
000019D8  E5C3E5C4 D8404040                       1387+         DC    CL8'VCVDQ'         instruction name
000019E0  00000010                                1388+         DC    A(16)              result length
000019E4  00001A64                                1389+         DC    A(RE12)            address of expected resul
000019E8  00000000 00000000                       1390+         DS    FD                 gap
000019F0  00000000 00000000                       1391+V1O12    DS    XL16               V1 output
000019F8  00000000 00000000
00001A00  00000000 00000000                       1392+         DS    FD                    gap
00001A08  40404040 40404040                       1393+         DC    CL8' '             was cross check skipped?
00001A10  00000000 00000000                       1394+         DS    FD                    gap
00001A18  00000000 00000000                       1395+XCO12    DS    XL16               Cross check Output
00001A20  00000000 00000000
00001A28  00000000 00000000                       1396+         DS    FD                    gap
00001A30  00000000 00000000                       1397+         DS    2F          extract PSW after test (has CC)
00001A38  00                                      1398+         DS    X           extracted cc
00001A39  00                                      1399+         DS    X           CC Check failed
                                                  1400+*
00001A3C                                          1401+X12      DS    0F
00001A3C  E720 8F56 0006                00001156  1402+         VL    V2,V1FUDGE
00001A42  E320 500C 0014                000019D4  1403+         LGF   R2,V2_12            get v2 address
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001A48  E722 0000 0006                00000000  1404+          VL    V2,0(R2)

00001A4E                                          1407+          DS    0H                            E64A VCVDQ
00001A4E  E6                                      1408+          DC    X'E6'                          VECTOR CONVERT TO DECIMAL (128)
00001A4F  22                                      1409+          DC    X'22'                          v1, v2
00001A50  00                                      1410+          DC    X'00'                         reserved
00001A51  39F0                                    1411+          DC    HL2'14832'                     m4, i3, rXB
00001A53  4A                                      1412+          DC    X'4A'

00001A54  B98D 0020                               1414+          EPSW  R2,R0             exptract psw
00001A58  5020 5068                     00000068  1415+          ST    R2,CCPSW              to save CC
00001A5C  E720 5028 000E                000019F0  1416+          VST   V2,V1O12          save instruction result
00001A62  07FB                                    1417+          BR    R11               return
00001A64                                          1418+RE12      DS    0F                expected 16 byte result
00001A64                                          1419+          DROP  R5
00001A64  00000000 00000000                       1420           DC    XL16'0000000000000000 000000000000001F'      v1
00001A6C  00000000 0000001F
00001A74  FFFFFFFF FFFFFFFF                       1421           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
00001A7C  FFFFFFFF FFFFFFFF
                                                  1422
                                                  1423           VRI_J VCVDQ,X'9F',X'3',0,N
00001A88                                          1424+          DS    0FD
00001A88                       00001A88           1425+          USING *,R5              base for test data and test routine
00001A88  00001AFC                                1426+T13       DC    A(X13)            address of test routine
00001A8C  000D                                    1427+          DC    H'13'             test number
00001A8E  00                                      1428+          DC    X'00'
00001A8F  D5                                      1429+          DC    CL1'N'            S or Y = skip cross check
00001A90  9F                                      1430+          DC    X'9F'             i3
00001A91  03                                      1431+          DC    X'3'              m4
00001A92  00                                      1432+          DC    HL1'0'            cc
00001A93  07                                      1433+          DC    HL1'7'            cc failed mask
00001A94  00001B34                                1434+V2_13     DC    A(RE13+16)        address of v2: 16-byte packed decimal
00001A98  E5C3E5C4 D8404040                       1435+          DC    CL8'VCVDQ'        instruction name
00001AA0  00000010                                1436+          DC    A(16)             result length
00001AA4  00001B24                                1437+          DC    A(RE13)           address of expected resul
00001AA8  00000000 00000000                       1438+          DS    FD                gap
00001AB0  00000000 00000000                       1439+V1O13     DS    XL16              V1 output
00001AB8  00000000 00000000
00001AC0  00000000 00000000                       1440+          DS    FD                   gap
00001AC8  40404040 40404040                       1441+          DC    CL8' '            was cross check skipped?
00001AD0  00000000 00000000                       1442+          DS    FD                   gap
00001AD8  00000000 00000000                       1443+XCO13     DS    XL16              Cross check Output
00001AE0  00000000 00000000
00001AE8  00000000 00000000                       1444+          DS    FD                   gap
00001AF0  00000000 00000000                       1445+          DS    2F          extract PSW after test (has CC)
00001AF8  00                                      1446+          DS    X           extracted cc
00001AF9  00                                      1447+          DS    X           CC Check failed
                                                  1448+*
00001AFC                                          1449+X13       DS    0F
00001AFC  E720 8F56 0006                00001156  1450+          VL    V2,V1FUDGE
00001B02  E320 500C 0014                00001A94  1451+          LGF   R2,V2_13           get v2 address
00001B08  E722 0000 0006                00000000  1452+          VL    V2,0(R2)

00001B0E                                          1455+          DS    0H                            E64A VCVDQ
00001B0E  E6                                      1456+          DC    X'E6'                          VECTOR CONVERT TO DECIMAL (128)
00001B0F  22                                      1457+          DC    X'22'                          v1, v2
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00001B10  00                                      1458+          DC    X'00'                        reserved
00001B11  39F0                                    1459+          DC    HL2'14832'                    m4, i3, rXB
00001B13  4A                                      1460+          DC    X'4A'

00001B14  B98D 0020                               1462+          EPSW  R2,R0              exptract psw
00001B18  5020 5068                     00000068  1463+          ST    R2,CCPSW               to save CC
00001B1C  E720 5028 000E                00001AB0  1464+          VST   V2,V1O13           save instruction result
00001B22  07FB                                    1465+          BR    R11                return
00001B24                                          1466+RE13      DS    0F                 expected 16 byte result
00001B24                                          1467+          DROP  R5
00001B24  00000000 00009223                       1468           DC    XL16'0000000000009223 372036854775807F'      v1
00001B2C  37203685 4775807F
00001B34  00000000 00000000                       1469           DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
00001B3C  7FFFFFFF FFFFFFFF
                                                  1470
                                                  1471           VRI_J VCVDQ,X'9F',X'3',0,S
00001B48                                          1472+          DS    0FD
00001B48                      00001B48            1473+          USING *,R5               base for test data and test routine
00001B48  00001BBC                                1474+T14       DC    A(X14)             address of test routine
00001B4C  000E                                    1475+          DC    H'14'              test number
00001B4E  00                                      1476+          DC    X'00'
00001B4F  E2                                      1477+          DC    CL1'S'             S or Y = skip cross check
00001B50  9F                                      1478+          DC    X'9F'              i3
00001B51  03                                      1479+          DC    X'3'               m4
00001B52  00                                      1480+          DC    HL1'0'             cc
00001B53  07                                      1481+          DC    HL1'7'             cc failed mask
00001B54  00001BF4                                1482+V2_14     DC    A(RE14+16)         address of v2: 16-byte packed decimal
00001B58  E5C3E5C4 D8404040                       1483+          DC    CL8'VCVDQ'         instruction name
00001B60  00000010                                1484+          DC    A(16)              result length
00001B64  00001BE4                                1485+          DC    A(RE14)            address of expected resul
00001B68  00000000 00000000                       1486+          DS    FD                 gap
00001B70  00000000 00000000                       1487+V1O14     DS    XL16               V1 output
00001B78  00000000 00000000
00001B80  00000000 00000000                       1488+          DS    FD                    gap
00001B88  40404040 40404040                       1489+          DC    CL8' '             was cross check skipped?
00001B90  00000000 00000000                       1490+          DS    FD                    gap
00001B98  00000000 00000000                       1491+XC014     DS    XL16               Cross check Output
00001BA0  00000000 00000000
00001BA8  00000000 00000000                       1492+          DS    FD                    gap
00001BB0  00000000 00000000                       1493+          DS    2F           extract PSW after test (has CC)
00001BB8  00                                      1494+          DS    X            extracted cc
00001BB9  00                                      1495+          DS    X            CC Check failed
                                                  1496+*
00001BBC                                          1497+X14       DS    0F
00001BBC  E720 8F56 0006                00001156  1498+          VL    V2,V1FUDGE
00001BC2  E320 500C 0014                00001B54  1499+          LGF   R2,V2_14            get v2 address
00001BC8  E722 0000 0006                00000000  1500+          VL    V2,0(R2)

00001BCE                                          1503+          DS    0H                           E64A VCVDQ
00001BCE  E6                                      1504+          DC    X'E6'                         VECTOR CONVERT TO DECIMAL (128)
00001BCF  22                                      1505+          DC    X'22'                         v1, v2
00001BD0  00                                      1506+          DC    X'00'                        reserved
00001BD1  39F0                                    1507+          DC    HL2'14832'                    m4, i3, rXB
00001BD3  4A                                      1508+          DC    X'4A'

00001BD4  B98D 0020                               1510+          EPSW  R2,R0              exptract psw
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

00001BD8  5020 5068                   00000068  1511+          ST    R2,CCPSW              to save CC
00001BDC  E720 5028 000E              00001B70  1512+          VST   V2,V1O14          save instruction result
00001BE2  07FB                                  1513+          BR    R11               return
00001BE4                                        1514+RE14      DS    0F                expected 16 byte result
00001BE4                                        1515+          DROP  R5
00001BE4  00000000 00009223                     1516           DC    XL16'0000000000009223 372036854775807F'      v1
00001BEC  37203685 4775807F
00001BF4  FFFFFFFF FFFFFFFF                     1517           DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
00001BFC  80000000 00000001
                                                1518
                                                1519           VRI_J VCVDQ,X'9F',X'3',0,N
00001C08                                        1520+          DS    0FD
00001C08                     00001C08           1521+          USING *,R5              base for test data and test routine
00001C08  00001C7C                              1522+T15       DC    A(X15)            address of test routine
00001C0C  000F                                  1523+          DC    H'15'             test number
00001C0E  00                                    1524+          DC    X'00'
00001C0F  D5                                    1525+          DC    CL1'N'            S or Y = skip cross check
00001C10  9F                                    1526+          DC    X'9F'             i3
00001C11  03                                    1527+          DC    X'3'              m4
00001C12  00                                    1528+          DC    HL1'0'            cc
00001C13  07                                    1529+          DC    HL1'7'            cc failed mask
00001C14  00001CB4                              1530+V2_15     DC    A(RE15+16)        address of v2: 16-byte packed decimal
00001C18  E5C3E5C4 D8404040                     1531+          DC    CL8'VCVDQ'        instruction name
00001C20  00000010                              1532+          DC    A(16)             result length
00001C24  00001CA4                              1533+          DC    A(RE15)           address of expected resul
00001C28  00000000 00000000                     1534+          DS    FD                gap
00001C30  00000000 00000000                     1535+V1O15     DS    XL16              V1 output
00001C38  00000000 00000000
00001C40  00000000 00000000                     1536+          DS    FD                   gap
00001C48  40404040 40404040                     1537+          DC    CL8' '            was cross check skipped?
00001C50  00000000 00000000                     1538+          DS    FD                   gap
00001C58  00000000 00000000                     1539+XCO15     DS    XL16              Cross check Output
00001C60  00000000 00000000
00001C68  00000000 00000000                     1540+          DS    FD                   gap
00001C70  00000000 00000000                     1541+          DS    2F          extract PSW after test (has CC)
00001C78  00                                    1542+          DS    X           extracted cc
00001C79  00                                    1543+          DS    X           CC Check failed
                                                1544+*
00001C7C                                        1545+X15       DS    0F
00001C7C  E720 8F56 0006              00001156  1546+          VL    V2,V1FUDGE
00001C82  E320 500C 0014              00001C14  1547+          LGF   R2,V2_15           get v2 address
00001C88  E722 0000 0006              00000000  1548+          VL    V2,0(R2)

00001C8E                                        1551+          DS    0H                      E64A VCVDQ
00001C8E  E6                                    1552+          DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
00001C8F  22                                    1553+          DC    X'22'                    v1, v2
00001C90  00                                    1554+          DC    X'00'                   reserved
00001C91  39F0                                  1555+          DC    HL2'14832'               m4, i3, rXB
00001C93  4A                                    1556+          DC    X'4A'

00001C94  B98D 0020                             1558+          EPSW  R2,R0             exptract psw
00001C98  5020 5068                   00000068  1559+          ST    R2,CCPSW              to save CC
00001C9C  E720 5028 000E              00001C30  1560+          VST   V2,V1O15          save instruction result
00001CA2  07FB                                  1561+          BR    R11               return
00001CA4                                        1562+RE15      DS    0F                expected 16 byte result
00001CA4                                        1563+          DROP  R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001CA4  99999999 99999999                        1564          DC    XL16'9999999999999999 999999999999999F'      v1
00001CAC  99999999 9999999F
00001CB4  0000007E 37BE2022                        1565          DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
00001CBC  C0914B26 7FFFFFFF
                                                   1566
                                                   1567          VRI_J VCVDQ,X'9F',X'3',0,S
00001CC8                                           1568+         DS    0FD
00001CC8                     00001CC8              1569+         USING *,R5              base for test data and test routine
00001CC8  00001D3C                                 1570+T16      DC    A(X16)            address of test routine
00001CCC  0010                                     1571+         DC    H'16'             test number
00001CCE  00                                       1572+         DC    X'00'
00001CCF  E2                                       1573+         DC    CL1'S'            S or Y = skip cross check
00001CD0  9F                                       1574+         DC    X'9F'             i3
00001CD1  03                                       1575+         DC    X'3'              m4
00001CD2  00                                       1576+         DC    HL1'0'            cc
00001CD3  07                                       1577+         DC    HL1'7'            cc failed mask
00001CD4  00001D74                                 1578+V2_16    DC    A(RE16+16)        address of v2: 16-byte packed decimal
00001CD8  E5C3E5C4 D8404040                        1579+         DC    CL8'VCVDQ'        instruction name
00001CE0  00000010                                 1580+         DC    A(16)             result length
00001CE4  00001D64                                 1581+         DC    A(RE16)           address of expected resul
00001CE8  00000000 00000000                        1582+         DS    FD                gap
00001CF0  00000000 00000000                        1583+V1O16    DS    XL16              V1 output
00001CF8  00000000 00000000
00001D00  00000000 00000000                        1584+         DS    FD                   gap
00001D08  40404040 40404040                        1585+         DC    CL8' '            was cross check skipped?
00001D10  00000000 00000000                        1586+         DS    FD                   gap
00001D18  00000000 00000000                        1587+XCO16    DS    XL16              Cross check Output
00001D20  00000000 00000000
00001D28  00000000 00000000                        1588+         DS    FD                   gap
00001D30  00000000 00000000                        1589+         DS    2F          extract PSW after test (has CC)
00001D38  00                                       1590+         DS    X           extracted cc
00001D39  00                                       1591+         DS    X           CC Check failed
                                                   1592+*
00001D3C                                           1593+X16      DS    0F
00001D3C  E720 8F56 0006             00001156      1594+         VL    V2,V1FUDGE
00001D42  E320 500C 0014             00001CD4      1595+         LGF   R2,V2_16           get v2 address
00001D48  E722 0000 0006             00000000      1596+         VL    V2,0(R2)

00001D4E                                           1599+         DS    0H                      E64A VCVDQ
00001D4E  E6                                       1600+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
00001D4F  22                                       1601+         DC    X'22'                    v1, v2
00001D50  00                                       1602+         DC    X'00'                   reserved
00001D51  39F0                                     1603+         DC    HL2'14832'               m4, i3, rXB
00001D53  4A                                       1604+         DC    X'4A'

00001D54  B98D 0020                                1606+         EPSW  R2,R0             exptract psw
00001D58  5020 5068                  00000068      1607+         ST    R2,CCPSW              to save CC
00001D5C  E720 5028 000E             00001CF0      1608+         VST   V2,V1O16          save instruction result
00001D62  07FB                                     1609+         BR    R11               return
00001D64                                           1610+RE16     DS    0F                expected 16 byte result
00001D64                                           1611+         DROP  R5
00001D64  99999999 99999999                        1612          DC    XL16'9999999999999999 999999999999999F'      v1
00001D6C  99999999 9999999F
00001D74  FFFFFF81 C841DFDD                        1613          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
00001D7C  3F6EB4D9 80000001
                                                   1614 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1615 * overflow
                                                   1616 *
                                                   1617          VRI_J VCVDQ,X'9F',X'3',3,N
00001D88                                           1618+         DS    0FD
00001D88                      00001D88             1619+         USING *,R5                base for test data and test routine
00001D88  00001DFC                                 1620+T17      DC    A(X17)              address of test routine
00001D8C  0011                                     1621+         DC    H'17'               test number
00001D8E  00                                       1622+         DC    X'00'
00001D8F  D5                                       1623+         DC    CL1'N'              S or Y = skip cross check
00001D90  9F                                       1624+         DC    X'9F'               i3
00001D91  03                                       1625+         DC    X'3'                m4
00001D92  03                                       1626+         DC    HL1'3'              cc
00001D93  0E                                       1627+         DC    HL1'14'             cc failed mask
00001D94  00001E34                                 1628+V2_17    DC    A(RE17+16)          address of v2: 16-byte packed decimal
00001D98  E5C3E5C4 D8404040                        1629+         DC    CL8'VCVDQ'          instruction name
00001DA0  00000010                                 1630+         DC    A(16)               result length
00001DA4  00001E24                                 1631+         DC    A(RE17)             address of expected resul
00001DA8  00000000 00000000                        1632+         DS    FD                  gap
00001DB0  00000000 00000000                        1633+V1O17    DS    XL16                V1 output
00001DB8  00000000 00000000
00001DC0  00000000 00000000                        1634+         DS    FD                     gap
00001DC8  40404040 40404040                        1635+         DC    CL8' '              was cross check skipped?
00001DD0  00000000 00000000                        1636+         DS    FD                     gap
00001DD8  00000000 00000000                        1637+XCO17    DS    XL16                Cross check Output
00001DE0  00000000 00000000
00001DE8  00000000 00000000                        1638+         DS    FD                     gap
00001DF0  00000000 00000000                        1639+         DS    2F            extract PSW after test (has CC)
00001DF8  00                                       1640+         DS    X             extracted cc
00001DF9  00                                       1641+         DS    X             CC Check failed
                                                   1642+*
00001DFC                                           1643+X17      DS    0F
00001DFC  E720 8F56 0006                00001156   1644+         VL    V2,V1FUDGE
00001E02  E320 500C 0014                00001D94   1645+         LGF   R2,V2_17             get v2 address
00001E08  E722 0000 0006                00000000   1646+         VL    V2,0(R2)

00001E0E                                           1649+         DS    0H                        E64A VCVDQ
00001E0E  E6                                       1650+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00001E0F  22                                       1651+         DC    X'22'                      v1, v2
00001E10  00                                       1652+         DC    X'00'                     reserved
00001E11  39F0                                     1653+         DC    HL2'14832'                 m4, i3, rXB
00001E13  4A                                       1654+         DC    X'4A'

00001E14  B98D 0020                                1656+         EPSW  R2,R0               exptract psw
00001E18  5020 5068                     00000068   1657+         ST    R2,CCPSW                to save CC
00001E1C  E720 5028 000E                00001DB0   1658+         VST   V2,V1O17            save instruction result
00001E22  07FB                                     1659+         BR    R11                 return
00001E24                                           1660+RE17     DS    0F                  expected 16 byte result
00001E24                                           1661+         DROP  R5
00001E24  00000000 00000000                        1662          DC    XL16'0000000000000000 000000000000000F'      v1
00001E2C  00000000 0000000F
00001E34  0000007E 37BE2022                        1663          DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
00001E3C  C0914B26 80000000
                                                   1664
                                                   1665          VRI_J VCVDQ,X'9F',X'3',3,N
00001E48                                           1666+         DS    0FD
00001E48                      00001E48             1667+         USING *,R5                base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001E48  00001EBC                                 1668+T18       DC    A(X18)            address of test routine
00001E4C  0012                                     1669+          DC    H'18'             test number
00001E4E  00                                       1670+          DC    X'00'
00001E4F  D5                                       1671+          DC    CL1'N'            S or Y = skip cross check
00001E50  9F                                       1672+          DC    X'9F'             i3
00001E51  03                                       1673+          DC    X'3'              m4
00001E52  03                                       1674+          DC    HL1'3'            cc
00001E53  0E                                       1675+          DC    HL1'14'           cc failed mask
00001E54  00001EF4                                 1676+V2_18     DC    A(RE18+16)        address of v2: 16-byte packed decimal
00001E58  E5C3E5C4 D8404040                        1677+          DC    CL8'VCVDQ'        instruction name
00001E60  00000010                                 1678+          DC    A(16)             result length
00001E64  00001EE4                                 1679+          DC    A(RE18)           address of expected resul
00001E68  00000000 00000000                        1680+          DS    FD                gap
00001E70  00000000 00000000                        1681+V1O18     DS    XL16              V1 output
00001E78  00000000 00000000
00001E80  00000000 00000000                        1682+          DS    FD                   gap
00001E88  40404040 40404040                        1683+          DC    CL8' '            was cross check skipped?
00001E90  00000000 00000000                        1684+          DS    FD                   gap
00001E98  00000000 00000000                        1685+XCO18     DS    XL16              Cross check Output
00001EA0  00000000 00000000
00001EA8  00000000 00000000                        1686+          DS    FD                   gap
00001EB0  00000000 00000000                        1687+          DS    2F          extract PSW after test (has CC)
00001EB8  00                                       1688+          DS    X           extracted cc
00001EB9  00                                       1689+          DS    X           CC Check failed
                                                   1690+*
00001EBC                                           1691+X18       DS    0F
00001EBC  E720 8F56 0006             00001156      1692+          VL    V2,V1FUDGE
00001EC2  E320 500C 0014             00001E54      1693+          LGF   R2,V2_18           get v2 address
00001EC8  E722 0000 0006             00000000      1694+          VL    V2,0(R2)

00001ECE                                           1697+          DS    0H                      E64A VCVDQ
00001ECE  E6                                       1698+          DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
00001ECF  22                                       1699+          DC    X'22'                    v1, v2
00001ED0  00                                       1700+          DC    X'00'                   reserved
00001ED1  39F0                                     1701+          DC    HL2'14832'               m4, i3, rXB
00001ED3  4A                                       1702+          DC    X'4A'

00001ED4  B98D 0020                                1704+          EPSW  R2,R0             exptract psw
00001ED8  5020 5068                  00000068      1705+          ST    R2,CCPSW              to save CC
00001EDC  E720 5028 000E             00001E70      1706+          VST   V2,V1O18          save instruction result
00001EE2  07FB                                     1707+          BR    R11               return
00001EE4                                           1708+RE18      DS    0F                expected 16 byte result
00001EE4                                           1709+          DROP  R5
00001EE4  00000000 00000000                        1710           DC    XL16'0000000000000000 000000000000000F'      v1
00001EEC  00000000 0000000F
00001EF4  FFFFFF81 C841DFDD                        1711           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
00001EFC  3F6EB4D9 80000000
                                                   1712
                                                   1713 *---------------------------------------------------------------------
                                                   1714 *          i3:   x'9f"   (IOM=1, RDC=31)
                                                   1715 *          m4:   x'9'    (LB=1, P1=0, CS=1)
                                                   1716 *
                                                   1717           VRI_J VCVDQ,X'9F',X'9',0,N
00001F08                                           1718+          DS    0FD
00001F08                   00001F08                1719+          USING *,R5              base for test data and test routine
00001F08  00001F7C                                 1720+T19       DC    A(X19)            address of test routine
```

```
  LOC        OBJECT CODE        ADDR1     ADDR2     STMT

00001F0C  0013                                      1721+          DC    H'19'               test number
00001F0E  00                                        1722+          DC    X'00'
00001F0F  D5                                        1723+          DC    CL1'N'              S or Y = skip cross check
00001F10  9F                                        1724+          DC    X'9F'               i3
00001F11  09                                        1725+          DC    X'9'                m4
00001F12  00                                        1726+          DC    HL1'0'              cc
00001F13  07                                        1727+          DC    HL1'7'              cc failed mask
00001F14  00001FB4                                  1728+V2_19     DC    A(RE19+16)          address of v2: 16-byte packed decimal
00001F18  E5C3E5C4 D8404040                         1729+          DC    CL8'VCVDQ'          instruction name
00001F20  00000010                                  1730+          DC    A(16)               result length
00001F24  00001FA4                                  1731+          DC    A(RE19)             address of expected resul
00001F28  00000000 00000000                         1732+          DS    FD                  gap
00001F30  00000000 00000000                         1733+V1O19     DS    XL16                V1 output
00001F38  00000000 00000000
00001F40  00000000 00000000                         1734+          DS    FD                     gap
00001F48  40404040 40404040                         1735+          DC    CL8' '              was cross check skipped?
00001F50  00000000 00000000                         1736+          DS    FD                     gap
00001F58  00000000 00000000                         1737+XCO19     DS    XL16                Cross check Output
00001F60  00000000 00000000
00001F68  00000000 00000000                         1738+          DS    FD                     gap
00001F70  00000000 00000000                         1739+          DS    2F            extract PSW after test (has CC)
00001F78  00                                        1740+          DS    X             extracted cc
00001F79  00                                        1741+          DS    X             CC Check failed
                                                    1742+*
00001F7C                                            1743+X19       DS    0F
00001F7C  E720 8F56 0006                  00001156  1744+          VL    V2,V1FUDGE
00001F82  E320 500C 0014                  00001F14  1745+          LGF   R2,V2_19             get v2 address
00001F88  E722 0000 0006                  00000000  1746+          VL    V2,0(R2)

00001F8E                                            1749+          DS    0H                        E64A VCVDQ
00001F8E  E6                                        1750+          DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00001F8F  22                                        1751+          DC    X'22'                      v1, v2
00001F90  00                                        1752+          DC    X'00'                     reserved
00001F91  99F0                                      1753+          DC    HL2'39408'                 m4, i3, rXB
00001F93  4A                                        1754+          DC    X'4A'

00001F94  B98D 0020                                 1756+          EPSW  R2,R0               exptract psw
00001F98  5020 5068                       00000068  1757+          ST    R2,CCPSW                to save CC
00001F9C  E720 5028 000E                  00001F30  1758+          VST   V2,V1O19            save instruction result
00001FA2  07FB                                      1759+          BR    R11                 return
00001FA4                                            1760+RE19      DS    0F                  expected 16 byte result
00001FA4                                            1761+          DROP  R5
00001FA4  00000000 00000000                         1762           DC    XL16'0000000000000000 000000000000000C'      v1
00001FAC  00000000 0000000C
00001FB4  00000000 00000000                         1763           DC    XL16'0000000000000000 0000000000000000'      v2
00001FBC  00000000 00000000
                                                    1764
                                                    1765           VRI_J VCVDQ,X'9F',X'9',0,N
00001FC8                                            1766+          DS    0FD
00001FC8                        00001FC8            1767+          USING *,R5                base for test data and test routine
00001FC8  0000203C                                  1768+T20       DC    A(X20)              address of test routine
00001FCC  0014                                      1769+          DC    H'20'               test number
00001FCE  00                                        1770+          DC    X'00'
00001FCF  D5                                        1771+          DC    CL1'N'              S or Y = skip cross check
00001FD0  9F                                        1772+          DC    X'9F'               i3
00001FD1  09                                        1773+          DC    X'9'                m4
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00001FD2  00                                      1774+          DC    HL1'0'            cc
00001FD3  07                                      1775+          DC    HL1'7'            cc failed mask
00001FD4  00002074                                1776+V2_20     DC    A(RE20+16)        address of v2: 16-byte packed decimal
00001FD8  E5C3E5C4 D8404040                       1777+          DC    CL8'VCVDQ'        instruction name
00001FE0  00000010                                1778+          DC    A(16)             result length
00001FE4  00002064                                1779+          DC    A(RE20)           address of expected resul
00001FE8  00000000 00000000                       1780+          DS    FD                gap
00001FF0  00000000 00000000                       1781+V1O20     DS    XL16              V1 output
00001FF8  00000000 00000000
00002000  00000000 00000000                       1782+          DS    FD                   gap
00002008  40404040 40404040                       1783+          DC    CL8' '            was cross check skipped?
00002010  00000000 00000000                       1784+          DS    FD                   gap
00002018  00000000 00000000                       1785+XCO20     DS    XL16              Cross check Output
00002020  00000000 00000000
00002028  00000000 00000000                       1786+          DS    FD                   gap
00002030  00000000 00000000                       1787+          DS    2F          extract PSW after test (has CC)
00002038  00                                      1788+          DS    X           extracted cc
00002039  00                                      1789+          DS    X           CC Check failed
                                                  1790+*
0000203C                                          1791+X20       DS    0F
0000203C  E720 8F56 0006             00001156     1792+          VL    V2,V1FUDGE
00002042  E320 500C 0014             00001FD4     1793+          LGF   R2,V2_20           get v2 address
00002048  E722 0000 0006             00000000     1794+          VL    V2,0(R2)

0000204E                                          1797+          DS    0H                         E64A VCVDQ
0000204E  E6                                      1798+          DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
0000204F  22                                      1799+          DC    X'22'                       v1, v2
00002050  00                                      1800+          DC    X'00'                      reserved
00002051  99F0                                    1801+          DC    HL2'39408'                  m4, i3, rXB
00002053  4A                                      1802+          DC    X'4A'

00002054  B98D 0020                               1804+          EPSW  R2,R0             exptract psw
00002058  5020 5068                  00000068     1805+          ST    R2,CCPSW              to save CC
0000205C  E720 5028 000E             00001FF0     1806+          VST   V2,V1O20          save instruction result
00002062  07FB                                    1807+          BR    R11               return
00002064                                          1808+RE20      DS    0F                expected 16 byte result
00002064                                          1809+          DROP  R5
00002064  00000000 00000000                       1810           DC    XL16'0000000000000000 000000000000001C'      v1
0000206C  00000000 0000001C
00002074  00000000 00000000                       1811           DC    XL16'0000000000000000 0000000000000001'      v2
0000207C  00000000 00000001
                                                  1812
                                                  1813           VRI_J VCVDQ,X'9F',X'9',3,S
00002088                                          1814+          DS    0FD
00002088                    00002088              1815+          USING *,R5              base for test data and test routine
00002088  000020FC                                1816+T21       DC    A(X21)            address of test routine
0000208C  0015                                    1817+          DC    H'21'             test number
0000208E  00                                      1818+          DC    X'00'
0000208F  E2                                      1819+          DC    CL1'S'            S or Y = skip cross check
00002090  9F                                      1820+          DC    X'9F'             i3
00002091  09                                      1821+          DC    X'9'              m4
00002092  03                                      1822+          DC    HL1'3'            cc
00002093  0E                                      1823+          DC    HL1'14'           cc failed mask
00002094  00002134                                1824+V2_21     DC    A(RE21+16)        address of v2: 16-byte packed decimal
00002098  E5C3E5C4 D8404040                       1825+          DC    CL8'VCVDQ'        instruction name
000020A0  00000010                                1826+          DC    A(16)             result length
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000020A4  00002124                                 1827+          DC    A(RE21)            address of expected resul
000020A8  00000000 00000000                        1828+          DS    FD                 gap
000020B0  00000000 00000000                        1829+V1O21     DS    XL16               V1 output
000020B8  00000000 00000000
000020C0  00000000 00000000                        1830+          DS    FD                    gap
000020C8  40404040 40404040                        1831+          DC    CL8' '             was cross check skipped?
000020D0  00000000 00000000                        1832+          DS    FD                    gap
000020D8  00000000 00000000                        1833+XCO21     DS    XL16               Cross check Output
000020E0  00000000 00000000
000020E8  00000000 00000000                        1834+          DS    FD                    gap
000020F0  00000000 00000000                        1835+          DS    2F           extract PSW after test (has CC)
000020F8  00                                       1836+          DS    X            extracted cc
000020F9  00                                       1837+          DS    X            CC Check failed
                                                   1838+*
000020FC                                           1839+X21       DS    0F
000020FC  E720 8F56 0006              00001156     1840+          VL    V2,V1FUDGE
00002102  E320 500C 0014              00002094     1841+          LGF   R2,V2_21            get v2 address
00002108  E722 0000 0006              00000000     1842+          VL    V2,0(R2)

0000210E                                           1845+          DS    0H                       E64A VCVDQ
0000210E  E6                                       1846+          DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000210F  22                                       1847+          DC    X'22'                     v1, v2
00002110  00                                       1848+          DC    X'00'                    reserved
00002111  99F0                                     1849+          DC    HL2'39408'                m4, i3, rXB
00002113  4A                                       1850+          DC    X'4A'

00002114  B98D 0020                                1852+          EPSW  R2,R0              exptract psw
00002118  5020 5068                   00000068     1853+          ST    R2,CCPSW               to save CC
0000211C  E720 5028 000E              000020B0     1854+          VST   V2,V1O21           save instruction result
00002122  07FB                                     1855+          BR    R11                return
00002124                                           1856+RE21      DS    0F                 expected 16 byte result
00002124                                           1857+          DROP  R5
00002124  69209384 63463374                        1858           DC    XL16'6920938463463374 607431768211455C'      v1
0000212C  60743176 8211455C
00002134  FFFFFFFF FFFFFFFF                        1859           DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
0000213C  FFFFFFFF FFFFFFFF
                                                   1860
                                                   1861           VRI_J VCVDQ,X'9F',X'9',0,N
00002148                                           1862+          DS    0FD
00002148                    00002148               1863+          USING *,R5               base for test data and test routine
00002148  000021BC                                 1864+T22       DC    A(X22)             address of test routine
0000214C  0016                                     1865+          DC    H'22'              test number
0000214E  00                                       1866+          DC    X'00'
0000214F  D5                                       1867+          DC    CL1'N'             S or Y = skip cross check
00002150  9F                                       1868+          DC    X'9F'              i3
00002151  09                                       1869+          DC    X'9'               m4
00002152  00                                       1870+          DC    HL1'0'             cc
00002153  07                                       1871+          DC    HL1'7'             cc failed mask
00002154  000021F4                                 1872+V2_22     DC    A(RE22+16)         address of v2: 16-byte packed decimal
00002158  E5C3E5C4 D8404040                        1873+          DC    CL8'VCVDQ'         instruction name
00002160  00000010                                 1874+          DC    A(16)              result length
00002164  000021E4                                 1875+          DC    A(RE22)            address of expected resul
00002168  00000000 00000000                        1876+          DS    FD                 gap
00002170  00000000 00000000                        1877+V1O22     DS    XL16               V1 output
00002178  00000000 00000000
00002180  00000000 00000000                        1878+          DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002188  40404040 40404040                       1879+          DC    CL8' '              was cross check skipped?
00002190  00000000 00000000                       1880+          DS    FD                     gap
00002198  00000000 00000000                       1881+XC022     DS    XL16                Cross check Output
000021A0  00000000 00000000
000021A8  00000000 00000000                       1882+          DS    FD                     gap
000021B0  00000000 00000000                       1883+          DS    2F            extract PSW after test (has CC)
000021B8  00                                      1884+          DS    X             extracted cc
000021B9  00                                      1885+          DS    X             CC Check failed
                                                  1886+*
000021BC                                          1887+X22       DS    0F
000021BC  E720 8F56 0006              00001156    1888+          VL    V2,V1FUDGE
000021C2  E320 500C 0014              00002154    1889+          LGF   R2,V2_22             get v2 address
000021C8  E722 0000 0006              00000000    1890+          VL    V2,0(R2)

000021CE                                          1893+          DS    0H                          E64A VCVDQ
000021CE  E6                                      1894+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
000021CF  22                                      1895+          DC    X'22'                        v1, v2
000021D0  00                                      1896+          DC    X'00'                       reserved
000021D1  99F0                                    1897+          DC    HL2'39408'                   m4, i3, rXB
000021D3  4A                                      1898+          DC    X'4A'

000021D4  B98D 0020                               1900+          EPSW  R2,R0               exptract psw
000021D8  5020 5068                   00000068    1901+          ST    R2,CCPSW                to save CC
000021DC  E720 5028 000E              00002170    1902+          VST   V2,V1O22            save instruction result
000021E2  07FB                                    1903+          BR    R11                 return
000021E4                                          1904+RE22      DS    0F                  expected 16 byte result
000021E4                                          1905+          DROP  R5
000021E4  00000000 00009223                       1906           DC    XL16'0000000000009223 372036854775807C'      v1
000021EC  37203685 4775807C
000021F4  00000000 00000000                       1907           DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
000021FC  7FFFFFFF FFFFFFFF
                                                  1908
                                                  1909           VRI_J VCVDQ,X'9F',X'9',3,S
00002208                                          1910+          DS    0FD
00002208                         00002208         1911+          USING *,R5                base for test data and test routine
00002208  0000227C                                1912+T23       DC    A(X23)              address of test routine
0000220C  0017                                    1913+          DC    H'23'               test number
0000220E  00                                      1914+          DC    X'00'
0000220F  E2                                      1915+          DC    CL1'S'              S or Y = skip cross check
00002210  9F                                      1916+          DC    X'9F'               i3
00002211  09                                      1917+          DC    X'9'                m4
00002212  03                                      1918+          DC    HL1'3'              cc
00002213  0E                                      1919+          DC    HL1'14'             cc failed mask
00002214  000022B4                                1920+V2_23     DC    A(RE23+16)          address of v2: 16-byte packed decimal
00002218  E5C3E5C4 D8404040                       1921+          DC    CL8'VCVDQ'          instruction name
00002220  00000010                                1922+          DC    A(16)               result length
00002224  000022A4                                1923+          DC    A(RE23)             address of expected resul
00002228  00000000 00000000                       1924+          DS    FD                  gap
00002230  00000000 00000000                       1925+V1O23     DS    XL16                V1 output
00002238  00000000 00000000
00002240  00000000 00000000                       1926+          DS    FD                     gap
00002248  40404040 40404040                       1927+          DC    CL8' '              was cross check skipped?
00002250  00000000 00000000                       1928+          DS    FD                     gap
00002258  00000000 00000000                       1929+XC023     DS    XL16                Cross check Output
00002260  00000000 00000000
00002268  00000000 00000000                       1930+          DS    FD                     gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002270  00000000 00000000                       1931+         DS    2F          extract PSW after test (has CC)
00002278  00                                      1932+         DS    X           extracted cc
00002279  00                                      1933+         DS    X           CC Check failed
                                                  1934+*
0000227C                                          1935+X23      DS    0F
0000227C  E720 8F56 0006              00001156    1936+         VL    V2,V1FUDGE
00002282  E320 500C 0014              00002214    1937+         LGF   R2,V2_23             get v2 address
00002288  E722 0000 0006              00000000    1938+         VL    V2,0(R2)

0000228E                                          1941+         DS    0H                               E64A VCVDQ
0000228E  E6                                      1942+         DC    X'E6'                             VECTOR CONVERT TO DECIMAL (128)
0000228F  22                                      1943+         DC    X'22'                             v1, v2
00002290  00                                      1944+         DC    X'00'                            reserved
00002291  99F0                                    1945+         DC    HL2'39408'                        m4, i3, rXB
00002293  4A                                      1946+         DC    X'4A'

00002294  B98D 0020                               1948+         EPSW  R2,R0               exptract psw
00002298  5020 5068                   00000068    1949+         ST    R2,CCPSW                to save CC
0000229C  E720 5028 000E              00002230    1950+         VST   V2,V1O23            save instruction result
000022A2  07FB                                    1951+         BR    R11                 return
000022A4                                          1952+RE23     DS    0F                  expected 16 byte result
000022A4                                          1953+         DROP  R5
000022A4  69209384 63454151                       1954          DC    XL16'6920938463454151 235394913435649C'      v1
000022AC  23539491 3435649C
000022B4  FFFFFFFF FFFFFFFF                       1955          DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
000022BC  80000000 00000001
                                                  1956
                                                  1957          VRI_J VCVDQ,X'9F',X'9',0,N
000022C8                                          1958+         DS    0FD
000022C8                    000022C8              1959+         USING *,R5                base for test data and test routine
000022C8  0000233C                                1960+T24      DC    A(X24)              address of test routine
000022CC  0018                                    1961+         DC    H'24'               test number
000022CE  00                                      1962+         DC    X'00'
000022CF  D5                                      1963+         DC    CL1'N'              S or Y = skip cross check
000022D0  9F                                      1964+         DC    X'9F'               i3
000022D1  09                                      1965+         DC    X'9'                m4
000022D2  00                                      1966+         DC    HL1'0'              cc
000022D3  07                                      1967+         DC    HL1'7'              cc failed mask
000022D4  00002374                                1968+V2_24    DC    A(RE24+16)          address of v2: 16-byte packed decimal
000022D8  E5C3E5C4 D8404040                       1969+         DC    CL8'VCVDQ'          instruction name
000022E0  00000010                                1970+         DC    A(16)               result length
000022E4  00002364                                1971+         DC    A(RE24)             address of expected resul
000022E8  00000000 00000000                       1972+         DS    FD                  gap
000022F0  00000000 00000000                       1973+V1O24    DS    XL16                V1 output
000022F8  00000000 00000000
00002300  00000000 00000000                       1974+         DS    FD                     gap
00002308  40404040 40404040                       1975+         DC    CL8' '              was cross check skipped?
00002310  00000000 00000000                       1976+         DS    FD                     gap
00002318  00000000 00000000                       1977+XCO24    DS    XL16                Cross check Output
00002320  00000000 00000000
00002328  00000000 00000000                       1978+         DS    FD                     gap
00002330  00000000 00000000                       1979+         DS    2F          extract PSW after test (has CC)
00002338  00                                      1980+         DS    X           extracted cc
00002339  00                                      1981+         DS    X           CC Check failed
                                                  1982+*
0000233C                                          1983+X24      DS    0F
```

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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000233C  E720 8F56 0006                00001156  1984+         VL    V2,V1FUDGE
00002342  E320 500C 0014                000022D4  1985+         LGF   R2,V2_24           get v2 address
00002348  E722 0000 0006                00000000  1986+         VL    V2,0(R2)

0000234E                                          1989+         DS    0H                      E64A VCVDQ
0000234E  E6                                      1990+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000234F  22                                      1991+         DC    X'22'                    v1, v2
00002350  00                                      1992+         DC    X'00'                   reserved
00002351  99F0                                    1993+         DC    HL2'39408'               m4, i3, rXB
00002353  4A                                      1994+         DC    X'4A'

00002354  B98D 0020                               1996+         EPSW  R2,R0             exptract psw
00002358  5020 5068                     00000068  1997+         ST    R2,CCPSW              to save CC
0000235C  E720 5028 000E                000022F0  1998+         VST   V2,V1O24          save instruction result
00002362  07FB                                    1999+         BR    R11               return
00002364                                          2000+RE24     DS    0F                expected 16 byte result
00002364                                          2001+         DROP  R5
00002364  99999999 99999999                       2002          DC    XL16'9999999999999999 999999999999999C'      v1
0000236C  99999999 9999999C
00002374  0000007E 37BE2022                       2003          DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
0000237C  C0914B26 7FFFFFFF
                                                  2004
                                                  2005          VRI_J VCVDQ,X'9F',X'9',3,S
00002388                                          2006+         DS    0FD
00002388                       00002388           2007+         USING *,R5              base for test data and test routine
00002388  000023FC                                2008+T25      DC    A(X25)            address of test routine
0000238C  0019                                    2009+         DC    H'25'             test number
0000238E  00                                      2010+         DC    X'00'
0000238F  E2                                      2011+         DC    CL1'S'            S or Y = skip cross check
00002390  9F                                      2012+         DC    X'9F'             i3
00002391  09                                      2013+         DC    X'9'              m4
00002392  03                                      2014+         DC    HL1'3'            cc
00002393  0E                                      2015+         DC    HL1'14'           cc failed mask
00002394  00002434                                2016+V2_25    DC    A(RE25+16)        address of v2: 16-byte packed decimal
00002398  E5C3E5C4 D8404040                       2017+         DC    CL8'VCVDQ'        instruction name
000023A0  00000010                                2018+         DC    A(16)             result length
000023A4  00002424                                2019+         DC    A(RE25)           address of expected resul
000023A8  00000000 00000000                       2020+         DS    FD                gap
000023B0  00000000 00000000                       2021+V1O25    DS    XL16              V1 output
000023B8  00000000 00000000
000023C0  00000000 00000000                       2022+         DS    FD                   gap
000023C8  40404040 40404040                       2023+         DC    CL8' '            was cross check skipped?
000023D0  00000000 00000000                       2024+         DS    FD                   gap
000023D8  00000000 00000000                       2025+XCO25    DS    XL16              Cross check Output
000023E0  00000000 00000000
000023E8  00000000 00000000                       2026+         DS    FD                   gap
000023F0  00000000 00000000                       2027+         DS    2F          extract PSW after test (has CC)
000023F8  00                                      2028+         DS    X           extracted cc
000023F9  00                                      2029+         DS    X           CC Check failed
                                                  2030+*
000023FC                                          2031+X25      DS    0F
000023FC  E720 8F56 0006                00001156  2032+         VL    V2,V1FUDGE
00002402  E320 500C 0014                00002394  2033+         LGF   R2,V2_25           get v2 address
00002408  E722 0000 0006                00000000  2034+         VL    V2,0(R2)

0000240E                                          2037+         DS    0H                      E64A VCVDQ
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000240E  E6                                       2038+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
0000240F  22                                       2039+          DC    X'22'                        v1, v2
00002410  00                                       2040+          DC    X'00'                       reserved
00002411  99F0                                     2041+          DC    HL2'39408'                   m4, i3, rXB
00002413  4A                                       2042+          DC    X'4A'

00002414  B98D 0020                                2044+          EPSW  R2,R0                 exptract psw
00002418  5020 5068                      00000068  2045+          ST    R2,CCPSW                  to save CC
0000241C  E720 5028 000E                 000023B0  2046+          VST   V2,V1O25              save instruction result
00002422  07FB                                     2047+          BR    R11                   return
00002424                                           2048+RE25      DS    0F                    expected 16 byte result
00002424                                           2049+          DROP  R5
00002424  69209384 63463374                        2050           DC    XL16'6920938463463374 607431768211457C'      v1
0000242C  60743176 8211457C
00002434  FFFFFF81 C841DFDD                        2051           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
0000243C  3F6EB4D9 80000001
                                                   2052 *
                                                   2053 * overflow
                                                   2054 *
                                                   2055           VRI_J VCVDQ,X'9F',X'9',3,N
00002448                                           2056+          DS    0FD
00002448                       00002448            2057+          USING *,R5                  base for test data and test routine
00002448  000024BC                                 2058+T26       DC    A(X26)                address of test routine
0000244C  001A                                     2059+          DC    H'26'                 test number
0000244E  00                                       2060+          DC    X'00'
0000244F  D5                                       2061+          DC    CL1'N'                S or Y = skip cross check
00002450  9F                                       2062+          DC    X'9F'                 i3
00002451  09                                       2063+          DC    X'9'                  m4
00002452  03                                       2064+          DC    HL1'3'                cc
00002453  0E                                       2065+          DC    HL1'14'               cc failed mask
00002454  000024F4                                 2066+V2_26     DC    A(RE26+16)            address of v2: 16-byte packed decimal
00002458  E5C3E5C4 D8404040                        2067+          DC    CL8'VCVDQ'            instruction name
00002460  00000010                                 2068+          DC    A(16)                 result length
00002464  000024E4                                 2069+          DC    A(RE26)               address of expected resul
00002468  00000000 00000000                        2070+          DS    FD                    gap
00002470  00000000 00000000                        2071+V1O26     DS    XL16                  V1 output
00002478  00000000 00000000
00002480  00000000 00000000                        2072+          DS    FD                       gap
00002488  40404040 40404040                        2073+          DC    CL8' '                was cross check skipped?
00002490  00000000 00000000                        2074+          DS    FD                       gap
00002498  00000000 00000000                        2075+XCO26     DS    XL16                  Cross check Output
000024A0  00000000 00000000
000024A8  00000000 00000000                        2076+          DS    FD                       gap
000024B0  00000000 00000000                        2077+          DS    2F              extract PSW after test (has CC)
000024B8  00                                       2078+          DS    X               extracted cc
000024B9  00                                       2079+          DS    X               CC Check failed
                                                   2080+*
000024BC                                           2081+X26       DS    0F
000024BC  E720 8F56 0006                 00001156  2082+          VL    V2,V1FUDGE
000024C2  E320 500C 0014                 00002454  2083+          LGF   R2,V2_26                get v2 address
000024C8  E722 0000 0006                 00000000  2084+          VL    V2,0(R2)

000024CE                                           2087+          DS    0H                          E64A VCVDQ
000024CE  E6                                       2088+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
000024CF  22                                       2089+          DC    X'22'                        v1, v2
000024D0  00                                       2090+          DC    X'00'                       reserved
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000024D1  99F0                                     2091+         DC    HL2'39408'                m4, i3, rXB
000024D3  4A                                       2092+         DC    X'4A'

000024D4  B98D 0020                                2094+         EPSW  R2,R0             exptract psw
000024D8  5020 5068                      00000068  2095+         ST    R2,CCPSW              to save CC
000024DC  E720 5028 000E                 00002470  2096+         VST   V2,V1O26          save instruction result
000024E2  07FB                                     2097+         BR    R11               return
000024E4                                           2098+RE26     DS    0F                expected 16 byte result
000024E4                                           2099+         DROP  R5
000024E4  00000000 00000000                        2100          DC    XL16'0000000000000000 000000000000000C'      v1
000024EC  00000000 0000000C
000024F4  0000007E 37BE2022                        2101          DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
000024FC  C0914B26 80000000
                                                   2102
                                                   2103          VRI_J VCVDQ,X'9F',X'9',3,N
00002508                                           2104+         DS    0FD
00002508                       00002508            2105+         USING *,R5              base for test data and test routine
00002508  0000257C                                 2106+T27      DC    A(X27)            address of test routine
0000250C  001B                                     2107+         DC    H'27'             test number
0000250E  00                                       2108+         DC    X'00'
0000250F  D5                                       2109+         DC    CL1'N'            S or Y = skip cross check
00002510  9F                                       2110+         DC    X'9F'             i3
00002511  09                                       2111+         DC    X'9'              m4
00002512  03                                       2112+         DC    HL1'3'            cc
00002513  0E                                       2113+         DC    HL1'14'           cc failed mask
00002514  000025B4                                 2114+V2_27    DC    A(RE27+16)        address of v2: 16-byte packed decimal
00002518  E5C3E5C4 D8404040                        2115+         DC    CL8'VCVDQ'        instruction name
00002520  00000010                                 2116+         DC    A(16)             result length
00002524  000025A4                                 2117+         DC    A(RE27)           address of expected resul
00002528  00000000 00000000                        2118+         DS    FD                gap
00002530  00000000 00000000                        2119+V1O27    DS    XL16              V1 output
00002538  00000000 00000000
00002540  00000000 00000000                        2120+         DS    FD                   gap
00002548  40404040 40404040                        2121+         DC    CL8' '            was cross check skipped?
00002550  00000000 00000000                        2122+         DS    FD                   gap
00002558  00000000 00000000                        2123+XCO27    DS    XL16              Cross check Output
00002560  00000000 00000000
00002568  00000000 00000000                        2124+         DS    FD                   gap
00002570  00000000 00000000                        2125+         DS    2F          extract PSW after test (has CC)
00002578  00                                       2126+         DS    X           extracted cc
00002579  00                                       2127+         DS    X           CC Check failed
                                                   2128+*
0000257C                                           2129+X27      DS    0F
0000257C  E720 8F56 0006                 00001156  2130+         VL    V2,V1FUDGE
00002582  E320 500C 0014                 00002514  2131+         LGF   R2,V2_27           get v2 address
00002588  E722 0000 0006                 00000000  2132+         VL    V2,0(R2)

0000258E                                           2135+         DS    0H                      E64A VCVDQ
0000258E  E6                                       2136+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000258F  22                                       2137+         DC    X'22'                    v1, v2
00002590  00                                       2138+         DC    X'00'                   reserved
00002591  99F0                                     2139+         DC    HL2'39408'               m4, i3, rXB
00002593  4A                                       2140+         DC    X'4A'

00002594  B98D 0020                                2142+         EPSW  R2,R0             exptract psw
00002598  5020 5068                      00000068  2143+         ST    R2,CCPSW              to save CC
```

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  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000259C  E720 5028 000E                00002530  2144+        VST   V2,V1O27          save instruction result
000025A2  07FB                                    2145+        BR    R11               return
000025A4                                          2146+RE27    DS    0F                expected 16 byte result
000025A4                                          2147+        DROP  R5
000025A4  69209384 63463374                       2148         DC    XL16'6920938463463374 607431768211456C'      v1
000025AC  60743176 8211456C
000025B4  FFFFFF81 C841DFDD                       2149         DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
000025BC  3F6EB4D9 80000000
                                                  2150
                                                  2151
                                                  2152 *----------------------------------------------------------------------
                                                  2153 *       i3:   x'9f"   (IOM=1, RDC=31)
                                                  2154 *       m4:   x'B'    (LB=1, P1=1, CS=1)
                                                  2155 *
                                                  2156         VRI_J VCVDQ,X'9F',X'B',0,N
000025C8                                          2157+        DS    0FD
000025C8                      000025C8            2158+        USING *,R5              base for test data and test routine
000025C8  0000263C                                2159+T28     DC    A(X28)            address of test routine
000025CC  001C                                    2160+        DC    H'28'             test number
000025CE  00                                      2161+        DC    X'00'
000025CF  D5                                      2162+        DC    CL1'N'            S or Y = skip cross check
000025D0  9F                                      2163+        DC    X'9F'             i3
000025D1  0B                                      2164+        DC    X'B'              m4
000025D2  00                                      2165+        DC    HL1'0'            cc
000025D3  07                                      2166+        DC    HL1'7'            cc failed mask
000025D4  00002674                                2167+V2_28   DC    A(RE28+16)        address of v2: 16-byte packed decimal
000025D8  E5C3E5C4 D8404040                       2168+        DC    CL8'VCVDQ'        instruction name
000025E0  00000010                                2169+        DC    A(16)             result length
000025E4  00002664                                2170+        DC    A(RE28)           address of expected resul
000025E8  00000000 00000000                       2171+        DS    FD                gap
000025F0  00000000 00000000                       2172+V1O28   DS    XL16              V1 output
000025F8  00000000 00000000
00002600  00000000 00000000                       2173+        DS    FD                   gap
00002608  40404040 40404040                       2174+        DC    CL8' '            was cross check skipped?
00002610  00000000 00000000                       2175+        DS    FD                   gap
00002618  00000000 00000000                       2176+XCO28   DS    XL16              Cross check Output
00002620  00000000 00000000
00002628  00000000 00000000                       2177+        DS    FD                   gap
00002630  00000000 00000000                       2178+        DS    2F          extract PSW after test (has CC)
00002638  00                                      2179+        DS    X           extracted cc
00002639  00                                      2180+        DS    X           CC Check failed
                                                  2181+*
0000263C                                          2182+X28     DS    0F
0000263C  E720 8F56 0006                00001156  2183+        VL    V2,V1FUDGE
00002642  E320 500C 0014                000025D4  2184+        LGF   R2,V2_28           get v2 address
00002648  E722 0000 0006                00000000  2185+        VL    V2,0(R2)

0000264E                                          2188+        DS    0H                      E64A VCVDQ
0000264E  E6                                      2189+        DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000264F  22                                      2190+        DC    X'22'                    v1, v2
00002650  00                                      2191+        DC    X'00'                   reserved
00002651  B9F0                                    2192+        DC    HL2'47600'               m4, i3, rXB
00002653  4A                                      2193+        DC    X'4A'

00002654  B98D 0020                               2195+        EPSW  R2,R0             exptract psw
00002658  5020 5068                     00000068  2196+        ST    R2,CCPSW             to save CC
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000265C  E720 5028 000E                000025F0  2197+         VST   V2,V1O28          save instruction result
00002662  07FB                                    2198+         BR    R11               return
00002664                                          2199+RE28     DS    0F                expected 16 byte result
00002664                                          2200+         DROP  R5
00002664  00000000 00000000                       2201          DC    XL16'0000000000000000 000000000000000F'      v1
0000266C  00000000 0000000F
00002674  00000000 00000000                       2202          DC    XL16'0000000000000000 0000000000000000'      v2
0000267C  00000000 00000000
                                                  2203
                                                  2204          VRI_J VCVDQ,X'9F',X'B',0,N
00002688                                          2205+         DS    0FD
00002688                     00002688             2206+         USING *,R5              base for test data and test routine
00002688  000026FC                                2207+T29      DC    A(X29)            address of test routine
0000268C  001D                                    2208+         DC    H'29'             test number
0000268E  00                                      2209+         DC    X'00'
0000268F  D5                                      2210+         DC    CL1'N'            S or Y = skip cross check
00002690  9F                                      2211+         DC    X'9F'             i3
00002691  0B                                      2212+         DC    X'B'              m4
00002692  00                                      2213+         DC    HL1'0'            cc
00002693  07                                      2214+         DC    HL1'7'            cc failed mask
00002694  00002734                                2215+V2_29    DC    A(RE29+16)        address of v2: 16-byte packed decimal
00002698  E5C3E5C4 D8404040                       2216+         DC    CL8'VCVDQ'        instruction name
000026A0  00000010                                2217+         DC    A(16)             result length
000026A4  00002724                                2218+         DC    A(RE29)           address of expected resul
000026A8  00000000 00000000                       2219+         DS    FD                gap
000026B0  00000000 00000000                       2220+V1O29    DS    XL16              V1 output
000026B8  00000000 00000000
000026C0  00000000 00000000                       2221+         DS    FD                   gap
000026C8  40404040 40404040                       2222+         DC    CL8' '            was cross check skipped?
000026D0  00000000 00000000                       2223+         DS    FD                   gap
000026D8  00000000 00000000                       2224+XCO29    DS    XL16              Cross check Output
000026E0  00000000 00000000
000026E8  00000000 00000000                       2225+         DS    FD                   gap
000026F0  00000000 00000000                       2226+         DS    2F          extract PSW after test (has CC)
000026F8  00                                      2227+         DS    X           extracted cc
000026F9  00                                      2228+         DS    X           CC Check failed
                                                  2229+*
000026FC                                          2230+X29      DS    0F
000026FC  E720 8F56 0006                00001156  2231+         VL    V2,V1FUDGE
00002702  E320 500C 0014                00002694  2232+         LGF   R2,V2_29           get v2 address
00002708  E722 0000 0006                00000000  2233+         VL    V2,0(R2)

0000270E                                          2236+         DS    0H                      E64A VCVDQ
0000270E  E6                                      2237+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000270F  22                                      2238+         DC    X'22'                    v1, v2
00002710  00                                      2239+         DC    X'00'                   reserved
00002711  B9F0                                    2240+         DC    HL2'47600'               m4, i3, rXB
00002713  4A                                      2241+         DC    X'4A'

00002714  B98D 0020                               2243+         EPSW  R2,R0             exptract psw
00002718  5020 5068                     00000068  2244+         ST    R2,CCPSW              to save CC
0000271C  E720 5028 000E                000026B0  2245+         VST   V2,V1O29          save instruction result
00002722  07FB                                    2246+         BR    R11               return
00002724                                          2247+RE29     DS    0F                expected 16 byte result
00002724                                          2248+         DROP  R5
00002724  00000000 00000000                       2249          DC    XL16'0000000000000000 000000000000001F'      v1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000272C  00000000 0000001F
00002734  00000000 00000000                        2250          DC    XL16'0000000000000000 0000000000000001'      v2
0000273C  00000000 00000001
                                                   2251
                                                   2252          VRI_J VCVDQ,X'9F',X'B',3,S
00002748                                           2253+         DS    0FD
00002748                     00002748              2254+         USING *,R5                 base for test data and test routine
00002748  000027BC                                 2255+T30      DC    A(X30)               address of test routine
0000274C  001E                                     2256+         DC    H'30'                test number
0000274E  00                                       2257+         DC    X'00'
0000274F  E2                                       2258+         DC    CL1'S'               S or Y = skip cross check
00002750  9F                                       2259+         DC    X'9F'                i3
00002751  0B                                       2260+         DC    X'B'                 m4
00002752  03                                       2261+         DC    HL1'3'               cc
00002753  0E                                       2262+         DC    HL1'14'              cc failed mask
00002754  000027F4                                 2263+V2_30    DC    A(RE30+16)           address of v2: 16-byte packed decimal
00002758  E5C3E5C4 D8404040                        2264+         DC    CL8'VCVDQ'           instruction name
00002760  00000010                                 2265+         DC    A(16)                result length
00002764  000027E4                                 2266+         DC    A(RE30)              address of expected resul
00002768  00000000 00000000                        2267+         DS    FD                   gap
00002770  00000000 00000000                        2268+V1O30    DS    XL16                 V1 output
00002778  00000000 00000000
00002780  00000000 00000000                        2269+         DS    FD                      gap
00002788  40404040 40404040                        2270+         DC    CL8' '               was cross check skipped?
00002790  00000000 00000000                        2271+         DS    FD                      gap
00002798  00000000 00000000                        2272+XCO30    DS    XL16                 Cross check Output
000027A0  00000000 00000000
000027A8  00000000 00000000                        2273+         DS    FD                      gap
000027B0  00000000 00000000                        2274+         DS    2F             extract PSW after test (has CC)
000027B8  00                                       2275+         DS    X              extracted cc
000027B9  00                                       2276+         DS    X              CC Check failed
                                                   2277+*
000027BC                                           2278+X30      DS    0F
000027BC  E720 8F56 0006               00001156    2279+         VL    V2,V1FUDGE
000027C2  E320 500C 0014               00002754    2280+         LGF   R2,V2_30               get v2 address
000027C8  E722 0000 0006               00000000    2281+         VL    V2,0(R2)

000027CE                                           2284+         DS    0H                         E64A VCVDQ
000027CE  E6                                       2285+         DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
000027CF  22                                       2286+         DC    X'22'                       v1, v2
000027D0  00                                       2287+         DC    X'00'                      reserved
000027D1  B9F0                                     2288+         DC    HL2'47600'                  m4, i3, rXB
000027D3  4A                                       2289+         DC    X'4A'

000027D4  B98D 0020                                2291+         EPSW  R2,R0                exptract psw
000027D8  5020 5068                    00000068    2292+         ST    R2,CCPSW                 to save CC
000027DC  E720 5028 000E               00002770    2293+         VST   V2,V1O30             save instruction result
000027E2  07FB                                     2294+         BR    R11                  return
000027E4                                           2295+RE30     DS    0F                   expected 16 byte result
000027E4                                           2296+         DROP  R5
000027E4  69209384 63463374                        2297          DC    XL16'6920938463463374 607431768211455F'      v1
000027EC  60743176 8211455F
000027F4  FFFFFFFF FFFFFFFF                        2298          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
000027FC  FFFFFFFF FFFFFFFF
                                                   2299
                                                   2300          VRI_J VCVDQ,X'9F',X'B',0,N
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002808                                           2301+          DS    0FD
00002808                       00002808            2302+          USING *,R5                base for test data and test routine
00002808  0000287C                                 2303+T31       DC    A(X31)              address of test routine
0000280C  001F                                     2304+          DC    H'31'               test number
0000280E  00                                       2305+          DC    X'00'
0000280F  D5                                       2306+          DC    CL1'N'              S or Y = skip cross check
00002810  9F                                       2307+          DC    X'9F'               i3
00002811  0B                                       2308+          DC    X'B'                m4
00002812  00                                       2309+          DC    HL1'0'              cc
00002813  07                                       2310+          DC    HL1'7'              cc failed mask
00002814  000028B4                                 2311+V2_31     DC    A(RE31+16)          address of v2: 16-byte packed decimal
00002818  E5C3E5C4 D8404040                        2312+          DC    CL8'VCVDQ'          instruction name
00002820  00000010                                 2313+          DC    A(16)               result length
00002824  000028A4                                 2314+          DC    A(RE31)             address of expected resul
00002828  00000000 00000000                        2315+          DS    FD                  gap
00002830  00000000 00000000                        2316+V1O31     DS    XL16                V1 output
00002838  00000000 00000000
00002840  00000000 00000000                        2317+          DS    FD                     gap
00002848  40404040 40404040                        2318+          DC    CL8' '              was cross check skipped?
00002850  00000000 00000000                        2319+          DS    FD                     gap
00002858  00000000 00000000                        2320+XCO31     DS    XL16                Cross check Output
00002860  00000000 00000000
00002868  00000000 00000000                        2321+          DS    FD                     gap
00002870  00000000 00000000                        2322+          DS    2F            extract PSW after test (has CC)
00002878  00                                       2323+          DS    X             extracted cc
00002879  00                                       2324+          DS    X             CC Check failed
                                                   2325+*
0000287C                                           2326+X31       DS    0F
0000287C  E720 8F56 0006             00001156      2327+          VL    V2,V1FUDGE
00002882  E320 500C 0014             00002814      2328+          LGF   R2,V2_31             get v2 address
00002888  E722 0000 0006             00000000      2329+          VL    V2,0(R2)

0000288E                                           2332+          DS    0H                        E64A VCVDQ
0000288E  E6                                       2333+          DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000288F  22                                       2334+          DC    X'22'                      v1, v2
00002890  00                                       2335+          DC    X'00'                     reserved
00002891  B9F0                                     2336+          DC    HL2'47600'                 m4, i3, rXB
00002893  4A                                       2337+          DC    X'4A'

00002894  B98D 0020                                2339+          EPSW  R2,R0               exptract psw
00002898  5020 5068                  00000068      2340+          ST    R2,CCPSW                to save CC
0000289C  E720 5028 000E             00002830      2341+          VST   V2,V1O31            save instruction result
000028A2  07FB                                     2342+          BR    R11                 return
000028A4                                           2343+RE31      DS    0F                  expected 16 byte result
000028A4                                           2344+          DROP  R5
000028A4  00000000 00009223                        2345           DC    XL16'0000000000009223 372036854775807F'      v1
000028AC  37203685 4775807F
000028B4  00000000 00000000                        2346           DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
000028BC  7FFFFFFF FFFFFFFF
                                                   2347
                                                   2348           VRI_J VCVDQ,X'9F',X'B',3,S
000028C8                                           2349+          DS    0FD
000028C8                       000028C8            2350+          USING *,R5                base for test data and test routine
000028C8  0000293C                                 2351+T32       DC    A(X32)              address of test routine
000028CC  0020                                     2352+          DC    H'32'               test number
000028CE  00                                       2353+          DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000028CF  E2                                      2354+         DC    CL1'S'              S or Y = skip cross check
000028D0  9F                                      2355+         DC    X'9F'               i3
000028D1  0B                                      2356+         DC    X'B'                m4
000028D2  03                                      2357+         DC    HL1'3'              cc
000028D3  0E                                      2358+         DC    HL1'14'             cc failed mask
000028D4  00002974                                2359+V2_32    DC    A(RE32+16)          address of v2: 16-byte packed decimal
000028D8  E5C3E5C4 D8404040                       2360+         DC    CL8'VCVDQ'          instruction name
000028E0  00000010                                2361+         DC    A(16)               result length
000028E4  00002964                                2362+         DC    A(RE32)             address of expected resul
000028E8  00000000 00000000                       2363+         DS    FD                  gap
000028F0  00000000 00000000                       2364+V1O32    DS    XL16                V1 output
000028F8  00000000 00000000
00002900  00000000 00000000                       2365+         DS    FD                     gap
00002908  40404040 40404040                       2366+         DC    CL8' '              was cross check skipped?
00002910  00000000 00000000                       2367+         DS    FD                     gap
00002918  00000000 00000000                       2368+XCO32    DS    XL16                Cross check Output
00002920  00000000 00000000
00002928  00000000 00000000                       2369+         DS    FD                     gap
00002930  00000000 00000000                       2370+         DS    2F            extract PSW after test (has CC)
00002938  00                                      2371+         DS    X             extracted cc
00002939  00                                      2372+         DS    X             CC Check failed
                                                  2373+*
0000293C                                          2374+X32      DS    0F
0000293C  E720 8F56 0006             00001156     2375+         VL    V2,V1FUDGE
00002942  E320 500C 0014             000028D4     2376+         LGF   R2,V2_32             get v2 address
00002948  E722 0000 0006             00000000     2377+         VL    V2,0(R2)

0000294E                                          2380+         DS    0H                        E64A VCVDQ
0000294E  E6                                      2381+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000294F  22                                      2382+         DC    X'22'                      v1, v2
00002950  00                                      2383+         DC    X'00'                     reserved
00002951  B9F0                                    2384+         DC    HL2'47600'                 m4, i3, rXB
00002953  4A                                      2385+         DC    X'4A'

00002954  B98D 0020                               2387+         EPSW  R2,R0               exptract psw
00002958  5020 5068                  00000068     2388+         ST    R2,CCPSW                to save CC
0000295C  E720 5028 000E             000028F0     2389+         VST   V2,V1O32            save instruction result
00002962  07FB                                    2390+         BR    R11                 return
00002964                                          2391+RE32     DS    0F                  expected 16 byte result
00002964                                          2392+         DROP  R5
00002964  69209384 63454151                       2393          DC    XL16'6920938463454151 235394913435649F'      v1
0000296C  23539491 3435649F
00002974  FFFFFFFF FFFFFFFF                       2394          DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
0000297C  80000000 00000001
                                                  2395
                                                  2396          VRI_J VCVDQ,X'9F',X'B',0,N
00002988                                          2397+         DS    0FD
00002988                        00002988          2398+         USING *,R5                base for test data and test routine
00002988  000029FC                                2399+T33      DC    A(X33)              address of test routine
0000298C  0021                                    2400+         DC    H'33'               test number
0000298E  00                                      2401+         DC    X'00'
0000298F  D5                                      2402+         DC    CL1'N'              S or Y = skip cross check
00002990  9F                                      2403+         DC    X'9F'               i3
00002991  0B                                      2404+         DC    X'B'                m4
00002992  00                                      2405+         DC    HL1'0'              cc
00002993  07                                      2406+         DC    HL1'7'              cc failed mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002994  00002A34                                2407+V2_33    DC    A(RE33+16)          address of v2: 16-byte packed decimal
00002998  E5C3E5C4 D8404040                       2408+         DC    CL8'VCVDQ'          instruction name
000029A0  00000010                                2409+         DC    A(16)               result length
000029A4  00002A24                                2410+         DC    A(RE33)             address of expected resul
000029A8  00000000 00000000                       2411+         DS    FD                  gap
000029B0  00000000 00000000                       2412+V1O33    DS    XL16                V1 output
000029B8  00000000 00000000
000029C0  00000000 00000000                       2413+         DS    FD                     gap
000029C8  40404040 40404040                       2414+         DC    CL8' '              was cross check skipped?
000029D0  00000000 00000000                       2415+         DS    FD                     gap
000029D8  00000000 00000000                       2416+XCO33    DS    XL16                Cross check Output
000029E0  00000000 00000000
000029E8  00000000 00000000                       2417+         DS    FD                     gap
000029F0  00000000 00000000                       2418+         DS    2F            extract PSW after test (has CC)
000029F8  00                                      2419+         DS    X             extracted cc
000029F9  00                                      2420+         DS    X             CC Check failed
                                                  2421+*
000029FC                                          2422+X33      DS    0F
000029FC  E720 8F56 0006                00001156  2423+         VL    V2,V1FUDGE
00002A02  E320 500C 0014                00002994  2424+         LGF   R2,V2_33             get v2 address
00002A08  E722 0000 0006                00000000  2425+         VL    V2,0(R2)

00002A0E                                          2428+         DS    0H                        E64A VCVDQ
00002A0E  E6                                      2429+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00002A0F  22                                      2430+         DC    X'22'                      v1, v2
00002A10  00                                      2431+         DC    X'00'                     reserved
00002A11  B9F0                                    2432+         DC    HL2'47600'                 m4, i3, rXB
00002A13  4A                                      2433+         DC    X'4A'

00002A14  B98D 0020                               2435+         EPSW  R2,R0               exptract psw
00002A18  5020 5068                     00000068  2436+         ST    R2,CCPSW                to save CC
00002A1C  E720 5028 000E                000029B0  2437+         VST   V2,V1O33            save instruction result
00002A22  07FB                                    2438+         BR    R11                 return
00002A24                                          2439+RE33     DS    0F                  expected 16 byte result
00002A24                                          2440+         DROP  R5
00002A24  99999999 99999999                       2441          DC    XL16'9999999999999999 999999999999999F'      v1
00002A2C  99999999 9999999F
00002A34  0000007E 37BE2022                       2442          DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
00002A3C  C0914B26 7FFFFFFF
                                                  2443
                                                  2444          VRI_J VCVDQ,X'9F',X'B',3,S
00002A48                                          2445+         DS    0FD
00002A48                      00002A48            2446+         USING *,R5                base for test data and test routine
00002A48  00002ABC                                2447+T34      DC    A(X34)              address of test routine
00002A4C  0022                                    2448+         DC    H'34'               test number
00002A4E  00                                      2449+         DC    X'00'
00002A4F  E2                                      2450+         DC    CL1'S'              S or Y = skip cross check
00002A50  9F                                      2451+         DC    X'9F'               i3
00002A51  0B                                      2452+         DC    X'B'                m4
00002A52  03                                      2453+         DC    HL1'3'              cc
00002A53  0E                                      2454+         DC    HL1'14'             cc failed mask
00002A54  00002AF4                                2455+V2_34    DC    A(RE34+16)          address of v2: 16-byte packed decimal
00002A58  E5C3E5C4 D8404040                       2456+         DC    CL8'VCVDQ'          instruction name
00002A60  00000010                                2457+         DC    A(16)               result length
00002A64  00002AE4                                2458+         DC    A(RE34)             address of expected resul
00002A68  00000000 00000000                       2459+         DS    FD                  gap
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002A70  00000000 00000000                        2460+V1O34    DS    XL16              V1 output
00002A78  00000000 00000000
00002A80  00000000 00000000                        2461+         DS    FD                   gap
00002A88  40404040 40404040                        2462+         DC    CL8' '            was cross check skipped?
00002A90  00000000 00000000                        2463+         DS    FD                   gap
00002A98  00000000 00000000                        2464+XCO34    DS    XL16              Cross check Output
00002AA0  00000000 00000000
00002AA8  00000000 00000000                        2465+         DS    FD                   gap
00002AB0  00000000 00000000                        2466+         DS    2F          extract PSW after test (has CC)
00002AB8  00                                       2467+         DS    X           extracted cc
00002AB9  00                                       2468+         DS    X           CC Check failed
                                                   2469+*
00002ABC                                           2470+X34      DS    0F
00002ABC  E720 8F56 0006             00001156      2471+         VL    V2,V1FUDGE
00002AC2  E320 500C 0014             00002A54      2472+         LGF   R2,V2_34           get v2 address
00002AC8  E722 0000 0006             00000000      2473+         VL    V2,0(R2)

00002ACE                                           2476+         DS    0H                      E64A VCVDQ
00002ACE  E6                                       2477+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
00002ACF  22                                       2478+         DC    X'22'                    v1, v2
00002AD0  00                                       2479+         DC    X'00'                   reserved
00002AD1  B9F0                                     2480+         DC    HL2'47600'               m4, i3, rXB
00002AD3  4A                                       2481+         DC    X'4A'

00002AD4  B98D 0020                                2483+         EPSW  R2,R0             exptract psw
00002AD8  5020 5068                  00000068      2484+         ST    R2,CCPSW              to save CC
00002ADC  E720 5028 000E             00002A70      2485+         VST   V2,V1O34          save instruction result
00002AE2  07FB                                     2486+         BR    R11               return
00002AE4                                           2487+RE34     DS    0F                expected 16 byte result
00002AE4                                           2488+         DROP  R5
00002AE4  69209384 63463374                        2489          DC    XL16'6920938463463374 607431768211457F'      v1
00002AEC  60743176 8211457F
00002AF4  FFFFFF81 C841DFDD                        2490          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
00002AFC  3F6EB4D9 80000001
                                                   2491 *
                                                   2492 * overflow
                                                   2493 *
                                                   2494          VRI_J VCVDQ,X'9F',X'B',3,N
00002B08                                           2495+         DS    0FD
00002B08                       00002B08            2496+         USING *,R5              base for test data and test routine
00002B08  00002B7C                                 2497+T35      DC    A(X35)            address of test routine
00002B0C  0023                                     2498+         DC    H'35'             test number
00002B0E  00                                       2499+         DC    X'00'
00002B0F  D5                                       2500+         DC    CL1'N'            S or Y = skip cross check
00002B10  9F                                       2501+         DC    X'9F'             i3
00002B11  0B                                       2502+         DC    X'B'              m4
00002B12  03                                       2503+         DC    HL1'3'            cc
00002B13  0E                                       2504+         DC    HL1'14'           cc failed mask
00002B14  00002BB4                                 2505+V2_35    DC    A(RE35+16)        address of v2: 16-byte packed decimal
00002B18  E5C3E5C4 D8404040                        2506+         DC    CL8'VCVDQ'        instruction name
00002B20  00000010                                 2507+         DC    A(16)             result length
00002B24  00002BA4                                 2508+         DC    A(RE35)           address of expected resul
00002B28  00000000 00000000                        2509+         DS    FD                gap
00002B30  00000000 00000000                        2510+V1O35    DS    XL16              V1 output
00002B38  00000000 00000000
00002B40  00000000 00000000                        2511+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002B48  40404040 40404040                     2512+          DC    CL8' '            was cross check skipped?
00002B50  00000000 00000000                     2513+          DS    FD                   gap
00002B58  00000000 00000000                     2514+XC035     DS    XL16              Cross check Output
00002B60  00000000 00000000
00002B68  00000000 00000000                     2515+          DS    FD                   gap
00002B70  00000000 00000000                     2516+          DS    2F          extract PSW after test (has CC)
00002B78  00                                    2517+          DS    X           extracted cc
00002B79  00                                    2518+          DS    X           CC Check failed
                                                2519+*
00002B7C                                        2520+X35       DS    0F
00002B7C  E720 8F56 0006             00001156   2521+          VL    V2,V1FUDGE
00002B82  E320 500C 0014             00002B14   2522+          LGF   R2,V2_35           get v2 address
00002B88  E722 0000 0006             00000000   2523+          VL    V2,0(R2)

00002B8E                                        2526+          DS    0H                     E64A VCVDQ
00002B8E  E6                                    2527+          DC    X'E6'                   VECTOR CONVERT TO DECIMAL (128)
00002B8F  22                                    2528+          DC    X'22'                   v1, v2
00002B90  00                                    2529+          DC    X'00'                  reserved
00002B91  B9F0                                  2530+          DC    HL2'47600'              m4, i3, rXB
00002B93  4A                                    2531+          DC    X'4A'

00002B94  B98D 0020                             2533+          EPSW  R2,R0             exptract psw
00002B98  5020 5068                  00000068   2534+          ST    R2,CCPSW              to save CC
00002B9C  E720 5028 000E             00002B30   2535+          VST   V2,V1035          save instruction result
00002BA2  07FB                                  2536+          BR    R11               return
00002BA4                                        2537+RE35      DS    0F                expected 16 byte result
00002BA4                                        2538+          DROP  R5
00002BA4  00000000 00000000                     2539           DC    XL16'0000000000000000 000000000000000F'      v1
00002BAC  00000000 0000000F
00002BB4  0000007E 37BE2022                     2540           DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
00002BBC  C0914B26 80000000
                                                2541
                                                2542           VRI_J VCVDQ,X'9F',X'B',3,N
00002BC8                                        2543+          DS    0FD
00002BC8                   00002BC8             2544+          USING *,R5              base for test data and test routine
00002BC8  00002C3C                              2545+T36       DC    A(X36)            address of test routine
00002BCC  0024                                  2546+          DC    H'36'             test number
00002BCE  00                                    2547+          DC    X'00'
00002BCF  D5                                    2548+          DC    CL1'N'            S or Y = skip cross check
00002BD0  9F                                    2549+          DC    X'9F'             i3
00002BD1  0B                                    2550+          DC    X'B'              m4
00002BD2  03                                    2551+          DC    HL1'3'            cc
00002BD3  0E                                    2552+          DC    HL1'14'           cc failed mask
00002BD4  00002C74                              2553+V2_36     DC    A(RE36+16)        address of v2: 16-byte packed decimal
00002BD8  E5C3E5C4 D8404040                     2554+          DC    CL8'VCVDQ'        instruction name
00002BE0  00000010                              2555+          DC    A(16)             result length
00002BE4  00002C64                              2556+          DC    A(RE36)           address of expected resul
00002BE8  00000000 00000000                     2557+          DS    FD                gap
00002BF0  00000000 00000000                     2558+V1036     DS    XL16              V1 output
00002BF8  00000000 00000000
00002C00  00000000 00000000                     2559+          DS    FD                   gap
00002C08  40404040 40404040                     2560+          DC    CL8' '            was cross check skipped?
00002C10  00000000 00000000                     2561+          DS    FD                   gap
00002C18  00000000 00000000                     2562+XC036     DS    XL16              Cross check Output
00002C20  00000000 00000000
00002C28  00000000 00000000                     2563+          DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002C30  00000000 00000000                        2564+         DS    2F          extract PSW after test (has CC)
00002C38  00                                       2565+         DS    X           extracted cc
00002C39  00                                       2566+         DS    X           CC Check failed
                                                   2567+*
00002C3C                                           2568+X36      DS    0F
00002C3C  E720 8F56 0006               00001156    2569+         VL    V2,V1FUDGE
00002C42  E320 500C 0014               00002BD4    2570+         LGF   R2,V2_36           get v2 address
00002C48  E722 0000 0006               00000000    2571+         VL    V2,0(R2)

00002C4E                                           2574+         DS    0H                       E64A VCVDQ
00002C4E  E6                                       2575+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
00002C4F  22                                       2576+         DC    X'22'                     v1, v2
00002C50  00                                       2577+         DC    X'00'                    reserved
00002C51  B9F0                                     2578+         DC    HL2'47600'                m4, i3, rXB
00002C53  4A                                       2579+         DC    X'4A'

00002C54  B98D 0020                                2581+         EPSW  R2,R0             exptract psw
00002C58  5020 5068                    00000068    2582+         ST    R2,CCPSW              to save CC
00002C5C  E720 5028 000E               00002BF0    2583+         VST   V2,V1O36          save instruction result
00002C62  07FB                                     2584+         BR    R11               return
00002C64                                           2585+RE36     DS    0F                expected 16 byte result
00002C64                                           2586+         DROP  R5
00002C64  69209384 63463374                        2587          DC    XL16'6920938463463374 607431768211456F'      v1
00002C6C  60743176 8211456F
00002C74  FFFFFF81 C841DFDD                        2588          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
00002C7C  3F6EB4D9 80000000
                                                   2589
                                                   2590 *--------------------------------------------------------------------
                                                   2591 *  VCVDQ    - VECTOR CONVERT TO DECIMAL (128)
                                                   2592 *--------------------------------------------------------------------
                                                   2593 *        i3:   x'8f"   (IOM=1, RDC=15)
                                                   2594 *        m4:   x'1'    (LB=0, P1=0, CS=1)
                                                   2595 *
                                                   2596          VRI_J VCVDQ,X'8F',X'1',0,N
00002C88                                           2597+         DS    0FD
00002C88                     00002C88              2598+         USING *,R5              base for test data and test routine
00002C88  00002CFC                                 2599+T37      DC    A(X37)            address of test routine
00002C8C  0025                                     2600+         DC    H'37'             test number
00002C8E  00                                       2601+         DC    X'00'
00002C8F  D5                                       2602+         DC    CL1'N'            S or Y = skip cross check
00002C90  8F                                       2603+         DC    X'8F'             i3
00002C91  01                                       2604+         DC    X'1'              m4
00002C92  00                                       2605+         DC    HL1'0'            cc
00002C93  07                                       2606+         DC    HL1'7'            cc failed mask
00002C94  00002D34                                 2607+V2_37    DC    A(RE37+16)        address of v2: 16-byte packed decimal
00002C98  E5C3E5C4 D8404040                        2608+         DC    CL8'VCVDQ'        instruction name
00002CA0  00000010                                 2609+         DC    A(16)             result length
00002CA4  00002D24                                 2610+         DC    A(RE37)           address of expected resul
00002CA8  00000000 00000000                        2611+         DS    FD                gap
00002CB0  00000000 00000000                        2612+V1O37    DS    XL16              V1 output
00002CB8  00000000 00000000
00002CC0  00000000 00000000                        2613+         DS    FD                   gap
00002CC8  40404040 40404040                        2614+         DC    CL8' '            was cross check skipped?
00002CD0  00000000 00000000                        2615+         DS    FD                   gap
00002CD8  00000000 00000000                        2616+XCO37    DS    XL16              Cross check Output
00002CE0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002CE8  00000000 00000000                        2617+          DS    FD                          gap
00002CF0  00000000 00000000                        2618+          DS    2F                 extract PSW after test (has CC)
00002CF8  00                                       2619+          DS    X                  extracted cc
00002CF9  00                                       2620+          DS    X                  CC Check failed
                                                   2621+*
00002CFC                                           2622+X37       DS    0F
00002CFC  E720 8F56 0006                 00001156  2623+          VL    V2,V1FUDGE
00002D02  E320 500C 0014                 00002C94  2624+          LGF   R2,V2_37                  get v2 address
00002D08  E722 0000 0006                 00000000  2625+          VL    V2,0(R2)

00002D0E                                           2628+          DS    0H                             E64A VCVDQ
00002D0E  E6                                       2629+          DC    X'E6'                           VECTOR CONVERT TO DECIMAL (128)
00002D0F  22                                       2630+          DC    X'22'                           v1, v2
00002D10  00                                       2631+          DC    X'00'                          reserved
00002D11  18F0                                     2632+          DC    HL2'6384'                       m4, i3, rXB
00002D13  4A                                       2633+          DC    X'4A'

00002D14  B98D 0020                                2635+          EPSW  R2,R0                    exptract psw
00002D18  5020 5068                      00000068  2636+          ST    R2,CCPSW                     to save CC
00002D1C  E720 5028 000E                 00002CB0  2637+          VST   V2,V1O37                 save instruction result
00002D22  07FB                                     2638+          BR    R11                      return
00002D24                                           2639+RE37      DS    0F                       expected 16 byte result
00002D24                                           2640+          DROP  R5
00002D24  00000000 00000000                        2641           DC    XL16'0000000000000000 000000000000000C'      v1
00002D2C  00000000 0000000C
00002D34  00000000 00000000                        2642           DC    XL16'0000000000000000 0000000000000000'      v2
00002D3C  00000000 00000000
                                                   2643
                                                   2644           VRI_J VCVDQ,X'8F',X'1',0,N
00002D48                                           2645+          DS    0FD
00002D48                       00002D48            2646+          USING *,R5                     base for test data and test routine
00002D48  00002DBC                                 2647+T38       DC    A(X38)                   address of test routine
00002D4C  0026                                     2648+          DC    H'38'                    test number
00002D4E  00                                       2649+          DC    X'00'
00002D4F  D5                                       2650+          DC    CL1'N'                   S or Y = skip cross check
00002D50  8F                                       2651+          DC    X'8F'                    i3
00002D51  01                                       2652+          DC    X'1'                     m4
00002D52  00                                       2653+          DC    HL1'0'                   cc
00002D53  07                                       2654+          DC    HL1'7'                   cc failed mask
00002D54  00002DF4                                 2655+V2_38     DC    A(RE38+16)               address of v2: 16-byte packed decimal
00002D58  E5C3E5C4 D8404040                        2656+          DC    CL8'VCVDQ'               instruction name
00002D60  00000010                                 2657+          DC    A(16)                    result length
00002D64  00002DE4                                 2658+          DC    A(RE38)                  address of expected resul
00002D68  00000000 00000000                        2659+          DS    FD                       gap
00002D70  00000000 00000000                        2660+V1O38     DS    XL16                     V1 output
00002D78  00000000 00000000
00002D80  00000000 00000000                        2661+          DS    FD                          gap
00002D88  40404040 40404040                        2662+          DC    CL8' '                   was cross check skipped?
00002D90  00000000 00000000                        2663+          DS    FD                          gap
00002D98  00000000 00000000                        2664+XCO38     DS    XL16                     Cross check Output
00002DA0  00000000 00000000
00002DA8  00000000 00000000                        2665+          DS    FD                          gap
00002DB0  00000000 00000000                        2666+          DS    2F                 extract PSW after test (has CC)
00002DB8  00                                       2667+          DS    X                  extracted cc
00002DB9  00                                       2668+          DS    X                  CC Check failed
                                                   2669+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002DBC                                          2670+X38      DS    0F
00002DBC  E720 8F56 0006                00001156  2671+         VL    V2,V1FUDGE
00002DC2  E320 500C 0014                00002D54  2672+         LGF   R2,V2_38           get v2 address
00002DC8  E722 0000 0006                00000000  2673+         VL    V2,0(R2)

00002DCE                                          2676+         DS    0H                          E64A VCVDQ
00002DCE  E6                                      2677+         DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
00002DCF  22                                      2678+         DC    X'22'                        v1, v2
00002DD0  00                                      2679+         DC    X'00'                       reserved
00002DD1  18F0                                    2680+         DC    HL2'6384'                    m4, i3, rXB
00002DD3  4A                                      2681+         DC    X'4A'

00002DD4  B98D 0020                               2683+         EPSW  R2,R0             exptract psw
00002DD8  5020 5068                     00000068  2684+         ST    R2,CCPSW              to save CC
00002DDC  E720 5028 000E                00002D70  2685+         VST   V2,V1O38          save instruction result
00002DE2  07FB                                    2686+         BR    R11               return
00002DE4                                          2687+RE38     DS    0F                expected 16 byte result
00002DE4                                          2688+         DROP  R5
00002DE4  00000000 00000000                       2689          DC    XL16'0000000000000000 000000000000001C'      v1
00002DEC  00000000 0000001C
00002DF4  00000000 00000000                       2690          DC    XL16'0000000000000000 0000000000000001'      v2
00002DFC  00000000 00000001
                                                  2691
                                                  2692          VRI_J VCVDQ,X'8F',X'1',0,N
00002E08                                          2693+         DS    0FD
00002E08                         00002E08         2694+         USING *,R5              base for test data and test routine
00002E08  00002E7C                                2695+T39      DC    A(X39)            address of test routine
00002E0C  0027                                    2696+         DC    H'39'             test number
00002E0E  00                                      2697+         DC    X'00'
00002E0F  D5                                      2698+         DC    CL1'N'            S or Y = skip cross check
00002E10  8F                                      2699+         DC    X'8F'             i3
00002E11  01                                      2700+         DC    X'1'              m4
00002E12  00                                      2701+         DC    HL1'0'            cc
00002E13  07                                      2702+         DC    HL1'7'            cc failed mask
00002E14  00002EB4                                2703+V2_39    DC    A(RE39+16)        address of v2: 16-byte packed decimal
00002E18  E5C3E5C4 D8404040                       2704+         DC    CL8'VCVDQ'        instruction name
00002E20  00000010                                2705+         DC    A(16)             result length
00002E24  00002EA4                                2706+         DC    A(RE39)           address of expected resul
00002E28  00000000 00000000                       2707+         DS    FD                gap
00002E30  00000000 00000000                       2708+V1O39    DS    XL16              V1 output
00002E38  00000000 00000000
00002E40  00000000 00000000                       2709+         DS    FD                   gap
00002E48  40404040 40404040                       2710+         DC    CL8' '            was cross check skipped?
00002E50  00000000 00000000                       2711+         DS    FD                   gap
00002E58  00000000 00000000                       2712+XCO39    DS    XL16              Cross check Output
00002E60  00000000 00000000
00002E68  00000000 00000000                       2713+         DS    FD                   gap
00002E70  00000000 00000000                       2714+         DS    2F          extract PSW after test (has CC)
00002E78  00                                      2715+         DS    X           extracted cc
00002E79  00                                      2716+         DS    X           CC Check failed
                                                  2717+*
00002E7C                                          2718+X39      DS    0F
00002E7C  E720 8F56 0006                00001156  2719+         VL    V2,V1FUDGE
00002E82  E320 500C 0014                00002E14  2720+         LGF   R2,V2_39           get v2 address
00002E88  E722 0000 0006                00000000  2721+         VL    V2,0(R2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002E8E                                           2724+          DS     0H                           E64A VCVDQ
00002E8E  E6                                       2725+          DC     X'E6'                         VECTOR CONVERT TO DECIMAL (128)
00002E8F  22                                       2726+          DC     X'22'                         v1, v2
00002E90  00                                       2727+          DC     X'00'                        reserved
00002E91  18F0                                     2728+          DC     HL2'6384'                     m4, i3, rXB
00002E93  4A                                       2729+          DC     X'4A'

00002E94  B98D 0020                                2731+          EPSW   R2,R0              exptract psw
00002E98  5020 5068                      00000068  2732+          ST     R2,CCPSW               to save CC
00002E9C  E720 5028 000E                 00002E30  2733+          VST    V2,V1O39           save instruction result
00002EA2  07FB                                     2734+          BR     R11                return
00002EA4                                           2735+RE39      DS     0F                 expected 16 byte result
00002EA4                                           2736+          DROP   R5
00002EA4  00000000 00000000                        2737           DC     XL16'0000000000000000 000000000000001D'      v1
00002EAC  00000000 0000001D
00002EB4  FFFFFFFF FFFFFFFF                        2738           DC     XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
00002EBC  FFFFFFFF FFFFFFFF
                                                   2739 *
                                                   2740 * overflow
                                                   2741 *
                                                   2742           VRI_J VCVDQ,X'8F',X'1',3,N
00002EC8                                           2743+          DS     0FD
00002EC8                     00002EC8              2744+          USING *,R5              base for test data and test routine
00002EC8  00002F3C                                 2745+T40       DC     A(X40)             address of test routine
00002ECC  0028                                     2746+          DC     H'40'              test number
00002ECE  00                                       2747+          DC     X'00'
00002ECF  D5                                       2748+          DC     CL1'N'             S or Y = skip cross check
00002ED0  8F                                       2749+          DC     X'8F'              i3
00002ED1  01                                       2750+          DC     X'1'               m4
00002ED2  03                                       2751+          DC     HL1'3'             cc
00002ED3  0E                                       2752+          DC     HL1'14'            cc failed mask
00002ED4  00002F74                                 2753+V2_40     DC     A(RE40+16)         address of v2: 16-byte packed decimal
00002ED8  E5C3E5C4 D8404040                        2754+          DC     CL8'VCVDQ'         instruction name
00002EE0  00000010                                 2755+          DC     A(16)              result length
00002EE4  00002F64                                 2756+          DC     A(RE40)            address of expected resul
00002EE8  00000000 00000000                        2757+          DS     FD                 gap
00002EF0  00000000 00000000                        2758+V1O40     DS     XL16               V1 output
00002EF8  00000000 00000000
00002F00  00000000 00000000                        2759+          DS     FD                    gap
00002F08  40404040 40404040                        2760+          DC     CL8' '             was cross check skipped?
00002F10  00000000 00000000                        2761+          DS     FD                    gap
00002F18  00000000 00000000                        2762+XCO40     DS     XL16               Cross check Output
00002F20  00000000 00000000
00002F28  00000000 00000000                        2763+          DS     FD                    gap
00002F30  00000000 00000000                        2764+          DS     2F           extract PSW after test (has CC)
00002F38  00                                       2765+          DS     X            extracted cc
00002F39  00                                       2766+          DS     X            CC Check failed
                                                   2767+*
00002F3C                                           2768+X40       DS     0F
00002F3C  E720 8F56 0006                 00001156  2769+          VL     V2,V1FUDGE
00002F42  E320 500C 0014                 00002ED4  2770+          LGF    R2,V2_40             get v2 address
00002F48  E722 0000 0006                 00000000  2771+          VL     V2,0(R2)

00002F4E                                           2774+          DS     0H                           E64A VCVDQ
00002F4E  E6                                       2775+          DC     X'E6'                         VECTOR CONVERT TO DECIMAL (128)
00002F4F  22                                       2776+          DC     X'22'                         v1, v2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002F50  00                                      2777+          DC    X'00'                     reserved
00002F51  18F0                                    2778+          DC    HL2'6384'                  m4, i3, rXB
00002F53  4A                                      2779+          DC    X'4A'

00002F54  B98D 0020                               2781+          EPSW  R2,R0             exptract psw
00002F58  5020 5068                     00000068  2782+          ST    R2,CCPSW              to save CC
00002F5C  E720 5028 000E                00002EF0  2783+          VST   V2,V1O40          save instruction result
00002F62  07FB                                    2784+          BR    R11               return
00002F64                                          2785+RE40      DS    0F                expected 16 byte result
00002F64                                          2786+          DROP  R5
00002F64  00000000 00000000                       2787           DC    XL16'0000000000000000 372036854775807C'      v1
00002F6C  37203685 4775807C
00002F74  00000000 00000000                       2788           DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
00002F7C  7FFFFFFF FFFFFFFF
                                                  2789
                                                  2790           VRI_J VCVDQ,X'8F',X'1',3,N
00002F88                                          2791+          DS    0FD
00002F88                      00002F88            2792+          USING *,R5              base for test data and test routine
00002F88  00002FFC                                2793+T41       DC    A(X41)            address of test routine
00002F8C  0029                                    2794+          DC    H'41'             test number
00002F8E  00                                      2795+          DC    X'00'
00002F8F  D5                                      2796+          DC    CL1'N'            S or Y = skip cross check
00002F90  8F                                      2797+          DC    X'8F'             i3
00002F91  01                                      2798+          DC    X'1'              m4
00002F92  03                                      2799+          DC    HL1'3'            cc
00002F93  0E                                      2800+          DC    HL1'14'           cc failed mask
00002F94  00003034                                2801+V2_41     DC    A(RE41+16)        address of v2: 16-byte packed decimal
00002F98  E5C3E5C4 D8404040                       2802+          DC    CL8'VCVDQ'        instruction name
00002FA0  00000010                                2803+          DC    A(16)             result length
00002FA4  00003024                                2804+          DC    A(RE41)           address of expected resul
00002FA8  00000000 00000000                       2805+          DS    FD                gap
00002FB0  00000000 00000000                       2806+V1O41     DS    XL16              V1 output
00002FB8  00000000 00000000
00002FC0  00000000 00000000                       2807+          DS    FD                   gap
00002FC8  40404040 40404040                       2808+          DC    CL8' '            was cross check skipped?
00002FD0  00000000 00000000                       2809+          DS    FD                   gap
00002FD8  00000000 00000000                       2810+XCO41     DS    XL16              Cross check Output
00002FE0  00000000 00000000
00002FE8  00000000 00000000                       2811+          DS    FD                   gap
00002FF0  00000000 00000000                       2812+          DS    2F          extract PSW after test (has CC)
00002FF8  00                                      2813+          DS    X           extracted cc
00002FF9  00                                      2814+          DS    X           CC Check failed
                                                  2815+*
00002FFC                                          2816+X41       DS    0F
00002FFC  E720 8F56 0006                00001156  2817+          VL    V2,V1FUDGE
00003002  E320 500C 0014                00002F94  2818+          LGF   R2,V2_41           get v2 address
00003008  E722 0000 0006                00000000  2819+          VL    V2,0(R2)

0000300E                                          2822+          DS    0H                     E64A VCVDQ
0000300E  E6                                      2823+          DC    X'E6'                   VECTOR CONVERT TO DECIMAL (128)
0000300F  22                                      2824+          DC    X'22'                   v1, v2
00003010  00                                      2825+          DC    X'00'                  reserved
00003011  18F0                                    2826+          DC    HL2'6384'               m4, i3, rXB
00003013  4A                                      2827+          DC    X'4A'

00003014  B98D 0020                               2829+          EPSW  R2,R0             exptract psw
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00003018  5020 5068                   00000068  2830+          ST    R2,CCPSW               to save CC
0000301C  E720 5028 000E              00002FB0  2831+          VST   V2,V1O41           save instruction result
00003022  07FB                                  2832+          BR    R11                return
00003024                                        2833+RE41      DS    0F                 expected 16 byte result
00003024                                        2834+          DROP  R5
00003024  00000000 00000000                     2835           DC    XL16'0000000000000000 372036854775807D'      v1
0000302C  37203685 4775807D
00003034  FFFFFFFF FFFFFFFF                     2836           DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
0000303C  80000000 00000001
                                                2837
                                                2838           VRI_J VCVDQ,X'8F',X'1',3,N
00003048                                        2839+          DS    0FD
00003048                      00003048          2840+          USING *,R5               base for test data and test routine
00003048  000030BC                              2841+T42       DC    A(X42)             address of test routine
0000304C  002A                                  2842+          DC    H'42'              test number
0000304E  00                                    2843+          DC    X'00'
0000304F  D5                                    2844+          DC    CL1'N'             S or Y = skip cross check
00003050  8F                                    2845+          DC    X'8F'              i3
00003051  01                                    2846+          DC    X'1'               m4
00003052  03                                    2847+          DC    HL1'3'             cc
00003053  0E                                    2848+          DC    HL1'14'            cc failed mask
00003054  000030F4                              2849+V2_42     DC    A(RE42+16)         address of v2: 16-byte packed decimal
00003058  E5C3E5C4 D8404040                     2850+          DC    CL8'VCVDQ'         instruction name
00003060  00000010                              2851+          DC    A(16)              result length
00003064  000030E4                              2852+          DC    A(RE42)            address of expected resul
00003068  00000000 00000000                     2853+          DS    FD                 gap
00003070  00000000 00000000                     2854+V1O42     DS    XL16               V1 output
00003078  00000000 00000000
00003080  00000000 00000000                     2855+          DS    FD                    gap
00003088  40404040 40404040                     2856+          DC    CL8' '             was cross check skipped?
00003090  00000000 00000000                     2857+          DS    FD                    gap
00003098  00000000 00000000                     2858+XCO42     DS    XL16               Cross check Output
000030A0  00000000 00000000
000030A8  00000000 00000000                     2859+          DS    FD                    gap
000030B0  00000000 00000000                     2860+          DS    2F           extract PSW after test (has CC)
000030B8  00                                    2861+          DS    X            extracted cc
000030B9  00                                    2862+          DS    X            CC Check failed
                                                2863+*
000030BC                                        2864+X42       DS    0F
000030BC  E720 8F56 0006              00001156  2865+          VL    V2,V1FUDGE
000030C2  E320 500C 0014              00003054  2866+          LGF   R2,V2_42            get v2 address
000030C8  E722 0000 0006              00000000  2867+          VL    V2,0(R2)

000030CE                                        2870+          DS    0H                       E64A VCVDQ
000030CE  E6                                    2871+          DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
000030CF  22                                    2872+          DC    X'22'                     v1, v2
000030D0  00                                    2873+          DC    X'00'                    reserved
000030D1  18F0                                  2874+          DC    HL2'6384'                 m4, i3, rXB
000030D3  4A                                    2875+          DC    X'4A'

000030D4  B98D 0020                             2877+          EPSW  R2,R0              exptract psw
000030D8  5020 5068                   00000068  2878+          ST    R2,CCPSW               to save CC
000030DC  E720 5028 000E              00003070  2879+          VST   V2,V1O42           save instruction result
000030E2  07FB                                  2880+          BR    R11                return
000030E4                                        2881+RE42      DS    0F                 expected 16 byte result
000030E4                                        2882+          DROP  R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000030E4  00000000 00000000                        2883          DC    XL16'0000000000000000 999999999999999C'      v1
000030EC  99999999 9999999C
000030F4  0000007E 37BE2022                        2884          DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
000030FC  C0914B26 7FFFFFFF
                                                   2885
                                                   2886          VRI_J VCVDQ,X'8F',X'1',3,N
00003108                                           2887+         DS    0FD
00003108                           00003108        2888+         USING *,R5                 base for test data and test routine
00003108  0000317C                                 2889+T43      DC    A(X43)               address of test routine
0000310C  002B                                     2890+         DC    H'43'                test number
0000310E  00                                       2891+         DC    X'00'
0000310F  D5                                       2892+         DC    CL1'N'               S or Y = skip cross check
00003110  8F                                       2893+         DC    X'8F'                i3
00003111  01                                       2894+         DC    X'1'                 m4
00003112  03                                       2895+         DC    HL1'3'               cc
00003113  0E                                       2896+         DC    HL1'14'              cc failed mask
00003114  000031B4                                 2897+V2_43    DC    A(RE43+16)           address of v2: 16-byte packed decimal
00003118  E5C3E5C4 D8404040                        2898+         DC    CL8'VCVDQ'           instruction name
00003120  00000010                                 2899+         DC    A(16)                result length
00003124  000031A4                                 2900+         DC    A(RE43)              address of expected resul
00003128  00000000 00000000                        2901+         DS    FD                   gap
00003130  00000000 00000000                        2902+V1O43    DS    XL16                 V1 output
00003138  00000000 00000000
00003140  00000000 00000000                        2903+         DS    FD                      gap
00003148  40404040 40404040                        2904+         DC    CL8' '               was cross check skipped?
00003150  00000000 00000000                        2905+         DS    FD                      gap
00003158  00000000 00000000                        2906+XCO43    DS    XL16                 Cross check Output
00003160  00000000 00000000
00003168  00000000 00000000                        2907+         DS    FD                      gap
00003170  00000000 00000000                        2908+         DS    2F             extract PSW after test (has CC)
00003178  00                                       2909+         DS    X              extracted cc
00003179  00                                       2910+         DS    X              CC Check failed
                                                   2911+*
0000317C                                           2912+X43      DS    0F
0000317C  E720 8F56 0006                 00001156  2913+         VL    V2,V1FUDGE
00003182  E320 500C 0014                 00003114  2914+         LGF   R2,V2_43              get v2 address
00003188  E722 0000 0006                 00000000  2915+         VL    V2,0(R2)

0000318E                                           2918+         DS    0H                         E64A VCVDQ
0000318E  E6                                       2919+         DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
0000318F  22                                       2920+         DC    X'22'                       v1, v2
00003190  00                                       2921+         DC    X'00'                      reserved
00003191  18F0                                     2922+         DC    HL2'6384'                   m4, i3, rXB
00003193  4A                                       2923+         DC    X'4A'

00003194  B98D 0020                                2925+         EPSW  R2,R0                exptract psw
00003198  5020 5068                      00000068  2926+         ST    R2,CCPSW                 to save CC
0000319C  E720 5028 000E                 00003130  2927+         VST   V2,V1O43             save instruction result
000031A2  07FB                                     2928+         BR    R11                  return
000031A4                                           2929+RE43     DS    0F                   expected 16 byte result
000031A4                                           2930+         DROP  R5
000031A4  00000000 00000000                        2931          DC    XL16'0000000000000000 999999999999999D'      v1
000031AC  99999999 9999999D
000031B4  FFFFFF81 C841DFDD                        2932          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
000031BC  3F6EB4D9 80000001
                                                   2933
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   2934          VRI_J VCVDQ,X'8F',X'1',3,N
000031C8                                           2935+         DS    0FD
000031C8                       000031C8            2936+         USING *,R5                 base for test data and test routine
000031C8  0000323C                                 2937+T44      DC    A(X44)               address of test routine
000031CC  002C                                     2938+         DC    H'44'                test number
000031CE  00                                       2939+         DC    X'00'
000031CF  D5                                       2940+         DC    CL1'N'               S or Y = skip cross check
000031D0  8F                                       2941+         DC    X'8F'                i3
000031D1  01                                       2942+         DC    X'1'                 m4
000031D2  03                                       2943+         DC    HL1'3'               cc
000031D3  0E                                       2944+         DC    HL1'14'              cc failed mask
000031D4  00003274                                 2945+V2_44    DC    A(RE44+16)           address of v2: 16-byte packed decimal
000031D8  E5C3E5C4 D8404040                        2946+         DC    CL8'VCVDQ'           instruction name
000031E0  00000010                                 2947+         DC    A(16)                result length
000031E4  00003264                                 2948+         DC    A(RE44)              address of expected resul
000031E8  00000000 00000000                        2949+         DS    FD                   gap
000031F0  00000000 00000000                        2950+V1O44    DS    XL16                 V1 output
000031F8  00000000 00000000
00003200  00000000 00000000                        2951+         DS    FD                      gap
00003208  40404040 40404040                        2952+         DC    CL8' '               was cross check skipped?
00003210  00000000 00000000                        2953+         DS    FD                      gap
00003218  00000000 00000000                        2954+XCO44    DS    XL16                 Cross check Output
00003220  00000000 00000000
00003228  00000000 00000000                        2955+         DS    FD                      gap
00003230  00000000 00000000                        2956+         DS    2F             extract PSW after test (has CC)
00003238  00                                       2957+         DS    X              extracted cc
00003239  00                                       2958+         DS    X              CC Check failed
                                                   2959+*
0000323C                                           2960+X44      DS    0F
0000323C  E720 8F56 0006              00001156     2961+         VL    V2,V1FUDGE
00003242  E320 500C 0014              000031D4     2962+         LGF   R2,V2_44               get v2 address
00003248  E722 0000 0006              00000000     2963+         VL    V2,0(R2)

0000324E                                           2966+         DS    0H                         E64A VCVDQ
0000324E  E6                                       2967+         DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
0000324F  22                                       2968+         DC    X'22'                       v1, v2
00003250  00                                       2969+         DC    X'00'                      reserved
00003251  18F0                                     2970+         DC    HL2'6384'                   m4, i3, rXB
00003253  4A                                       2971+         DC    X'4A'

00003254  B98D 0020                                2973+         EPSW  R2,R0                exptract psw
00003258  5020 5068                   00000068     2974+         ST    R2,CCPSW                 to save CC
0000325C  E720 5028 000E              000031F0     2975+         VST   V2,V1O44             save instruction result
00003262  07FB                                     2976+         BR    R11                  return
00003264                                           2977+RE44     DS    0F                   expected 16 byte result
00003264                                           2978+         DROP  R5
00003264  00000000 00000000                        2979          DC    XL16'0000000000000000 000000000000000C'      v1
0000326C  00000000 0000000C
00003274  0000007E 37BE2022                        2980          DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
0000327C  C0914B26 80000000
                                                   2981
                                                   2982          VRI_J VCVDQ,X'8F',X'1',3,N
00003288                                           2983+         DS    0FD
00003288                       00003288            2984+         USING *,R5                 base for test data and test routine
00003288  000032FC                                 2985+T45      DC    A(X45)               address of test routine
0000328C  002D                                     2986+         DC    H'45'                test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000328E  00                                     2987+          DC    X'00'
0000328F  D5                                     2988+          DC    CL1'N'                S or Y = skip cross check
00003290  8F                                     2989+          DC    X'8F'                 i3
00003291  01                                     2990+          DC    X'1'                  m4
00003292  03                                     2991+          DC    HL1'3'                cc
00003293  0E                                     2992+          DC    HL1'14'               cc failed mask
00003294  00003334                               2993+V2_45     DC    A(RE45+16)            address of v2: 16-byte packed decimal
00003298  E5C3E5C4 D8404040                      2994+          DC    CL8'VCVDQ'            instruction name
000032A0  00000010                               2995+          DC    A(16)                 result length
000032A4  00003324                               2996+          DC    A(RE45)               address of expected resul
000032A8  00000000 00000000                      2997+          DS    FD                    gap
000032B0  00000000 00000000                      2998+V1O45     DS    XL16                  V1 output
000032B8  00000000 00000000
000032C0  00000000 00000000                      2999+          DS    FD                       gap
000032C8  40404040 40404040                      3000+          DC    CL8' '                was cross check skipped?
000032D0  00000000 00000000                      3001+          DS    FD                       gap
000032D8  00000000 00000000                      3002+XCO45     DS    XL16                  Cross check Output
000032E0  00000000 00000000
000032E8  00000000 00000000                      3003+          DS    FD                       gap
000032F0  00000000 00000000                      3004+          DS    2F              extract PSW after test (has CC)
000032F8  00                                     3005+          DS    X               extracted cc
000032F9  00                                     3006+          DS    X               CC Check failed
                                                 3007+*
000032FC                                         3008+X45       DS    0F
000032FC  E720 8F56 0006               00001156  3009+          VL    V2,V1FUDGE
00003302  E320 500C 0014               00003294  3010+          LGF   R2,V2_45               get v2 address
00003308  E722 0000 0006               00000000  3011+          VL    V2,0(R2)

0000330E                                         3014+          DS    0H                         E64A VCVDQ
0000330E  E6                                     3015+          DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
0000330F  22                                     3016+          DC    X'22'                       v1, v2
00003310  00                                     3017+          DC    X'00'                      reserved
00003311  18F0                                   3018+          DC    HL2'6384'                   m4, i3, rXB
00003313  4A                                     3019+          DC    X'4A'

00003314  B98D 0020                              3021+          EPSW  R2,R0                 exptract psw
00003318  5020 5068                    00000068  3022+          ST    R2,CCPSW                  to save CC
0000331C  E720 5028 000E               000032B0  3023+          VST   V2,V1O45              save instruction result
00003322  07FB                                   3024+          BR    R11                   return
00003324                                         3025+RE45      DS    0F                    expected 16 byte result
00003324                                         3026+          DROP  R5
00003324  00000000 00000000                      3027           DC    XL16'0000000000000000 000000000000000D'      v1
0000332C  00000000 0000000D
00003334  FFFFFF81 C841DFDD                      3028           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
0000333C  3F6EB4D9 80000000
                                                 3029
                                                 3030 *--------------------------------------------------------------------
                                                 3031 *         i3:   x'9f"   (IOM=1, RDC=31)
                                                 3032 *         m4:   x'3'    (LB=0, P1=1, CS=1)
                                                 3033 *
                                                 3034           VRI_J VCVDQ,X'8F',X'3',0,N
00003348                                         3035+          DS    0FD
00003348                     00003348            3036+          USING *,R5                  base for test data and test routine
00003348  000033BC                               3037+T46       DC    A(X46)                address of test routine
0000334C  002E                                   3038+          DC    H'46'                 test number
0000334E  00                                     3039+          DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000334F  D5                                       3040+          DC     CL1'N'              S or Y = skip cross check
00003350  8F                                       3041+          DC     X'8F'               i3
00003351  03                                       3042+          DC     X'3'                m4
00003352  00                                       3043+          DC     HL1'0'              cc
00003353  07                                       3044+          DC     HL1'7'              cc failed mask
00003354  000033F4                                 3045+V2_46     DC     A(RE46+16)          address of v2: 16-byte packed decimal
00003358  E5C3E5C4 D8404040                        3046+          DC     CL8'VCVDQ'          instruction name
00003360  00000010                                 3047+          DC     A(16)               result length
00003364  000033E4                                 3048+          DC     A(RE46)             address of expected resul
00003368  00000000 00000000                        3049+          DS     FD                  gap
00003370  00000000 00000000                        3050+V1O46     DS     XL16                V1 output
00003378  00000000 00000000
00003380  00000000 00000000                        3051+          DS     FD                     gap
00003388  40404040 40404040                        3052+          DC     CL8' '              was cross check skipped?
00003390  00000000 00000000                        3053+          DS     FD                     gap
00003398  00000000 00000000                        3054+XCO46     DS     XL16                Cross check Output
000033A0  00000000 00000000
000033A8  00000000 00000000                        3055+          DS     FD                     gap
000033B0  00000000 00000000                        3056+          DS     2F            extract PSW after test (has CC)
000033B8  00                                       3057+          DS     X             extracted cc
000033B9  00                                       3058+          DS     X             CC Check failed
                                                   3059+*
000033BC                                           3060+X46       DS     0F
000033BC  E720 8F56 0006             00001156      3061+          VL     V2,V1FUDGE
000033C2  E320 500C 0014             00003354      3062+          LGF    R2,V2_46             get v2 address
000033C8  E722 0000 0006             00000000      3063+          VL     V2,0(R2)

000033CE                                           3066+          DS     0H                        E64A VCVDQ
000033CE  E6                                       3067+          DC     X'E6'                      VECTOR CONVERT TO DECIMAL (128)
000033CF  22                                       3068+          DC     X'22'                      v1, v2
000033D0  00                                       3069+          DC     X'00'                     reserved
000033D1  38F0                                     3070+          DC     HL2'14576'                 m4, i3, rXB
000033D3  4A                                       3071+          DC     X'4A'

000033D4  B98D 0020                                3073+          EPSW   R2,R0               exptract psw
000033D8  5020 5068                  00000068      3074+          ST     R2,CCPSW                to save CC
000033DC  E720 5028 000E             00003370      3075+          VST    V2,V1O46            save instruction result
000033E2  07FB                                     3076+          BR     R11                 return
000033E4                                           3077+RE46      DS     0F                  expected 16 byte result
000033E4                                           3078+          DROP   R5
000033E4  00000000 00000000                        3079           DC     XL16'0000000000000000 000000000000000F'      v1
000033EC  00000000 0000000F
000033F4  00000000 00000000                        3080           DC     XL16'0000000000000000 0000000000000000'      v2
000033FC  00000000 00000000
                                                   3081
                                                   3082           VRI_J  VCVDQ,X'8F',X'3',0,N
00003408                                           3083+          DS     0FD
00003408                       00003408            3084+          USING  *,R5                base for test data and test routine
00003408  0000347C                                 3085+T47       DC     A(X47)              address of test routine
0000340C  002F                                     3086+          DC     H'47'               test number
0000340E  00                                       3087+          DC     X'00'
0000340F  D5                                       3088+          DC     CL1'N'              S or Y = skip cross check
00003410  8F                                       3089+          DC     X'8F'               i3
00003411  03                                       3090+          DC     X'3'                m4
00003412  00                                       3091+          DC     HL1'0'              cc
00003413  07                                       3092+          DC     HL1'7'              cc failed mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00003414  000034B4                                3093+V2_47    DC    A(RE47+16)         address of v2: 16-byte packed decimal
00003418  E5C3E5C4 D8404040                       3094+         DC    CL8'VCVDQ'         instruction name
00003420  00000010                                3095+         DC    A(16)              result length
00003424  000034A4                                3096+         DC    A(RE47)            address of expected resul
00003428  00000000 00000000                       3097+         DS    FD                 gap
00003430  00000000 00000000                       3098+V1O47    DS    XL16               V1 output
00003438  00000000 00000000
00003440  00000000 00000000                       3099+         DS    FD                    gap
00003448  40404040 40404040                       3100+         DC    CL8' '             was cross check skipped?
00003450  00000000 00000000                       3101+         DS    FD                    gap
00003458  00000000 00000000                       3102+XCO47    DS    XL16               Cross check Output
00003460  00000000 00000000
00003468  00000000 00000000                       3103+         DS    FD                    gap
00003470  00000000 00000000                       3104+         DS    2F           extract PSW after test (has CC)
00003478  00                                      3105+         DS    X            extracted cc
00003479  00                                      3106+         DS    X            CC Check failed
                                                  3107+*
0000347C                                          3108+X47      DS    0F
0000347C  E720 8F56 0006             00001156     3109+         VL    V2,V1FUDGE
00003482  E320 500C 0014             00003414     3110+         LGF   R2,V2_47            get v2 address
00003488  E722 0000 0006             00000000     3111+         VL    V2,0(R2)

0000348E                                          3114+         DS    0H                       E64A VCVDQ
0000348E  E6                                      3115+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000348F  22                                      3116+         DC    X'22'                     v1, v2
00003490  00                                      3117+         DC    X'00'                    reserved
00003491  38F0                                    3118+         DC    HL2'14576'                m4, i3, rXB
00003493  4A                                      3119+         DC    X'4A'

00003494  B98D 0020                               3121+         EPSW  R2,R0              exptract psw
00003498  5020 5068                  00000068     3122+         ST    R2,CCPSW               to save CC
0000349C  E720 5028 000E             00003430     3123+         VST   V2,V1O47           save instruction result
000034A2  07FB                                    3124+         BR    R11                return
000034A4                                          3125+RE47     DS    0F                 expected 16 byte result
000034A4                                          3126+         DROP  R5
000034A4  00000000 00000000                       3127          DC    XL16'0000000000000000 000000000000001F'      v1
000034AC  00000000 0000001F
000034B4  00000000 00000000                       3128          DC    XL16'0000000000000000 0000000000000001'      v2
000034BC  00000000 00000001
                                                  3129
                                                  3130          VRI_J VCVDQ,X'8F',X'3',0,S
000034C8                                          3131+         DS    0FD
000034C8                     000034C8             3132+         USING *,R5               base for test data and test routine
000034C8  0000353C                                3133+T48      DC    A(X48)             address of test routine
000034CC  0030                                    3134+         DC    H'48'              test number
000034CE  00                                      3135+         DC    X'00'
000034CF  E2                                      3136+         DC    CL1'S'             S or Y = skip cross check
000034D0  8F                                      3137+         DC    X'8F'              i3
000034D1  03                                      3138+         DC    X'3'               m4
000034D2  00                                      3139+         DC    HL1'0'             cc
000034D3  07                                      3140+         DC    HL1'7'             cc failed mask
000034D4  00003574                                3141+V2_48    DC    A(RE48+16)         address of v2: 16-byte packed decimal
000034D8  E5C3E5C4 D8404040                       3142+         DC    CL8'VCVDQ'         instruction name
000034E0  00000010                                3143+         DC    A(16)              result length
000034E4  00003564                                3144+         DC    A(RE48)            address of expected resul
000034E8  00000000 00000000                       3145+         DS    FD                 gap
```

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  LOC       OBJECT CODE       ADDR1     ADDR2    STMT

000034F0  00000000 00000000                      3146+V1O48    DS    XL16              V1 output
000034F8  00000000 00000000
00003500  00000000 00000000                      3147+         DS    FD                   gap
00003508  40404040 40404040                      3148+         DC    CL8' '            was cross check skipped?
00003510  00000000 00000000                      3149+         DS    FD                   gap
00003518  00000000 00000000                      3150+XCO48    DS    XL16              Cross check Output
00003520  00000000 00000000
00003528  00000000 00000000                      3151+         DS    FD                   gap
00003530  00000000 00000000                      3152+         DS    2F          extract PSW after test (has CC)
00003538  00                                     3153+         DS    X           extracted cc
00003539  00                                     3154+         DS    X           CC Check failed
                                                 3155+*
0000353C                                         3156+X48      DS    0F
0000353C  E720 8F56 0006            00001156     3157+         VL    V2,V1FUDGE
00003542  E320 500C 0014            000034D4     3158+         LGF   R2,V2_48           get v2 address
00003548  E722 0000 0006            00000000     3159+         VL    V2,0(R2)

0000354E                                         3162+         DS    0H                       E64A VCVDQ
0000354E  E6                                     3163+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000354F  22                                     3164+         DC    X'22'                     v1, v2
00003550  00                                     3165+         DC    X'00'                    reserved
00003551  38F0                                   3166+         DC    HL2'14576'                m4, i3, rXB
00003553  4A                                     3167+         DC    X'4A'

00003554  B98D 0020                              3169+         EPSW  R2,R0             exptract psw
00003558  5020 5068                 00000068     3170+         ST    R2,CCPSW              to save CC
0000355C  E720 5028 000E            000034F0     3171+         VST   V2,V1O48          save instruction result
00003562  07FB                                   3172+         BR    R11               return
00003564                                         3173+RE48     DS    0F                expected 16 byte result
00003564                                         3174+         DROP  R5
00003564  00000000 00000000                      3175          DC    XL16'0000000000000000 000000000000001F'      v1
0000356C  00000000 0000001F
00003574  FFFFFFFF FFFFFFFF                      3176          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
0000357C  FFFFFFFF FFFFFFFF
                                                 3177 *
                                                 3178 * overflow
                                                 3179 *
                                                 3180          VRI_J VCVDQ,X'8F',X'3',3,N
00003588                                         3181+         DS    0FD
00003588                    00003588             3182+         USING *,R5              base for test data and test routine
00003588  000035FC                               3183+T49      DC    A(X49)            address of test routine
0000358C  0031                                   3184+         DC    H'49'             test number
0000358E  00                                     3185+         DC    X'00'
0000358F  D5                                     3186+         DC    CL1'N'            S or Y = skip cross check
00003590  8F                                     3187+         DC    X'8F'             i3
00003591  03                                     3188+         DC    X'3'              m4
00003592  03                                     3189+         DC    HL1'3'            cc
00003593  0E                                     3190+         DC    HL1'14'           cc failed mask
00003594  00003634                               3191+V2_49    DC    A(RE49+16)        address of v2: 16-byte packed decimal
00003598  E5C3E5C4 D8404040                      3192+         DC    CL8'VCVDQ'        instruction name
000035A0  00000010                               3193+         DC    A(16)             result length
000035A4  00003624                               3194+         DC    A(RE49)           address of expected resul
000035A8  00000000 00000000                      3195+         DS    FD                gap
000035B0  00000000 00000000                      3196+V1O49    DS    XL16              V1 output
000035B8  00000000 00000000
000035C0  00000000 00000000                      3197+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000035C8  40404040 40404040                        3198+         DC    CL8' '              was cross check skipped?
000035D0  00000000 00000000                        3199+         DS    FD                     gap
000035D8  00000000 00000000                        3200+XC049    DS    XL16                Cross check Output
000035E0  00000000 00000000
000035E8  00000000 00000000                        3201+         DS    FD                     gap
000035F0  00000000 00000000                        3202+         DS    2F            extract PSW after test (has CC)
000035F8  00                                       3203+         DS    X             extracted cc
000035F9  00                                       3204+         DS    X             CC Check failed
                                                   3205+*
000035FC                                           3206+X49      DS    0F
000035FC  E720 8F56 0006              00001156     3207+         VL    V2,V1FUDGE
00003602  E320 500C 0014              00003594     3208+         LGF   R2,V2_49             get v2 address
00003608  E722 0000 0006              00000000     3209+         VL    V2,0(R2)

0000360E                                           3212+         DS    0H                        E64A VCVDQ
0000360E  E6                                       3213+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000360F  22                                       3214+         DC    X'22'                      v1, v2
00003610  00                                       3215+         DC    X'00'                     reserved
00003611  38F0                                     3216+         DC    HL2'14576'                 m4, i3, rXB
00003613  4A                                       3217+         DC    X'4A'

00003614  B98D 0020                                3219+         EPSW  R2,R0               exptract psw
00003618  5020 5068                   00000068     3220+         ST    R2,CCPSW                to save CC
0000361C  E720 5028 000E              000035B0     3221+         VST   V2,V1O49            save instruction result
00003622  07FB                                     3222+         BR    R11                 return
00003624                                           3223+RE49     DS    0F                  expected 16 byte result
00003624                                           3224+         DROP  R5
00003624  00000000 00000000                        3225          DC    XL16'0000000000000000 372036854775807F'      v1
0000362C  37203685 4775807F
00003634  00000000 00000000                        3226          DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
0000363C  7FFFFFFF FFFFFFFF
                                                   3227
                                                   3228          VRI_J VCVDQ,X'8F',X'3',3,S
00003648                                           3229+         DS    0FD
00003648                     00003648              3230+         USING *,R5              base for test data and test routine
00003648  000036BC                                 3231+T50      DC    A(X50)              address of test routine
0000364C  0032                                     3232+         DC    H'50'               test number
0000364E  00                                       3233+         DC    X'00'
0000364F  E2                                       3234+         DC    CL1'S'              S or Y = skip cross check
00003650  8F                                       3235+         DC    X'8F'               i3
00003651  03                                       3236+         DC    X'3'                m4
00003652  03                                       3237+         DC    HL1'3'              cc
00003653  0E                                       3238+         DC    HL1'14'             cc failed mask
00003654  000036F4                                 3239+V2_50    DC    A(RE50+16)          address of v2: 16-byte packed decimal
00003658  E5C3E5C4 D8404040                        3240+         DC    CL8'VCVDQ'          instruction name
00003660  00000010                                 3241+         DC    A(16)               result length
00003664  000036E4                                 3242+         DC    A(RE50)             address of expected resul
00003668  00000000 00000000                        3243+         DS    FD                  gap
00003670  00000000 00000000                        3244+V1O50    DS    XL16                V1 output
00003678  00000000 00000000
00003680  00000000 00000000                        3245+         DS    FD                     gap
00003688  40404040 40404040                        3246+         DC    CL8' '              was cross check skipped?
00003690  00000000 00000000                        3247+         DS    FD                     gap
00003698  00000000 00000000                        3248+XC050    DS    XL16                Cross check Output
000036A0  00000000 00000000
000036A8  00000000 00000000                        3249+         DS    FD                     gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000036B0  00000000 00000000                        3250+          DS    2F          extract PSW after test (has CC)
000036B8  00                                       3251+          DS    X           extracted cc
000036B9  00                                       3252+          DS    X           CC Check failed
                                                   3253+*
000036BC                                           3254+X50       DS    0F
000036BC  E720 8F56 0006             00001156      3255+          VL    V2,V1FUDGE
000036C2  E320 500C 0014             00003654      3256+          LGF   R2,V2_50           get v2 address
000036C8  E722 0000 0006             00000000      3257+          VL    V2,0(R2)

000036CE                                           3260+          DS    0H                          E64A VCVDQ
000036CE  E6                                       3261+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
000036CF  22                                       3262+          DC    X'22'                        v1, v2
000036D0  00                                       3263+          DC    X'00'                       reserved
000036D1  38F0                                     3264+          DC    HL2'14576'                   m4, i3, rXB
000036D3  4A                                       3265+          DC    X'4A'

000036D4  B98D 0020                                3267+          EPSW  R2,R0             exptract psw
000036D8  5020 5068                  00000068      3268+          ST    R2,CCPSW              to save CC
000036DC  E720 5028 000E             00003670      3269+          VST   V2,V1O50          save instruction result
000036E2  07FB                                     3270+          BR    R11               return
000036E4                                           3271+RE50      DS    0F                expected 16 byte result
000036E4                                           3272+          DROP  R5
000036E4  00000000 00000000                        3273           DC    XL16'0000000000000000 372036854775807F'      v1
000036EC  37203685 4775807F
000036F4  FFFFFFFF FFFFFFFF                        3274           DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
000036FC  80000000 00000001
                                                   3275
                                                   3276           VRI_J VCVDQ,X'8F',X'3',3,N
00003708                                           3277+          DS    0FD
00003708                           00003708        3278+          USING *,R5              base for test data and test routine
00003708  0000377C                                 3279+T51       DC    A(X51)            address of test routine
0000370C  0033                                     3280+          DC    H'51'             test number
0000370E  00                                       3281+          DC    X'00'
0000370F  D5                                       3282+          DC    CL1'N'            S or Y = skip cross check
00003710  8F                                       3283+          DC    X'8F'             i3
00003711  03                                       3284+          DC    X'3'              m4
00003712  03                                       3285+          DC    HL1'3'            cc
00003713  0E                                       3286+          DC    HL1'14'           cc failed mask
00003714  000037B4                                 3287+V2_51     DC    A(RE51+16)        address of v2: 16-byte packed decimal
00003718  E5C3E5C4 D8404040                        3288+          DC    CL8'VCVDQ'        instruction name
00003720  00000010                                 3289+          DC    A(16)             result length
00003724  000037A4                                 3290+          DC    A(RE51)           address of expected resul
00003728  00000000 00000000                        3291+          DS    FD                gap
00003730  00000000 00000000                        3292+V1O51     DS    XL16              V1 output
00003738  00000000 00000000
00003740  00000000 00000000                        3293+          DS    FD                   gap
00003748  40404040 40404040                        3294+          DC    CL8' '            was cross check skipped?
00003750  00000000 00000000                        3295+          DS    FD                   gap
00003758  00000000 00000000                        3296+XCO51     DS    XL16              Cross check Output
00003760  00000000 00000000
00003768  00000000 00000000                        3297+          DS    FD                   gap
00003770  00000000 00000000                        3298+          DS    2F          extract PSW after test (has CC)
00003778  00                                       3299+          DS    X           extracted cc
00003779  00                                       3300+          DS    X           CC Check failed
                                                   3301+*
0000377C                                           3302+X51       DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000377C  E720 8F56 0006                00001156  3303+          VL     V2,V1FUDGE
00003782  E320 500C 0014                00003714  3304+          LGF    R2,V2_51            get v2 address
00003788  E722 0000 0006                00000000  3305+          VL     V2,0(R2)

0000378E                                          3308+          DS     0H                           E64A VCVDQ
0000378E  E6                                      3309+          DC     X'E6'                         VECTOR CONVERT TO DECIMAL (128)
0000378F  22                                      3310+          DC     X'22'                         v1, v2
00003790  00                                      3311+          DC     X'00'                        reserved
00003791  38F0                                    3312+          DC     HL2'14576'                    m4, i3, rXB
00003793  4A                                      3313+          DC     X'4A'

00003794  B98D 0020                               3315+          EPSW   R2,R0              exptract psw
00003798  5020 5068                     00000068  3316+          ST     R2,CCPSW               to save CC
0000379C  E720 5028 000E                00003730  3317+          VST    V2,V1O51           save instruction result
000037A2  07FB                                    3318+          BR     R11                return
000037A4                                          3319+RE51      DS     0F                 expected 16 byte result
000037A4                                          3320+          DROP   R5
000037A4  00000000 00000000                       3321           DC     XL16'0000000000000000 999999999999999F'      v1
000037AC  99999999 9999999F
000037B4  0000007E 37BE2022                       3322           DC     XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
000037BC  C0914B26 7FFFFFFF
                                                  3323
                                                  3324           VRI_J VCVDQ,X'8F',X'3',3,S
000037C8                                          3325+          DS     0FD
000037C8                       000037C8           3326+          USING  *,R5               base for test data and test routine
000037C8  0000383C                                3327+T52       DC     A(X52)             address of test routine
000037CC  0034                                    3328+          DC     H'52'              test number
000037CE  00                                      3329+          DC     X'00'
000037CF  E2                                      3330+          DC     CL1'S'             S or Y = skip cross check
000037D0  8F                                      3331+          DC     X'8F'              i3
000037D1  03                                      3332+          DC     X'3'               m4
000037D2  03                                      3333+          DC     HL1'3'             cc
000037D3  0E                                      3334+          DC     HL1'14'            cc failed mask
000037D4  00003874                                3335+V2_52     DC     A(RE52+16)         address of v2: 16-byte packed decimal
000037D8  E5C3E5C4 D8404040                       3336+          DC     CL8'VCVDQ'         instruction name
000037E0  00000010                                3337+          DC     A(16)              result length
000037E4  00003864                                3338+          DC     A(RE52)            address of expected resul
000037E8  00000000 00000000                       3339+          DS     FD                 gap
000037F0  00000000 00000000                       3340+V1O52     DS     XL16               V1 output
000037F8  00000000 00000000
00003800  00000000 00000000                       3341+          DS     FD                    gap
00003808  40404040 40404040                       3342+          DC     CL8' '             was cross check skipped?
00003810  00000000 00000000                       3343+          DS     FD                    gap
00003818  00000000 00000000                       3344+XCO52     DS     XL16               Cross check Output
00003820  00000000 00000000
00003828  00000000 00000000                       3345+          DS     FD                    gap
00003830  00000000 00000000                       3346+          DS     2F           extract PSW after test (has CC)
00003838  00                                      3347+          DS     X            extracted cc
00003839  00                                      3348+          DS     X            CC Check failed
                                                  3349+*
0000383C                                          3350+X52       DS     0F
0000383C  E720 8F56 0006                00001156  3351+          VL     V2,V1FUDGE
00003842  E320 500C 0014                000037D4  3352+          LGF    R2,V2_52            get v2 address
00003848  E722 0000 0006                00000000  3353+          VL     V2,0(R2)

0000384E                                          3356+          DS     0H                           E64A VCVDQ
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000384E  E6                                       3357+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
0000384F  22                                       3358+          DC    X'22'                        v1, v2
00003850  00                                       3359+          DC    X'00'                       reserved
00003851  38F0                                     3360+          DC    HL2'14576'                   m4, i3, rXB
00003853  4A                                       3361+          DC    X'4A'

00003854  B98D 0020                                3363+          EPSW  R2,R0             exptract psw
00003858  5020 5068                      00000068  3364+          ST    R2,CCPSW              to save CC
0000385C  E720 5028 000E                 000037F0  3365+          VST   V2,V1O52          save instruction result
00003862  07FB                                     3366+          BR    R11               return
00003864                                           3367+RE52      DS    0F                expected 16 byte result
00003864                                           3368+          DROP  R5
00003864  00000000 00000000                        3369           DC    XL16'0000000000000000 999999999999999F'      v1
0000386C  99999999 9999999F
00003874  FFFFFF81 C841DFDD                        3370           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
0000387C  3F6EB4D9 80000001
                                                   3371
                                                   3372           VRI_J VCVDQ,X'8F',X'3',3,N
00003888                                           3373+          DS    0FD
00003888                       00003888            3374+          USING *,R5              base for test data and test routine
00003888  000038FC                                 3375+T53       DC    A(X53)            address of test routine
0000388C  0035                                     3376+          DC    H'53'             test number
0000388E  00                                       3377+          DC    X'00'
0000388F  D5                                       3378+          DC    CL1'N'            S or Y = skip cross check
00003890  8F                                       3379+          DC    X'8F'             i3
00003891  03                                       3380+          DC    X'3'              m4
00003892  03                                       3381+          DC    HL1'3'            cc
00003893  0E                                       3382+          DC    HL1'14'           cc failed mask
00003894  00003934                                 3383+V2_53     DC    A(RE53+16)        address of v2: 16-byte packed decimal
00003898  E5C3E5C4 D8404040                        3384+          DC    CL8'VCVDQ'        instruction name
000038A0  00000010                                 3385+          DC    A(16)             result length
000038A4  00003924                                 3386+          DC    A(RE53)           address of expected resul
000038A8  00000000 00000000                        3387+          DS    FD                gap
000038B0  00000000 00000000                        3388+V1O53     DS    XL16              V1 output
000038B8  00000000 00000000
000038C0  00000000 00000000                        3389+          DS    FD                   gap
000038C8  40404040 40404040                        3390+          DC    CL8' '            was cross check skipped?
000038D0  00000000 00000000                        3391+          DS    FD                   gap
000038D8  00000000 00000000                        3392+XCO53     DS    XL16              Cross check Output
000038E0  00000000 00000000
000038E8  00000000 00000000                        3393+          DS    FD                   gap
000038F0  00000000 00000000                        3394+          DS    2F          extract PSW after test (has CC)
000038F8  00                                       3395+          DS    X           extracted cc
000038F9  00                                       3396+          DS    X           CC Check failed
                                                   3397+*
000038FC                                           3398+X53       DS    0F
000038FC  E720 8F56 0006                 00001156  3399+          VL    V2,V1FUDGE
00003902  E320 500C 0014                 00003894  3400+          LGF   R2,V2_53           get v2 address
00003908  E722 0000 0006                 00000000  3401+          VL    V2,0(R2)

0000390E                                           3404+          DS    0H                      E64A VCVDQ
0000390E  E6                                       3405+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
0000390F  22                                       3406+          DC    X'22'                        v1, v2
00003910  00                                       3407+          DC    X'00'                       reserved
00003911  38F0                                     3408+          DC    HL2'14576'                   m4, i3, rXB
00003913  4A                                       3409+          DC    X'4A'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00003914  B98D 0020                               3411+          EPSW  R2,R0                exptract psw
00003918  5020 5068                      00000068 3412+          ST    R2,CCPSW                 to save CC
0000391C  E720 5028 000E                 000038B0 3413+          VST   V2,V1O53             save instruction result
00003922  07FB                                    3414+          BR    R11                  return
00003924                                          3415+RE53      DS    0F                   expected 16 byte result
00003924                                          3416+          DROP  R5
00003924  00000000 00000000                       3417           DC    XL16'0000000000000000 000000000000000F'      v1
0000392C  00000000 0000000F
00003934  0000007E 37BE2022                       3418           DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
0000393C  C0914B26 80000000
                                                  3419
                                                  3420           VRI_J VCVDQ,X'8F',X'3',3,N
00003948                                          3421+          DS    0FD
00003948                         00003948         3422+          USING *,R5                 base for test data and test routine
00003948  000039BC                                3423+T54       DC    A(X54)               address of test routine
0000394C  0036                                    3424+          DC    H'54'                test number
0000394E  00                                      3425+          DC    X'00'
0000394F  D5                                      3426+          DC    CL1'N'               S or Y = skip cross check
00003950  8F                                      3427+          DC    X'8F'                i3
00003951  03                                      3428+          DC    X'3'                 m4
00003952  03                                      3429+          DC    HL1'3'               cc
00003953  0E                                      3430+          DC    HL1'14'              cc failed mask
00003954  000039F4                                3431+V2_54     DC    A(RE54+16)           address of v2: 16-byte packed decimal
00003958  E5C3E5C4 D8404040                       3432+          DC    CL8'VCVDQ'           instruction name
00003960  00000010                                3433+          DC    A(16)                result length
00003964  000039E4                                3434+          DC    A(RE54)              address of expected resul
00003968  00000000 00000000                       3435+          DS    FD                   gap
00003970  00000000 00000000                       3436+V1O54     DS    XL16                 V1 output
00003978  00000000 00000000
00003980  00000000 00000000                       3437+          DS    FD                      gap
00003988  40404040 40404040                       3438+          DC    CL8' '               was cross check skipped?
00003990  00000000 00000000                       3439+          DS    FD                      gap
00003998  00000000 00000000                       3440+XCO54     DS    XL16                 Cross check Output
000039A0  00000000 00000000
000039A8  00000000 00000000                       3441+          DS    FD                      gap
000039B0  00000000 00000000                       3442+          DS    2F             extract PSW after test (has CC)
000039B8  00                                      3443+          DS    X              extracted cc
000039B9  00                                      3444+          DS    X              CC Check failed
                                                  3445+*
000039BC                                          3446+X54       DS    0F
000039BC  E720 8F56 0006                 00001156 3447+          VL    V2,V1FUDGE
000039C2  E320 500C 0014                 00003954 3448+          LGF   R2,V2_54              get v2 address
000039C8  E722 0000 0006                 00000000 3449+          VL    V2,0(R2)

000039CE                                          3452+          DS    0H                         E64A VCVDQ
000039CE  E6                                      3453+          DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
000039CF  22                                      3454+          DC    X'22'                       v1, v2
000039D0  00                                      3455+          DC    X'00'                      reserved
000039D1  38F0                                    3456+          DC    HL2'14576'                  m4, i3, rXB
000039D3  4A                                      3457+          DC    X'4A'

000039D4  B98D 0020                               3459+          EPSW  R2,R0                exptract psw
000039D8  5020 5068                      00000068 3460+          ST    R2,CCPSW                 to save CC
000039DC  E720 5028 000E                 00003970 3461+          VST   V2,V1O54             save instruction result
000039E2  07FB                                    3462+          BR    R11                  return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000039E4                                          3463+RE54     DS    0F                  expected 16 byte result
000039E4                                          3464+         DROP  R5
000039E4  00000000 00000000                       3465          DC    XL16'0000000000000000 000000000000000F'      v1
000039EC  00000000 0000000F
000039F4  FFFFFF81 C841DFDD                       3466          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
000039FC  3F6EB4D9 80000000
                                                  3467
                                                  3468 *-----------------------------------------------------------------------
                                                  3469 *        i3:   x'9f"   (IOM=1, RDC=31)
                                                  3470 *        m4:   x'9'    (LB=1, P1=0, CS=1)
                                                  3471 *
                                                  3472          VRI_J VCVDQ,X'8F',X'9',0,N
00003A08                                          3473+         DS    0FD
00003A08                        00003A08          3474+         USING *,R5                base for test data and test routine
00003A08  00003A7C                                3475+T55      DC    A(X55)              address of test routine
00003A0C  0037                                    3476+         DC    H'55'               test number
00003A0E  00                                      3477+         DC    X'00'
00003A0F  D5                                      3478+         DC    CL1'N'              S or Y = skip cross check
00003A10  8F                                      3479+         DC    X'8F'               i3
00003A11  09                                      3480+         DC    X'9'                m4
00003A12  00                                      3481+         DC    HL1'0'              cc
00003A13  07                                      3482+         DC    HL1'7'              cc failed mask
00003A14  00003AB4                                3483+V2_55    DC    A(RE55+16)          address of v2: 16-byte packed decimal
00003A18  E5C3E5C4 D8404040                       3484+         DC    CL8'VCVDQ'          instruction name
00003A20  00000010                                3485+         DC    A(16)               result length
00003A24  00003AA4                                3486+         DC    A(RE55)             address of expected resul
00003A28  00000000 00000000                       3487+         DS    FD                  gap
00003A30  00000000 00000000                       3488+V1O55    DS    XL16                V1 output
00003A38  00000000 00000000
00003A40  00000000 00000000                       3489+         DS    FD                     gap
00003A48  40404040 40404040                       3490+         DC    CL8' '              was cross check skipped?
00003A50  00000000 00000000                       3491+         DS    FD                     gap
00003A58  00000000 00000000                       3492+XCO55    DS    XL16                Cross check Output
00003A60  00000000 00000000
00003A68  00000000 00000000                       3493+         DS    FD                     gap
00003A70  00000000 00000000                       3494+         DS    2F            extract PSW after test (has CC)
00003A78  00                                      3495+         DS    X             extracted cc
00003A79  00                                      3496+         DS    X             CC Check failed
                                                  3497+*
00003A7C                                          3498+X55      DS    0F
00003A7C  E720 8F56 0006              00001156    3499+         VL    V2,V1FUDGE
00003A82  E320 500C 0014              00003A14    3500+         LGF   R2,V2_55             get v2 address
00003A88  E722 0000 0006              00000000    3501+         VL    V2,0(R2)

00003A8E                                          3504+         DS    0H                        E64A VCVDQ
00003A8E  E6                                      3505+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00003A8F  22                                      3506+         DC    X'22'                      v1, v2
00003A90  00                                      3507+         DC    X'00'                     reserved
00003A91  98F0                                    3508+         DC    HL2'39152'                 m4, i3, rXB
00003A93  4A                                      3509+         DC    X'4A'

00003A94  B98D 0020                               3511+         EPSW  R2,R0               exptract psw
00003A98  5020 5068                   00000068    3512+         ST    R2,CCPSW               to save CC
00003A9C  E720 5028 000E              00003A30    3513+         VST   V2,V1O55            save instruction result
00003AA2  07FB                                    3514+         BR    R11                 return
00003AA4                                          3515+RE55     DS    0F                  expected 16 byte result
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003AA4                                           3516+         DROP  R5
00003AA4  00000000 00000000                        3517          DC    XL16'0000000000000000 000000000000000C'      v1
00003AAC  00000000 0000000C
00003AB4  00000000 00000000                        3518          DC    XL16'0000000000000000 0000000000000000'      v2
00003ABC  00000000 00000000
                                                   3519
                                                   3520          VRI_J VCVDQ,X'8F',X'9',0,N
00003AC8                                           3521+         DS    0FD
00003AC8                     00003AC8              3522+         USING *,R5                base for test data and test routine
00003AC8  00003B3C                                 3523+T56      DC    A(X56)              address of test routine
00003ACC  0038                                     3524+         DC    H'56'               test number
00003ACE  00                                       3525+         DC    X'00'
00003ACF  D5                                       3526+         DC    CL1'N'              S or Y = skip cross check
00003AD0  8F                                       3527+         DC    X'8F'               i3
00003AD1  09                                       3528+         DC    X'9'                m4
00003AD2  00                                       3529+         DC    HL1'0'              cc
00003AD3  07                                       3530+         DC    HL1'7'              cc failed mask
00003AD4  00003B74                                 3531+V2_56    DC    A(RE56+16)          address of v2: 16-byte packed decimal
00003AD8  E5C3E5C4 D8404040                        3532+         DC    CL8'VCVDQ'          instruction name
00003AE0  00000010                                 3533+         DC    A(16)               result length
00003AE4  00003B64                                 3534+         DC    A(RE56)             address of expected resul
00003AE8  00000000 00000000                        3535+         DS    FD                  gap
00003AF0  00000000 00000000                        3536+V1O56    DS    XL16                V1 output
00003AF8  00000000 00000000
00003B00  00000000 00000000                        3537+         DS    FD                     gap
00003B08  40404040 40404040                        3538+         DC    CL8' '              was cross check skipped?
00003B10  00000000 00000000                        3539+         DS    FD                     gap
00003B18  00000000 00000000                        3540+XCO56    DS    XL16                Cross check Output
00003B20  00000000 00000000
00003B28  00000000 00000000                        3541+         DS    FD                     gap
00003B30  00000000 00000000                        3542+         DS    2F            extract PSW after test (has CC)
00003B38  00                                       3543+         DS    X             extracted cc
00003B39  00                                       3544+         DS    X             CC Check failed
                                                   3545+*
00003B3C                                           3546+X56      DS    0F
00003B3C  E720 8F56 0006               00001156    3547+         VL    V2,V1FUDGE
00003B42  E320 500C 0014               00003AD4    3548+         LGF   R2,V2_56             get v2 address
00003B48  E722 0000 0006               00000000    3549+         VL    V2,0(R2)

00003B4E                                           3552+         DS    0H                        E64A VCVDQ
00003B4E  E6                                       3553+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00003B4F  22                                       3554+         DC    X'22'                      v1, v2
00003B50  00                                       3555+         DC    X'00'                     reserved
00003B51  98F0                                     3556+         DC    HL2'39152'                 m4, i3, rXB
00003B53  4A                                       3557+         DC    X'4A'

00003B54  B98D 0020                                3559+         EPSW  R2,R0               exptract psw
00003B58  5020 5068                    00000068    3560+         ST    R2,CCPSW                to save CC
00003B5C  E720 5028 000E               00003AF0    3561+         VST   V2,V1O56            save instruction result
00003B62  07FB                                     3562+         BR    R11                 return
00003B64                                           3563+RE56     DS    0F                  expected 16 byte result
00003B64                                           3564+         DROP  R5
00003B64  00000000 00000000                        3565          DC    XL16'0000000000000000 000000000000001C'      v1
00003B6C  00000000 0000001C
00003B74  00000000 00000000                        3566          DC    XL16'0000000000000000 0000000000000001'      v2
00003B7C  00000000 00000001
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   3567 *
                                                   3568 * overflow
                                                   3569 *
                                                   3570          VRI_J VCVDQ,X'8F',X'9',3,S
00003B88                                           3571+         DS    0FD
00003B88                     00003B88              3572+         USING *,R5                 base for test data and test routine
00003B88  00003BFC                                 3573+T57      DC    A(X57)               address of test routine
00003B8C  0039                                     3574+         DC    H'57'                test number
00003B8E  00                                       3575+         DC    X'00'
00003B8F  E2                                       3576+         DC    CL1'S'               S or Y = skip cross check
00003B90  8F                                       3577+         DC    X'8F'                i3
00003B91  09                                       3578+         DC    X'9'                 m4
00003B92  03                                       3579+         DC    HL1'3'               cc
00003B93  0E                                       3580+         DC    HL1'14'              cc failed mask
00003B94  00003C34                                 3581+V2_57    DC    A(RE57+16)           address of v2: 16-byte packed decimal
00003B98  E5C3E5C4 D8404040                        3582+         DC    CL8'VCVDQ'           instruction name
00003BA0  00000010                                 3583+         DC    A(16)                result length
00003BA4  00003C24                                 3584+         DC    A(RE57)              address of expected resul
00003BA8  00000000 00000000                        3585+         DS    FD                   gap
00003BB0  00000000 00000000                        3586+V1O57    DS    XL16                 V1 output
00003BB8  00000000 00000000
00003BC0  00000000 00000000                        3587+         DS    FD                      gap
00003BC8  40404040 40404040                        3588+         DC    CL8' '               was cross check skipped?
00003BD0  00000000 00000000                        3589+         DS    FD                      gap
00003BD8  00000000 00000000                        3590+XC057    DS    XL16                 Cross check Output
00003BE0  00000000 00000000
00003BE8  00000000 00000000                        3591+         DS    FD                      gap
00003BF0  00000000 00000000                        3592+         DS    2F             extract PSW after test (has CC)
00003BF8  00                                       3593+         DS    X              extracted cc
00003BF9  00                                       3594+         DS    X              CC Check failed
                                                   3595+*
00003BFC                                           3596+X57      DS    0F
00003BFC  E720 8F56 0006             00001156      3597+         VL    V2,V1FUDGE
00003C02  E320 500C 0014             00003B94      3598+         LGF   R2,V2_57              get v2 address
00003C08  E722 0000 0006             00000000      3599+         VL    V2,0(R2)

00003C0E                                           3602+         DS    0H                         E64A VCVDQ
00003C0E  E6                                       3603+         DC    X'E6'                       VECTOR CONVERT TO DECIMAL (128)
00003C0F  22                                       3604+         DC    X'22'                       v1, v2
00003C10  00                                       3605+         DC    X'00'                      reserved
00003C11  98F0                                     3606+         DC    HL2'39152'                  m4, i3, rXB
00003C13  4A                                       3607+         DC    X'4A'

00003C14  B98D 0020                                3609+         EPSW  R2,R0                exptract psw
00003C18  5020 5068                  00000068      3610+         ST    R2,CCPSW                 to save CC
00003C1C  E720 5028 000E             00003BB0      3611+         VST   V2,V1O57             save instruction result
00003C22  07FB                                     3612+         BR    R11                  return
00003C24                                           3613+RE57     DS    0F                   expected 16 byte result
00003C24                                           3614+         DROP  R5
00003C24  00000000 00000000                        3615          DC    XL16'0000000000000000 607431768211455C'      v1
00003C2C  60743176 8211455C
00003C34  FFFFFFFF FFFFFFFF                        3616          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
00003C3C  FFFFFFFF FFFFFFFF
                                                   3617
                                                   3618          VRI_J VCVDQ,X'8F',X'9',3,N
00003C48                                           3619+         DS    0FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003C48                     00003C48             3620+          USING *,R5                base for test data and test routine
00003C48  00003CBC                                3621+T58       DC    A(X58)              address of test routine
00003C4C  003A                                    3622+          DC    H'58'               test number
00003C4E  00                                      3623+          DC    X'00'
00003C4F  D5                                      3624+          DC    CL1'N'              S or Y = skip cross check
00003C50  8F                                      3625+          DC    X'8F'               i3
00003C51  09                                      3626+          DC    X'9'                m4
00003C52  03                                      3627+          DC    HL1'3'              cc
00003C53  0E                                      3628+          DC    HL1'14'             cc failed mask
00003C54  00003CF4                                3629+V2_58     DC    A(RE58+16)          address of v2: 16-byte packed decimal
00003C58  E5C3E5C4 D8404040                       3630+          DC    CL8'VCVDQ'          instruction name
00003C60  00000010                                3631+          DC    A(16)               result length
00003C64  00003CE4                                3632+          DC    A(RE58)             address of expected resul
00003C68  00000000 00000000                       3633+          DS    FD                  gap
00003C70  00000000 00000000                       3634+V1O58     DS    XL16                V1 output
00003C78  00000000 00000000
00003C80  00000000 00000000                       3635+          DS    FD                     gap
00003C88  40404040 40404040                       3636+          DC    CL8' '              was cross check skipped?
00003C90  00000000 00000000                       3637+          DS    FD                     gap
00003C98  00000000 00000000                       3638+XCO58     DS    XL16                Cross check Output
00003CA0  00000000 00000000
00003CA8  00000000 00000000                       3639+          DS    FD                     gap
00003CB0  00000000 00000000                       3640+          DS    2F            extract PSW after test (has CC)
00003CB8  00                                      3641+          DS    X             extracted cc
00003CB9  00                                      3642+          DS    X             CC Check failed
                                                  3643+*
00003CBC                                          3644+X58       DS    0F
00003CBC  E720 8F56 0006               00001156   3645+          VL    V2,V1FUDGE
00003CC2  E320 500C 0014               00003C54   3646+          LGF   R2,V2_58             get v2 address
00003CC8  E722 0000 0006               00000000   3647+          VL    V2,0(R2)

00003CCE                                          3650+          DS    0H                        E64A VCVDQ
00003CCE  E6                                      3651+          DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
00003CCF  22                                      3652+          DC    X'22'                      v1, v2
00003CD0  00                                      3653+          DC    X'00'                     reserved
00003CD1  98F0                                    3654+          DC    HL2'39152'                 m4, i3, rXB
00003CD3  4A                                      3655+          DC    X'4A'

00003CD4  B98D 0020                               3657+          EPSW  R2,R0               exptract psw
00003CD8  5020 5068                    00000068   3658+          ST    R2,CCPSW                to save CC
00003CDC  E720 5028 000E               00003C70   3659+          VST   V2,V1O58            save instruction result
00003CE2  07FB                                    3660+          BR    R11                 return
00003CE4                                          3661+RE58      DS    0F                  expected 16 byte result
00003CE4                                          3662+          DROP  R5
00003CE4  00000000 00000000                       3663           DC    XL16'0000000000000000 372036854775807C'      v1
00003CEC  37203685 4775807C
00003CF4  00000000 00000000                       3664           DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
00003CFC  7FFFFFFF FFFFFFFF
                                                  3665
                                                  3666           VRI_J VCVDQ,X'8F',X'9',3,S
00003D08                                          3667+          DS    0FD
00003D08                     00003D08             3668+          USING *,R5                base for test data and test routine
00003D08  00003D7C                                3669+T59       DC    A(X59)              address of test routine
00003D0C  003B                                    3670+          DC    H'59'               test number
00003D0E  00                                      3671+          DC    X'00'
00003D0F  E2                                      3672+          DC    CL1'S'              S or Y = skip cross check
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00003D10  8F                                      3673+         DC    X'8F'              i3
00003D11  09                                      3674+         DC    X'9'               m4
00003D12  03                                      3675+         DC    HL1'3'             cc
00003D13  0E                                      3676+         DC    HL1'14'            cc failed mask
00003D14  00003DB4                                3677+V2_59    DC    A(RE59+16)         address of v2: 16-byte packed decimal
00003D18  E5C3E5C4 D8404040                       3678+         DC    CL8'VCVDQ'         instruction name
00003D20  00000010                                3679+         DC    A(16)              result length
00003D24  00003DA4                                3680+         DC    A(RE59)            address of expected resul
00003D28  00000000 00000000                       3681+         DS    FD                 gap
00003D30  00000000 00000000                       3682+V1O59    DS    XL16               V1 output
00003D38  00000000 00000000
00003D40  00000000 00000000                       3683+         DS    FD                    gap
00003D48  40404040 40404040                       3684+         DC    CL8' '             was cross check skipped?
00003D50  00000000 00000000                       3685+         DS    FD                    gap
00003D58  00000000 00000000                       3686+XCO59    DS    XL16               Cross check Output
00003D60  00000000 00000000
00003D68  00000000 00000000                       3687+         DS    FD                    gap
00003D70  00000000 00000000                       3688+         DS    2F           extract PSW after test (has CC)
00003D78  00                                      3689+         DS    X            extracted cc
00003D79  00                                      3690+         DS    X            CC Check failed
                                                  3691+*
00003D7C                                          3692+X59      DS    0F
00003D7C  E720 8F56 0006             00001156     3693+         VL    V2,V1FUDGE
00003D82  E320 500C 0014             00003D14     3694+         LGF   R2,V2_59            get v2 address
00003D88  E722 0000 0006             00000000     3695+         VL    V2,0(R2)

00003D8E                                          3698+         DS    0H                       E64A VCVDQ
00003D8E  E6                                      3699+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
00003D8F  22                                      3700+         DC    X'22'                     v1, v2
00003D90  00                                      3701+         DC    X'00'                    reserved
00003D91  98F0                                    3702+         DC    HL2'39152'                m4, i3, rXB
00003D93  4A                                      3703+         DC    X'4A'

00003D94  B98D 0020                               3705+         EPSW  R2,R0              exptract psw
00003D98  5020 5068                  00000068     3706+         ST    R2,CCPSW               to save CC
00003D9C  E720 5028 000E             00003D30     3707+         VST   V2,V1O59           save instruction result
00003DA2  07FB                                    3708+         BR    R11                return
00003DA4                                          3709+RE59     DS    0F                 expected 16 byte result
00003DA4                                          3710+         DROP  R5
00003DA4  00000000 00000000                       3711          DC    XL16'0000000000000000 235394913435649C'      v1
00003DAC  23539491 3435649C
00003DB4  FFFFFFFF FFFFFFFF                       3712          DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
00003DBC  80000000 00000001
                                                  3713
                                                  3714          VRI_J VCVDQ,X'8F',X'9',3,N
00003DC8                                          3715+         DS    0FD
00003DC8                    00003DC8              3716+         USING *,R5               base for test data and test routine
00003DC8  00003E3C                                3717+T60      DC    A(X60)             address of test routine
00003DCC  003C                                    3718+         DC    H'60'              test number
00003DCE  00                                      3719+         DC    X'00'
00003DCF  D5                                      3720+         DC    CL1'N'             S or Y = skip cross check
00003DD0  8F                                      3721+         DC    X'8F'              i3
00003DD1  09                                      3722+         DC    X'9'               m4
00003DD2  03                                      3723+         DC    HL1'3'             cc
00003DD3  0E                                      3724+         DC    HL1'14'            cc failed mask
00003DD4  00003E74                                3725+V2_60    DC    A(RE60+16)         address of v2: 16-byte packed decimal
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003DD8  E5C3E5C4 D8404040                    3726+         DC    CL8'VCVDQ'        instruction name
00003DE0  00000010                             3727+         DC    A(16)             result length
00003DE4  00003E64                             3728+         DC    A(RE60)           address of expected resul
00003DE8  00000000 00000000                    3729+         DS    FD                gap
00003DF0  00000000 00000000                    3730+V1O60    DS    XL16              V1 output
00003DF8  00000000 00000000
00003E00  00000000 00000000                    3731+         DS    FD                   gap
00003E08  40404040 40404040                    3732+         DC    CL8' '            was cross check skipped?
00003E10  00000000 00000000                    3733+         DS    FD                   gap
00003E18  00000000 00000000                    3734+XC060    DS    XL16              Cross check Output
00003E20  00000000 00000000
00003E28  00000000 00000000                    3735+         DS    FD                   gap
00003E30  00000000 00000000                    3736+         DS    2F          extract PSW after test (has CC)
00003E38  00                                   3737+         DS    X           extracted cc
00003E39  00                                   3738+         DS    X           CC Check failed
                                               3739+*
00003E3C                                       3740+X60      DS    0F
00003E3C  E720 8F56 0006             00001156  3741+         VL    V2,V1FUDGE
00003E42  E320 500C 0014             00003DD4  3742+         LGF   R2,V2_60           get v2 address
00003E48  E722 0000 0006             00000000  3743+         VL    V2,0(R2)

00003E4E                                       3746+         DS    0H                      E64A VCVDQ
00003E4E  E6                                   3747+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
00003E4F  22                                   3748+         DC    X'22'                    v1, v2
00003E50  00                                   3749+         DC    X'00'                   reserved
00003E51  98F0                                 3750+         DC    HL2'39152'               m4, i3, rXB
00003E53  4A                                   3751+         DC    X'4A'

00003E54  B98D 0020                            3753+         EPSW  R2,R0             exptract psw
00003E58  5020 5068                  00000068  3754+         ST    R2,CCPSW              to save CC
00003E5C  E720 5028 000E             00003DF0  3755+         VST   V2,V1O60          save instruction result
00003E62  07FB                                 3756+         BR    R11               return
00003E64                                       3757+RE60     DS    0F                expected 16 byte result
00003E64                                       3758+         DROP  R5
00003E64  00000000 00000000                    3759          DC    XL16'00000000000000009999999999999999C'      v1
00003E6C  99999999 9999999C
00003E74  0000007E 37BE2022                    3760          DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
00003E7C  C0914B26 7FFFFFFF
                                               3761
                                               3762          VRI_J VCVDQ,X'8F',X'9',3,S
00003E88                                       3763+         DS    0FD
00003E88                   00003E88            3764+         USING *,R5              base for test data and test routine
00003E88  00003EFC                             3765+T61      DC    A(X61)            address of test routine
00003E8C  003D                                 3766+         DC    H'61'             test number
00003E8E  00                                   3767+         DC    X'00'
00003E8F  E2                                   3768+         DC    CL1'S'            S or Y = skip cross check
00003E90  8F                                   3769+         DC    X'8F'             i3
00003E91  09                                   3770+         DC    X'9'              m4
00003E92  03                                   3771+         DC    HL1'3'            cc
00003E93  0E                                   3772+         DC    HL1'14'           cc failed mask
00003E94  00003F34                             3773+V2_61    DC    A(RE61+16)        address of v2: 16-byte packed decimal
00003E98  E5C3E5C4 D8404040                    3774+         DC    CL8'VCVDQ'        instruction name
00003EA0  00000010                             3775+         DC    A(16)             result length
00003EA4  00003F24                             3776+         DC    A(RE61)           address of expected resul
00003EA8  00000000 00000000                    3777+         DS    FD                gap
00003EB0  00000000 00000000                    3778+V1O61    DS    XL16              V1 output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003EB8  00000000 00000000
00003EC0  00000000 00000000                        3779+          DS    FD                     gap
00003EC8  40404040 40404040                        3780+          DC    CL8' '              was cross check skipped?
00003ED0  00000000 00000000                        3781+          DS    FD                     gap
00003ED8  00000000 00000000                        3782+XC061     DS    XL16                Cross check Output
00003EE0  00000000 00000000
00003EE8  00000000 00000000                        3783+          DS    FD                     gap
00003EF0  00000000 00000000                        3784+          DS    2F            extract PSW after test (has CC)
00003EF8  00                                       3785+          DS    X             extracted cc
00003EF9  00                                       3786+          DS    X             CC Check failed
                                                   3787+*
00003EFC                                           3788+X61       DS    0F
00003EFC  E720 8F56 0006             00001156      3789+          VL    V2,V1FUDGE
00003F02  E320 500C 0014             00003E94      3790+          LGF   R2,V2_61               get v2 address
00003F08  E722 0000 0006             00000000      3791+          VL    V2,0(R2)

00003F0E                                           3794+          DS    0H                          E64A VCVDQ
00003F0E  E6                                       3795+          DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
00003F0F  22                                       3796+          DC    X'22'                        v1, v2
00003F10  00                                       3797+          DC    X'00'                       reserved
00003F11  98F0                                     3798+          DC    HL2'39152'                   m4, i3, rXB
00003F13  4A                                       3799+          DC    X'4A'

00003F14  B98D 0020                                3801+          EPSW  R2,R0               exptract psw
00003F18  5020 5068                  00000068      3802+          ST    R2,CCPSW                to save CC
00003F1C  E720 5028 000E             00003EB0      3803+          VST   V2,V1O61            save instruction result
00003F22  07FB                                     3804+          BR    R11                 return
00003F24                                           3805+RE61      DS    0F                  expected 16 byte result
00003F24                                           3806+          DROP  R5
00003F24  00000000 00000000                        3807           DC    XL16'0000000000000000 607431768211457C'      v1
00003F2C  60743176 8211457C
00003F34  FFFFFF81 C841DFDD                        3808           DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
00003F3C  3F6EB4D9 80000001
                                                   3809
                                                   3810           VRI_J VCVDQ,X'8F',X'9',3,N
00003F48                                           3811+          DS    0FD
00003F48                       00003F48            3812+          USING *,R5              base for test data and test routine
00003F48  00003FBC                                 3813+T62       DC    A(X62)              address of test routine
00003F4C  003E                                     3814+          DC    H'62'               test number
00003F4E  00                                       3815+          DC    X'00'
00003F4F  D5                                       3816+          DC    CL1'N'              S or Y = skip cross check
00003F50  8F                                       3817+          DC    X'8F'               i3
00003F51  09                                       3818+          DC    X'9'                m4
00003F52  03                                       3819+          DC    HL1'3'              cc
00003F53  0E                                       3820+          DC    HL1'14'             cc failed mask
00003F54  00003FF4                                 3821+V2_62     DC    A(RE62+16)          address of v2: 16-byte packed decimal
00003F58  E5C3E5C4 D8404040                        3822+          DC    CL8'VCVDQ'          instruction name
00003F60  00000010                                 3823+          DC    A(16)               result length
00003F64  00003FE4                                 3824+          DC    A(RE62)             address of expected resul
00003F68  00000000 00000000                        3825+          DS    FD                  gap
00003F70  00000000 00000000                        3826+V1O62     DS    XL16                V1 output
00003F78  00000000 00000000
00003F80  00000000 00000000                        3827+          DS    FD                     gap
00003F88  40404040 40404040                        3828+          DC    CL8' '              was cross check skipped?
00003F90  00000000 00000000                        3829+          DS    FD                     gap
00003F98  00000000 00000000                        3830+XC062     DS    XL16                Cross check Output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00003FA0  00000000 00000000
00003FA8  00000000 00000000                        3831+          DS    FD                    gap
00003FB0  00000000 00000000                        3832+          DS    2F          extract PSW after test (has CC)
00003FB8  00                                       3833+          DS    X           extracted cc
00003FB9  00                                       3834+          DS    X           CC Check failed
                                                   3835+*
00003FBC                                           3836+X62       DS    0F
00003FBC  E720 8F56 0006                00001156  3837+          VL    V2,V1FUDGE
00003FC2  E320 500C 0014                00003F54  3838+          LGF   R2,V2_62            get v2 address
00003FC8  E722 0000 0006                00000000  3839+          VL    V2,0(R2)

00003FCE                                           3842+          DS    0H                       E64A VCVDQ
00003FCE  E6                                       3843+          DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
00003FCF  22                                       3844+          DC    X'22'                     v1, v2
00003FD0  00                                       3845+          DC    X'00'                    reserved
00003FD1  98F0                                     3846+          DC    HL2'39152'                m4, i3, rXB
00003FD3  4A                                       3847+          DC    X'4A'

00003FD4  B98D 0020                                3849+          EPSW  R2,R0             exptract psw
00003FD8  5020 5068                     00000068  3850+          ST    R2,CCPSW              to save CC
00003FDC  E720 5028 000E                00003F70  3851+          VST   V2,V1O62          save instruction result
00003FE2  07FB                                     3852+          BR    R11               return
00003FE4                                           3853+RE62      DS    0F                expected 16 byte result
00003FE4                                           3854+          DROP  R5
00003FE4  00000000 00000000                        3855           DC    XL16'0000000000000000 000000000000000C'      v1
00003FEC  00000000 0000000C
00003FF4  0000007E 37BE2022                        3856           DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
00003FFC  C0914B26 80000000
                                                   3857
                                                   3858           VRI_J VCVDQ,X'8F',X'9',3,N
00004008                                           3859+          DS    0FD
00004008                     00004008              3860+          USING *,R5              base for test data and test routine
00004008  0000407C                                 3861+T63       DC    A(X63)            address of test routine
0000400C  003F                                     3862+          DC    H'63'             test number
0000400E  00                                       3863+          DC    X'00'
0000400F  D5                                       3864+          DC    CL1'N'            S or Y = skip cross check
00004010  8F                                       3865+          DC    X'8F'             i3
00004011  09                                       3866+          DC    X'9'              m4
00004012  03                                       3867+          DC    HL1'3'            cc
00004013  0E                                       3868+          DC    HL1'14'           cc failed mask
00004014  000040B4                                 3869+V2_63     DC    A(RE63+16)        address of v2: 16-byte packed decimal
00004018  E5C3E5C4 D8404040                        3870+          DC    CL8'VCVDQ'        instruction name
00004020  00000010                                 3871+          DC    A(16)             result length
00004024  000040A4                                 3872+          DC    A(RE63)           address of expected resul
00004028  00000000 00000000                        3873+          DS    FD                gap
00004030  00000000 00000000                        3874+V1O63     DS    XL16              V1 output
00004038  00000000 00000000
00004040  00000000 00000000                        3875+          DS    FD                   gap
00004048  40404040 40404040                        3876+          DC    CL8' '            was cross check skipped?
00004050  00000000 00000000                        3877+          DS    FD                   gap
00004058  00000000 00000000                        3878+XCO63     DS    XL16              Cross check Output
00004060  00000000 00000000
00004068  00000000 00000000                        3879+          DS    FD                   gap
00004070  00000000 00000000                        3880+          DS    2F          extract PSW after test (has CC)
00004078  00                                       3881+          DS    X           extracted cc
00004079  00                                       3882+          DS    X           CC Check failed
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  3883+*
0000407C                                          3884+X63      DS    0F
0000407C  E720 8F56 0006                00001156  3885+         VL    V2,V1FUDGE
00004082  E320 500C 0014                00004014  3886+         LGF   R2,V2_63           get v2 address
00004088  E722 0000 0006                00000000  3887+         VL    V2,0(R2)

0000408E                                          3890+         DS    0H                      E64A VCVDQ
0000408E  E6                                      3891+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000408F  22                                      3892+         DC    X'22'                    v1, v2
00004090  00                                      3893+         DC    X'00'                   reserved
00004091  98F0                                    3894+         DC    HL2'39152'               m4, i3, rXB
00004093  4A                                      3895+         DC    X'4A'

00004094  B98D 0020                               3897+         EPSW  R2,R0             exptract psw
00004098  5020 5068                     00000068  3898+         ST    R2,CCPSW              to save CC
0000409C  E720 5028 000E                00004030  3899+         VST   V2,V1O63          save instruction result
000040A2  07FB                                    3900+         BR    R11               return
000040A4                                          3901+RE63     DS    0F                expected 16 byte result
000040A4                                          3902+         DROP  R5
000040A4  00000000 00000000                       3903          DC    XL16'0000000000000000 607431768211456C'      v1
000040AC  60743176 8211456C
000040B4  FFFFFF81 C841DFDD                       3904          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
000040BC  3F6EB4D9 80000000
                                                  3905
                                                  3906
                                                  3907 *-----------------------------------------------------------------------
                                                  3908 *        i3:   x'9f"   (IOM=1, RDC=31)
                                                  3909 *        m4:   x'B'    (LB=1, P1=1, CS=1)
                                                  3910 *
                                                  3911          VRI_J VCVDQ,X'8F',X'B',0,N
000040C8                                          3912+         DS    0FD
000040C8                       000040C8           3913+         USING *,R5              base for test data and test routine
000040C8  0000413C                                3914+T64      DC    A(X64)            address of test routine
000040CC  0040                                    3915+         DC    H'64'             test number
000040CE  00                                      3916+         DC    X'00'
000040CF  D5                                      3917+         DC    CL1'N'            S or Y = skip cross check
000040D0  8F                                      3918+         DC    X'8F'             i3
000040D1  0B                                      3919+         DC    X'B'              m4
000040D2  00                                      3920+         DC    HL1'0'            cc
000040D3  07                                      3921+         DC    HL1'7'            cc failed mask
000040D4  00004174                                3922+V2_64    DC    A(RE64+16)        address of v2: 16-byte packed decimal
000040D8  E5C3E5C4 D8404040                       3923+         DC    CL8'VCVDQ'        instruction name
000040E0  00000010                                3924+         DC    A(16)             result length
000040E4  00004164                                3925+         DC    A(RE64)           address of expected resul
000040E8  00000000 00000000                       3926+         DS    FD                gap
000040F0  00000000 00000000                       3927+V1O64    DS    XL16              V1 output
000040F8  00000000 00000000
00004100  00000000 00000000                       3928+         DS    FD                   gap
00004108  40404040 40404040                       3929+         DC    CL8' '            was cross check skipped?
00004110  00000000 00000000                       3930+         DS    FD                   gap
00004118  00000000 00000000                       3931+XCO64    DS    XL16              Cross check Output
00004120  00000000 00000000
00004128  00000000 00000000                       3932+         DS    FD                   gap
00004130  00000000 00000000                       3933+         DS    2F          extract PSW after test (has CC)
00004138  00                                      3934+         DS    X           extracted cc
00004139  00                                      3935+         DS    X           CC Check failed
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   3936+*
0000413C                                           3937+X64      DS    0F
0000413C  E720 8F56 0006                00001156   3938+         VL    V2,V1FUDGE
00004142  E320 500C 0014                000040D4   3939+         LGF   R2,V2_64           get v2 address
00004148  E722 0000 0006                00000000   3940+         VL    V2,0(R2)

0000414E                                           3943+         DS    0H                          E64A VCVDQ
0000414E  E6                                       3944+         DC    X'E6'                        VECTOR CONVERT TO DECIMAL (128)
0000414F  22                                       3945+         DC    X'22'                        v1, v2
00004150  00                                       3946+         DC    X'00'                       reserved
00004151  B8F0                                     3947+         DC    HL2'47344'                   m4, i3, rXB
00004153  4A                                       3948+         DC    X'4A'

00004154  B98D 0020                                3950+         EPSW  R2,R0             exptract psw
00004158  5020 5068                     00000068   3951+         ST    R2,CCPSW              to save CC
0000415C  E720 5028 000E                000040F0   3952+         VST   V2,V1O64          save instruction result
00004162  07FB                                     3953+         BR    R11               return
00004164                                           3954+RE64     DS    0F                expected 16 byte result
00004164                                           3955+         DROP  R5
00004164  00000000 00000000                        3956          DC    XL16'0000000000000000 000000000000000F'      v1
0000416C  00000000 0000000F
00004174  00000000 00000000                        3957          DC    XL16'0000000000000000 0000000000000000'      v2
0000417C  00000000 00000000
                                                   3958
                                                   3959          VRI_J VCVDQ,X'8F',X'B',0,N
00004188                                           3960+         DS    0FD
00004188                      00004188             3961+         USING *,R5              base for test data and test routine
00004188  000041FC                                 3962+T65      DC    A(X65)            address of test routine
0000418C  0041                                     3963+         DC    H'65'             test number
0000418E  00                                       3964+         DC    X'00'
0000418F  D5                                       3965+         DC    CL1'N'            S or Y = skip cross check
00004190  8F                                       3966+         DC    X'8F'             i3
00004191  0B                                       3967+         DC    X'B'              m4
00004192  00                                       3968+         DC    HL1'0'            cc
00004193  07                                       3969+         DC    HL1'7'            cc failed mask
00004194  00004234                                 3970+V2_65    DC    A(RE65+16)        address of v2: 16-byte packed decimal
00004198  E5C3E5C4 D8404040                        3971+         DC    CL8'VCVDQ'        instruction name
000041A0  00000010                                 3972+         DC    A(16)             result length
000041A4  00004224                                 3973+         DC    A(RE65)           address of expected resul
000041A8  00000000 00000000                        3974+         DS    FD                gap
000041B0  00000000 00000000                        3975+V1O65    DS    XL16              V1 output
000041B8  00000000 00000000
000041C0  00000000 00000000                        3976+         DS    FD                   gap
000041C8  40404040 40404040                        3977+         DC    CL8' '            was cross check skipped?
000041D0  00000000 00000000                        3978+         DS    FD                   gap
000041D8  00000000 00000000                        3979+XCO65    DS    XL16              Cross check Output
000041E0  00000000 00000000
000041E8  00000000 00000000                        3980+         DS    FD                   gap
000041F0  00000000 00000000                        3981+         DS    2F          extract PSW after test (has CC)
000041F8  00                                       3982+         DS    X           extracted cc
000041F9  00                                       3983+         DS    X           CC Check failed
                                                   3984+*
000041FC                                           3985+X65      DS    0F
000041FC  E720 8F56 0006                00001156   3986+         VL    V2,V1FUDGE
00004202  E320 500C 0014                00004194   3987+         LGF   R2,V2_65           get v2 address
00004208  E722 0000 0006                00000000   3988+         VL    V2,0(R2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT


0000420E                                           3991+          DS    0H                            E64A VCVDQ
0000420E  E6                                       3992+          DC    X'E6'                          VECTOR CONVERT TO DECIMAL (128)
0000420F  22                                       3993+          DC    X'22'                          v1, v2
00004210  00                                       3994+          DC    X'00'                         reserved
00004211  B8F0                                     3995+          DC    HL2'47344'                     m4, i3, rXB
00004213  4A                                       3996+          DC    X'4A'

00004214  B98D 0020                                3998+          EPSW  R2,R0                 exptract psw
00004218  5020 5068                      00000068  3999+          ST    R2,CCPSW                  to save CC
0000421C  E720 5028 000E                 000041B0  4000+          VST   V2,V1O65              save instruction result
00004222  07FB                                     4001+          BR    R11                   return
00004224                                           4002+RE65      DS    0F                    expected 16 byte result
00004224                                           4003+          DROP  R5
00004224  00000000 00000000                        4004           DC    XL16'0000000000000000 000000000000001F'      v1
0000422C  00000000 0000001F
00004234  00000000 00000000                        4005           DC    XL16'0000000000000000 0000000000000001'      v2
0000423C  00000000 00000001
                                                   4006 *
                                                   4007 * overflow
                                                   4008 *
                                                   4009           VRI_J VCVDQ,X'8F',X'B',3,S
00004248                                           4010+          DS    0FD
00004248                       00004248            4011+          USING *,R5                  base for test data and test routine
00004248  000042BC                                 4012+T66       DC    A(X66)                address of test routine
0000424C  0042                                     4013+          DC    H'66'                 test number
0000424E  00                                       4014+          DC    X'00'
0000424F  E2                                       4015+          DC    CL1'S'                S or Y = skip cross check
00004250  8F                                       4016+          DC    X'8F'                 i3
00004251  0B                                       4017+          DC    X'B'                  m4
00004252  03                                       4018+          DC    HL1'3'                cc
00004253  0E                                       4019+          DC    HL1'14'               cc failed mask
00004254  000042F4                                 4020+V2_66     DC    A(RE66+16)            address of v2: 16-byte packed decimal
00004258  E5C3E5C4 D8404040                        4021+          DC    CL8'VCVDQ'            instruction name
00004260  00000010                                 4022+          DC    A(16)                 result length
00004264  000042E4                                 4023+          DC    A(RE66)               address of expected resul
00004268  00000000 00000000                        4024+          DS    FD                    gap
00004270  00000000 00000000                        4025+V1O66     DS    XL16                  V1 output
00004278  00000000 00000000
00004280  00000000 00000000                        4026+          DS    FD                       gap
00004288  40404040 40404040                        4027+          DC    CL8' '                was cross check skipped?
00004290  00000000 00000000                        4028+          DS    FD                       gap
00004298  00000000 00000000                        4029+XC066     DS    XL16                  Cross check Output
000042A0  00000000 00000000
000042A8  00000000 00000000                        4030+          DS    FD                       gap
000042B0  00000000 00000000                        4031+          DS    2F              extract PSW after test (has CC)
000042B8  00                                       4032+          DS    X               extracted cc
000042B9  00                                       4033+          DS    X               CC Check failed
                                                   4034+*
000042BC                                           4035+X66       DS    0F
000042BC  E720 8F56 0006                 00001156  4036+          VL    V2,V1FUDGE
000042C2  E320 500C 0014                 00004254  4037+          LGF   R2,V2_66                get v2 address
000042C8  E722 0000 0006                 00000000  4038+          VL    V2,0(R2)

000042CE                                           4041+          DS    0H                            E64A VCVDQ
000042CE  E6                                       4042+          DC    X'E6'                          VECTOR CONVERT TO DECIMAL (128)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000042CF  22                                       4043+         DC    X'22'                     v1, v2
000042D0  00                                       4044+         DC    X'00'                    reserved
000042D1  B8F0                                     4045+         DC    HL2'47344'                m4, i3, rXB
000042D3  4A                                       4046+         DC    X'4A'

000042D4  B98D 0020                                4048+         EPSW  R2,R0             exptract psw
000042D8  5020 5068                      00000068  4049+         ST    R2,CCPSW              to save CC
000042DC  E720 5028 000E                 00004270  4050+         VST   V2,V1O66          save instruction result
000042E2  07FB                                     4051+         BR    R11               return
000042E4                                           4052+RE66     DS    0F                expected 16 byte result
000042E4                                           4053+         DROP  R5
000042E4  00000000 00000000                        4054          DC    XL16'0000000000000000 607431768211455F'      v1
000042EC  60743176 8211455F
000042F4  FFFFFFFF FFFFFFFF                        4055          DC    XL16'FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF'      v2
000042FC  FFFFFFFF FFFFFFFF
                                                   4056
                                                   4057          VRI_J VCVDQ,X'8F',X'B',3,N
00004308                                           4058+         DS    0FD
00004308                       00004308            4059+         USING *,R5              base for test data and test routine
00004308  0000437C                                 4060+T67      DC    A(X67)            address of test routine
0000430C  0043                                     4061+         DC    H'67'             test number
0000430E  00                                       4062+         DC    X'00'
0000430F  D5                                       4063+         DC    CL1'N'            S or Y = skip cross check
00004310  8F                                       4064+         DC    X'8F'             i3
00004311  0B                                       4065+         DC    X'B'              m4
00004312  03                                       4066+         DC    HL1'3'            cc
00004313  0E                                       4067+         DC    HL1'14'           cc failed mask
00004314  000043B4                                 4068+V2_67    DC    A(RE67+16)        address of v2: 16-byte packed decimal
00004318  E5C3E5C4 D8404040                        4069+         DC    CL8'VCVDQ'        instruction name
00004320  00000010                                 4070+         DC    A(16)             result length
00004324  000043A4                                 4071+         DC    A(RE67)           address of expected resul
00004328  00000000 00000000                        4072+         DS    FD                gap
00004330  00000000 00000000                        4073+V1O67    DS    XL16              V1 output
00004338  00000000 00000000
00004340  00000000 00000000                        4074+         DS    FD                   gap
00004348  40404040 40404040                        4075+         DC    CL8' '            was cross check skipped?
00004350  00000000 00000000                        4076+         DS    FD                   gap
00004358  00000000 00000000                        4077+XCO67    DS    XL16              Cross check Output
00004360  00000000 00000000
00004368  00000000 00000000                        4078+         DS    FD                   gap
00004370  00000000 00000000                        4079+         DS    2F          extract PSW after test (has CC)
00004378  00                                       4080+         DS    X           extracted cc
00004379  00                                       4081+         DS    X           CC Check failed
                                                   4082+*
0000437C                                           4083+X67      DS    0F
0000437C  E720 8F56 0006                 00001156  4084+         VL    V2,V1FUDGE
00004382  E320 500C 0014                 00004314  4085+         LGF   R2,V2_67            get v2 address
00004388  E722 0000 0006                 00000000  4086+         VL    V2,0(R2)

0000438E                                           4089+         DS    0H                       E64A VCVDQ
0000438E  E6                                       4090+         DC    X'E6'                     VECTOR CONVERT TO DECIMAL (128)
0000438F  22                                       4091+         DC    X'22'                     v1, v2
00004390  00                                       4092+         DC    X'00'                    reserved
00004391  B8F0                                     4093+         DC    HL2'47344'                m4, i3, rXB
00004393  4A                                       4094+         DC    X'4A'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004394  B98D 0020                             4096+         EPSW  R2,R0             exptract psw
00004398  5020 5068                   00000068  4097+         ST    R2,CCPSW              to save CC
0000439C  E720 5028 000E              00004330  4098+         VST   V2,V1O67          save instruction result
000043A2  07FB                                  4099+         BR    R11               return
000043A4                                        4100+RE67     DS    0F                expected 16 byte result
000043A4                                        4101+         DROP  R5
000043A4  00000000 00000000                     4102          DC    XL16'0000000000000000 372036854775807F'      v1
000043AC  37203685 4775807F
000043B4  00000000 00000000                     4103          DC    XL16'0000000000000000 7FFFFFFFFFFFFFFF'      v2
000043BC  7FFFFFFF FFFFFFFF
                                                4104
                                                4105          VRI_J VCVDQ,X'8F',X'B',3,S
000043C8                                        4106+         DS    0FD
000043C8                    000043C8            4107+         USING *,R5              base for test data and test routine
000043C8  0000443C                              4108+T68      DC    A(X68)            address of test routine
000043CC  0044                                  4109+         DC    H'68'             test number
000043CE  00                                    4110+         DC    X'00'
000043CF  E2                                    4111+         DC    CL1'S'            S or Y = skip cross check
000043D0  8F                                    4112+         DC    X'8F'             i3
000043D1  0B                                    4113+         DC    X'B'              m4
000043D2  03                                    4114+         DC    HL1'3'            cc
000043D3  0E                                    4115+         DC    HL1'14'           cc failed mask
000043D4  00004474                              4116+V2_68    DC    A(RE68+16)        address of v2: 16-byte packed decimal
000043D8  E5C3E5C4 D8404040                     4117+         DC    CL8'VCVDQ'        instruction name
000043E0  00000010                              4118+         DC    A(16)             result length
000043E4  00004464                              4119+         DC    A(RE68)           address of expected resul
000043E8  00000000 00000000                     4120+         DS    FD                gap
000043F0  00000000 00000000                     4121+V1O68    DS    XL16              V1 output
000043F8  00000000 00000000
00004400  00000000 00000000                     4122+         DS    FD                   gap
00004408  40404040 40404040                     4123+         DC    CL8' '            was cross check skipped?
00004410  00000000 00000000                     4124+         DS    FD                   gap
00004418  00000000 00000000                     4125+XCO68    DS    XL16              Cross check Output
00004420  00000000 00000000
00004428  00000000 00000000                     4126+         DS    FD                   gap
00004430  00000000 00000000                     4127+         DS    2F          extract PSW after test (has CC)
00004438  00                                    4128+         DS    X           extracted cc
00004439  00                                    4129+         DS    X           CC Check failed
                                                4130+*
0000443C                                        4131+X68      DS    0F
0000443C  E720 8F56 0006              00001156  4132+         VL    V2,V1FUDGE
00004442  E320 500C 0014              000043D4  4133+         LGF   R2,V2_68           get v2 address
00004448  E722 0000 0006              00000000  4134+         VL    V2,0(R2)

0000444E                                        4137+         DS    0H                      E64A VCVDQ
0000444E  E6                                    4138+         DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000444F  22                                    4139+         DC    X'22'                    v1, v2
00004450  00                                    4140+         DC    X'00'                   reserved
00004451  B8F0                                  4141+         DC    HL2'47344'               m4, i3, rXB
00004453  4A                                    4142+         DC    X'4A'

00004454  B98D 0020                             4144+         EPSW  R2,R0             exptract psw
00004458  5020 5068                   00000068  4145+         ST    R2,CCPSW              to save CC
0000445C  E720 5028 000E              000043F0  4146+         VST   V2,V1O68          save instruction result
00004462  07FB                                  4147+         BR    R11               return
00004464                                        4148+RE68     DS    0F                expected 16 byte result
```

```
  LOC       OBJECT CODE       ADDR1     ADDR2     STMT

00004464                                          4149+          DROP  R5
00004464  00000000 00000000                       4150           DC    XL16'0000000000000000 235394913435649F'      v1
0000446C  23539491 3435649F
00004474  FFFFFFFF FFFFFFFF                       4151           DC    XL16'FFFFFFFFFFFFFFFF 8000000000000001'      v2
0000447C  80000000 00000001
                                                  4152
                                                  4153           VRI_J VCVDQ,X'8F',X'B',3,N
00004488                                          4154+          DS    0FD
00004488                     00004488             4155+          USING *,R5              base for test data and test routine
00004488  000044FC                                4156+T69       DC    A(X69)            address of test routine
0000448C  0045                                    4157+          DC    H'69'             test number
0000448E  00                                      4158+          DC    X'00'
0000448F  D5                                      4159+          DC    CL1'N'            S or Y = skip cross check
00004490  8F                                      4160+          DC    X'8F'             i3
00004491  0B                                      4161+          DC    X'B'              m4
00004492  03                                      4162+          DC    HL1'3'            cc
00004493  0E                                      4163+          DC    HL1'14'           cc failed mask
00004494  00004534                                4164+V2_69     DC    A(RE69+16)        address of v2: 16-byte packed decimal
00004498  E5C3E5C4 D8404040                       4165+          DC    CL8'VCVDQ'        instruction name
000044A0  00000010                                4166+          DC    A(16)             result length
000044A4  00004524                                4167+          DC    A(RE69)           address of expected resul
000044A8  00000000 00000000                       4168+          DS    FD                gap
000044B0  00000000 00000000                       4169+V1O69     DS    XL16              V1 output
000044B8  00000000 00000000
000044C0  00000000 00000000                       4170+          DS    FD                   gap
000044C8  40404040 40404040                       4171+          DC    CL8' '            was cross check skipped?
000044D0  00000000 00000000                       4172+          DS    FD                   gap
000044D8  00000000 00000000                       4173+XCO69     DS    XL16              Cross check Output
000044E0  00000000 00000000
000044E8  00000000 00000000                       4174+          DS    FD                   gap
000044F0  00000000 00000000                       4175+          DS    2F          extract PSW after test (has CC)
000044F8  00                                      4176+          DS    X           extracted cc
000044F9  00                                      4177+          DS    X           CC Check failed
                                                  4178+*
000044FC                                          4179+X69       DS    0F
000044FC  E720 8F56 0006               00001156   4180+          VL    V2,V1FUDGE
00004502  E320 500C 0014               00004494   4181+          LGF   R2,V2_69           get v2 address
00004508  E722 0000 0006               00000000   4182+          VL    V2,0(R2)

0000450E                                          4185+          DS    0H                      E64A VCVDQ
0000450E  E6                                      4186+          DC    X'E6'                    VECTOR CONVERT TO DECIMAL (128)
0000450F  22                                      4187+          DC    X'22'                    v1, v2
00004510  00                                      4188+          DC    X'00'                   reserved
00004511  B8F0                                    4189+          DC    HL2'47344'               m4, i3, rXB
00004513  4A                                      4190+          DC    X'4A'

00004514  B98D 0020                               4192+          EPSW  R2,R0             exptract psw
00004518  5020 5068                    00000068   4193+          ST    R2,CCPSW              to save CC
0000451C  E720 5028 000E               000044B0   4194+          VST   V2,V1O69          save instruction result
00004522  07FB                                    4195+          BR    R11               return
00004524                                          4196+RE69      DS    0F                expected 16 byte result
00004524                                          4197+          DROP  R5
00004524  00000000 00000000                       4198           DC    XL16'0000000000000000 999999999999999F'      v1
0000452C  99999999 9999999F
00004534  0000007E 37BE2022                       4199           DC    XL16'0000007E37BE2022 C0914B267FFFFFFF'      v2
0000453C  C0914B26 7FFFFFFF
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                  4200
                                                  4201          VRI_J VCVDQ,X'8F',X'B',3,S
00004548                                          4202+         DS    0FD
00004548                    00004548              4203+         USING *,R5                base for test data and test routine
00004548  000045BC                                4204+T70      DC    A(X70)              address of test routine
0000454C  0046                                    4205+         DC    H'70'               test number
0000454E  00                                      4206+         DC    X'00'
0000454F  E2                                      4207+         DC    CL1'S'              S or Y = skip cross check
00004550  8F                                      4208+         DC    X'8F'               i3
00004551  0B                                      4209+         DC    X'B'                m4
00004552  03                                      4210+         DC    HL1'3'              cc
00004553  0E                                      4211+         DC    HL1'14'             cc failed mask
00004554  000045F4                                4212+V2_70    DC    A(RE70+16)          address of v2: 16-byte packed decimal
00004558  E5C3E5C4 D8404040                       4213+         DC    CL8'VCVDQ'          instruction name
00004560  00000010                                4214+         DC    A(16)               result length
00004564  000045E4                                4215+         DC    A(RE70)             address of expected resul
00004568  00000000 00000000                       4216+         DS    FD                  gap
00004570  00000000 00000000                       4217+V1O70    DS    XL16                V1 output
00004578  00000000 00000000
00004580  00000000 00000000                       4218+         DS    FD                     gap
00004588  40404040 40404040                       4219+         DC    CL8' '              was cross check skipped?
00004590  00000000 00000000                       4220+         DS    FD                     gap
00004598  00000000 00000000                       4221+XCO70    DS    XL16                Cross check Output
000045A0  00000000 00000000
000045A8  00000000 00000000                       4222+         DS    FD                     gap
000045B0  00000000 00000000                       4223+         DS    2F            extract PSW after test (has CC)
000045B8  00                                      4224+         DS    X             extracted cc
000045B9  00                                      4225+         DS    X             CC Check failed
                                                  4226+*
000045BC                                          4227+X70      DS    0F
000045BC  E720 8F56 0006             00001156     4228+         VL    V2,V1FUDGE
000045C2  E320 500C 0014             00004554     4229+         LGF   R2,V2_70             get v2 address
000045C8  E722 0000 0006             00000000     4230+         VL    V2,0(R2)

000045CE                                          4233+         DS    0H                        E64A VCVDQ
000045CE  E6                                      4234+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
000045CF  22                                      4235+         DC    X'22'                      v1, v2
000045D0  00                                      4236+         DC    X'00'                     reserved
000045D1  B8F0                                    4237+         DC    HL2'47344'                 m4, i3, rXB
000045D3  4A                                      4238+         DC    X'4A'

000045D4  B98D 0020                               4240+         EPSW  R2,R0               exptract psw
000045D8  5020 5068                  00000068     4241+         ST    R2,CCPSW                to save CC
000045DC  E720 5028 000E             00004570     4242+         VST   V2,V1O70            save instruction result
000045E2  07FB                                    4243+         BR    R11                 return
000045E4                                          4244+RE70     DS    0F                  expected 16 byte result
000045E4                                          4245+         DROP  R5
000045E4  00000000 00000000                       4246          DC    XL16'0000000000000000 607431768211457F'      v1
000045EC  60743176 8211457F
000045F4  FFFFFF81 C841DFDD                       4247          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000001'      v2
000045FC  3F6EB4D9 80000001
                                                  4248
                                                  4249          VRI_J VCVDQ,X'8F',X'B',3,N
00004608                                          4250+         DS    0FD
00004608                    00004608              4251+         USING *,R5                base for test data and test routine
00004608  0000467C                                4252+T71      DC    A(X71)              address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000460C  0047                                     4253+          DC    H'71'               test number
0000460E  00                                       4254+          DC    X'00'
0000460F  D5                                       4255+          DC    CL1'N'              S or Y = skip cross check
00004610  8F                                       4256+          DC    X'8F'               i3
00004611  0B                                       4257+          DC    X'B'                m4
00004612  03                                       4258+          DC    HL1'3'              cc
00004613  0E                                       4259+          DC    HL1'14'             cc failed mask
00004614  000046B4                                 4260+V2_71     DC    A(RE71+16)          address of v2: 16-byte packed decimal
00004618  E5C3E5C4 D8404040                        4261+          DC    CL8'VCVDQ'          instruction name
00004620  00000010                                 4262+          DC    A(16)               result length
00004624  000046A4                                 4263+          DC    A(RE71)             address of expected resul
00004628  00000000 00000000                        4264+          DS    FD                  gap
00004630  00000000 00000000                        4265+V1O71     DS    XL16                V1 output
00004638  00000000 00000000
00004640  00000000 00000000                        4266+          DS    FD                     gap
00004648  40404040 40404040                        4267+          DC    CL8' '              was cross check skipped?
00004650  00000000 00000000                        4268+          DS    FD                     gap
00004658  00000000 00000000                        4269+XCO71     DS    XL16                Cross check Output
00004660  00000000 00000000
00004668  00000000 00000000                        4270+          DS    FD                     gap
00004670  00000000 00000000                        4271+          DS    2F          extract PSW after test (has CC)
00004678  00                                       4272+          DS    X           extracted cc
00004679  00                                       4273+          DS    X           CC Check failed
                                                   4274+*
0000467C                                           4275+X71       DS    0F
0000467C  E720 8F56 0006                 00001156  4276+          VL    V2,V1FUDGE
00004682  E320 500C 0014                 00004614  4277+          LGF   R2,V2_71             get v2 address
00004688  E722 0000 0006                 00000000  4278+          VL    V2,0(R2)

0000468E                                           4281+          DS    0H                        E64A VCVDQ
0000468E  E6                                       4282+          DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000468F  22                                       4283+          DC    X'22'                      v1, v2
00004690  00                                       4284+          DC    X'00'                     reserved
00004691  B8F0                                     4285+          DC    HL2'47344'                 m4, i3, rXB
00004693  4A                                       4286+          DC    X'4A'

00004694  B98D 0020                                4288+          EPSW  R2,R0               exptract psw
00004698  5020 5068                      00000068  4289+          ST    R2,CCPSW                to save CC
0000469C  E720 5028 000E                 00004630  4290+          VST   V2,V1O71            save instruction result
000046A2  07FB                                     4291+          BR    R11                 return
000046A4                                           4292+RE71      DS    0F                  expected 16 byte result
000046A4                                           4293+          DROP  R5
000046A4  00000000 00000000                        4294           DC    XL16'0000000000000000 000000000000000F'      v1
000046AC  00000000 0000000F
000046B4  0000007E 37BE2022                        4295           DC    XL16'0000007E37BE2022 C0914B2680000000'      v2
000046BC  C0914B26 80000000
                                                   4296
                                                   4297           VRI_J VCVDQ,X'8F',X'B',3,N
000046C8                                           4298+          DS    0FD
000046C8                       000046C8            4299+          USING *,R5              base for test data and test routine
000046C8  0000473C                                 4300+T72       DC    A(X72)              address of test routine
000046CC  0048                                     4301+          DC    H'72'               test number
000046CE  00                                       4302+          DC    X'00'
000046CF  D5                                       4303+          DC    CL1'N'              S or Y = skip cross check
000046D0  8F                                       4304+          DC    X'8F'               i3
000046D1  0B                                       4305+          DC    X'B'                m4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000046D2  03                                      4306+         DC    HL1'3'              cc
000046D3  0E                                      4307+         DC    HL1'14'             cc failed mask
000046D4  00004774                                4308+V2_72    DC    A(RE72+16)          address of v2: 16-byte packed decimal
000046D8  E5C3E5C4 D8404040                       4309+         DC    CL8'VCVDQ'          instruction name
000046E0  00000010                                4310+         DC    A(16)               result length
000046E4  00004764                                4311+         DC    A(RE72)             address of expected resul
000046E8  00000000 00000000                       4312+         DS    FD                  gap
000046F0  00000000 00000000                       4313+V1O72    DS    XL16                V1 output
000046F8  00000000 00000000
00004700  00000000 00000000                       4314+         DS    FD                     gap
00004708  40404040 40404040                       4315+         DC    CL8' '              was cross check skipped?
00004710  00000000 00000000                       4316+         DS    FD                     gap
00004718  00000000 00000000                       4317+XCO72    DS    XL16                Cross check Output
00004720  00000000 00000000
00004728  00000000 00000000                       4318+         DS    FD                     gap
00004730  00000000 00000000                       4319+         DS    2F            extract PSW after test (has CC)
00004738  00                                      4320+         DS    X             extracted cc
00004739  00                                      4321+         DS    X             CC Check failed
                                                  4322+*
0000473C                                          4323+X72      DS    0F
0000473C  E720 8F56 0006              00001156    4324+         VL    V2,V1FUDGE
00004742  E320 500C 0014              000046D4    4325+         LGF   R2,V2_72             get v2 address
00004748  E722 0000 0006              00000000    4326+         VL    V2,0(R2)

0000474E                                          4329+         DS    0H                        E64A VCVDQ
0000474E  E6                                      4330+         DC    X'E6'                      VECTOR CONVERT TO DECIMAL (128)
0000474F  22                                      4331+         DC    X'22'                      v1, v2
00004750  00                                      4332+         DC    X'00'                     reserved
00004751  B8F0                                    4333+         DC    HL2'47344'                 m4, i3, rXB
00004753  4A                                      4334+         DC    X'4A'

00004754  B98D 0020                               4336+         EPSW  R2,R0               exptract psw
00004758  5020 5068                   00000068    4337+         ST    R2,CCPSW                to save CC
0000475C  E720 5028 000E              000046F0    4338+         VST   V2,V1O72            save instruction result
00004762  07FB                                    4339+         BR    R11                 return
00004764                                          4340+RE72     DS    0F                  expected 16 byte result
00004764                                          4341+         DROP  R5
00004764  00000000 00000000                       4342          DC    XL16'0000000000000000 607431768211456F'      v1
0000476C  60743176 8211456F
00004774  FFFFFF81 C841DFDD                       4343          DC    XL16'FFFFFF81C841DFDD 3F6EB4D980000000'      v2
0000477C  3F6EB4D9 80000000
                                                  4344
                                                  4345
                                                  4346
                                                  4347
00004784  00000000                                4348          DC    F'0'     END OF TABLE
00004788  00000000                                4349          DC    F'0'
                                                  4350 *
                                                  4351 * table of pointers to individual tests
                                                  4352 *
0000478C                                          4353 E6TESTS  DS    0F
                                                  4354          PTTABLE
0000478C                                          4355+TTABLE   DS    0F
0000478C  00001188                                4356+         DC    A(T1)               TEST &CUR
00004790  00001248                                4357+         DC    A(T2)               TEST &CUR
00004794  00001308                                4358+         DC    A(T3)               TEST &CUR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00004798  000013C8                                4359+         DC    A(T4)             TEST &CUR
0000479C  00001488                                4360+         DC    A(T5)             TEST &CUR
000047A0  00001548                                4361+         DC    A(T6)             TEST &CUR
000047A4  00001608                                4362+         DC    A(T7)             TEST &CUR
000047A8  000016C8                                4363+         DC    A(T8)             TEST &CUR
000047AC  00001788                                4364+         DC    A(T9)             TEST &CUR
000047B0  00001848                                4365+         DC    A(T10)            TEST &CUR
000047B4  00001908                                4366+         DC    A(T11)            TEST &CUR
000047B8  000019C8                                4367+         DC    A(T12)            TEST &CUR
000047BC  00001A88                                4368+         DC    A(T13)            TEST &CUR
000047C0  00001B48                                4369+         DC    A(T14)            TEST &CUR
000047C4  00001C08                                4370+         DC    A(T15)            TEST &CUR
000047C8  00001CC8                                4371+         DC    A(T16)            TEST &CUR
000047CC  00001D88                                4372+         DC    A(T17)            TEST &CUR
000047D0  00001E48                                4373+         DC    A(T18)            TEST &CUR
000047D4  00001F08                                4374+         DC    A(T19)            TEST &CUR
000047D8  00001FC8                                4375+         DC    A(T20)            TEST &CUR
000047DC  00002088                                4376+         DC    A(T21)            TEST &CUR
000047E0  00002148                                4377+         DC    A(T22)            TEST &CUR
000047E4  00002208                                4378+         DC    A(T23)            TEST &CUR
000047E8  000022C8                                4379+         DC    A(T24)            TEST &CUR
000047EC  00002388                                4380+         DC    A(T25)            TEST &CUR
000047F0  00002448                                4381+         DC    A(T26)            TEST &CUR
000047F4  00002508                                4382+         DC    A(T27)            TEST &CUR
000047F8  000025C8                                4383+         DC    A(T28)            TEST &CUR
000047FC  00002688                                4384+         DC    A(T29)            TEST &CUR
00004800  00002748                                4385+         DC    A(T30)            TEST &CUR
00004804  00002808                                4386+         DC    A(T31)            TEST &CUR
00004808  000028C8                                4387+         DC    A(T32)            TEST &CUR
0000480C  00002988                                4388+         DC    A(T33)            TEST &CUR
00004810  00002A48                                4389+         DC    A(T34)            TEST &CUR
00004814  00002B08                                4390+         DC    A(T35)            TEST &CUR
00004818  00002BC8                                4391+         DC    A(T36)            TEST &CUR
0000481C  00002C88                                4392+         DC    A(T37)            TEST &CUR
00004820  00002D48                                4393+         DC    A(T38)            TEST &CUR
00004824  00002E08                                4394+         DC    A(T39)            TEST &CUR
00004828  00002EC8                                4395+         DC    A(T40)            TEST &CUR
0000482C  00002F88                                4396+         DC    A(T41)            TEST &CUR
00004830  00003048                                4397+         DC    A(T42)            TEST &CUR
00004834  00003108                                4398+         DC    A(T43)            TEST &CUR
00004838  000031C8                                4399+         DC    A(T44)            TEST &CUR
0000483C  00003288                                4400+         DC    A(T45)            TEST &CUR
00004840  00003348                                4401+         DC    A(T46)            TEST &CUR
00004844  00003408                                4402+         DC    A(T47)            TEST &CUR
00004848  000034C8                                4403+         DC    A(T48)            TEST &CUR
0000484C  00003588                                4404+         DC    A(T49)            TEST &CUR
00004850  00003648                                4405+         DC    A(T50)            TEST &CUR
00004854  00003708                                4406+         DC    A(T51)            TEST &CUR
00004858  000037C8                                4407+         DC    A(T52)            TEST &CUR
0000485C  00003888                                4408+         DC    A(T53)            TEST &CUR
00004860  00003948                                4409+         DC    A(T54)            TEST &CUR
00004864  00003A08                                4410+         DC    A(T55)            TEST &CUR
00004868  00003AC8                                4411+         DC    A(T56)            TEST &CUR
0000486C  00003B88                                4412+         DC    A(T57)            TEST &CUR
00004870  00003C48                                4413+         DC    A(T58)            TEST &CUR
00004874  00003D08                                4414+         DC    A(T59)            TEST &CUR
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004878  00003DC8                                 4415+          DC    A(T60)              TEST &CUR
0000487C  00003E88                                 4416+          DC    A(T61)              TEST &CUR
00004880  00003F48                                 4417+          DC    A(T62)              TEST &CUR
00004884  00004008                                 4418+          DC    A(T63)              TEST &CUR
00004888  000040C8                                 4419+          DC    A(T64)              TEST &CUR
0000488C  00004188                                 4420+          DC    A(T65)              TEST &CUR
00004890  00004248                                 4421+          DC    A(T66)              TEST &CUR
00004894  00004308                                 4422+          DC    A(T67)              TEST &CUR
00004898  000043C8                                 4423+          DC    A(T68)              TEST &CUR
0000489C  00004488                                 4424+          DC    A(T69)              TEST &CUR
000048A0  00004548                                 4425+          DC    A(T70)              TEST &CUR
000048A4  00004608                                 4426+          DC    A(T71)              TEST &CUR
000048A8  000046C8                                 4427+          DC    A(T72)              TEST &CUR
                                                   4428+*
000048AC  00000000                                 4429+          DC    A(0)                END OF TABLE
000048B0  00000000                                 4430+          DC    A(0)
                                                   4431
000048B4  00000000                                 4432           DC    F'0'     END OF TABLE
000048B8  00000000                                 4433           DC    F'0'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                  4435 *******************************************************************
                                                  4436 *        Register equates
                                                  4437 *******************************************************************

                                00000000  00000001  4439 R0       EQU   0
                                00000001  00000001  4440 R1       EQU   1
                                00000002  00000001  4441 R2       EQU   2
                                00000003  00000001  4442 R3       EQU   3
                                00000004  00000001  4443 R4       EQU   4
                                00000005  00000001  4444 R5       EQU   5
                                00000006  00000001  4445 R6       EQU   6
                                00000007  00000001  4446 R7       EQU   7
                                00000008  00000001  4447 R8       EQU   8
                                00000009  00000001  4448 R9       EQU   9
                                0000000A  00000001  4449 R10      EQU   10
                                0000000B  00000001  4450 R11      EQU   11
                                0000000C  00000001  4451 R12      EQU   12
                                0000000D  00000001  4452 R13      EQU   13
                                0000000E  00000001  4453 R14      EQU   14
                                0000000F  00000001  4454 R15      EQU   15

                                                  4456 *******************************************************************
                                                  4457 *        Register equates
                                                  4458 *******************************************************************

                                00000000  00000001  4460 V0       EQU   0
                                00000001  00000001  4461 V1       EQU   1
                                00000002  00000001  4462 V2       EQU   2
                                00000003  00000001  4463 V3       EQU   3
                                00000004  00000001  4464 V4       EQU   4
                                00000005  00000001  4465 V5       EQU   5
                                00000006  00000001  4466 V6       EQU   6
                                00000007  00000001  4467 V7       EQU   7
                                00000008  00000001  4468 V8       EQU   8
                                00000009  00000001  4469 V9       EQU   9
                                0000000A  00000001  4470 V10      EQU   10
                                0000000B  00000001  4471 V11      EQU   11
                                0000000C  00000001  4472 V12      EQU   12
                                0000000D  00000001  4473 V13      EQU   13
                                0000000E  00000001  4474 V14      EQU   14
                                0000000F  00000001  4475 V15      EQU   15
                                00000010  00000001  4476 V16      EQU   16
                                00000011  00000001  4477 V17      EQU   17
                                00000012  00000001  4478 V18      EQU   18
                                00000013  00000001  4479 V19      EQU   19
                                00000014  00000001  4480 V20      EQU   20
                                00000015  00000001  4481 V21      EQU   21
                                00000016  00000001  4482 V22      EQU   22
                                00000017  00000001  4483 V23      EQU   23
                                00000018  00000001  4484 V24      EQU   24
                                00000019  00000001  4485 V25      EQU   25
                                0000001A  00000001  4486 V26      EQU   26
                                0000001B  00000001  4487 V27      EQU   27
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                0000001C  00000001  4488 V28      EQU   28
                                0000001D  00000001  4489 V29      EQU   29
                                0000001E  00000001  4490 V30      EQU   30
                                0000001F  00000001  4491 V31      EQU   31
                                                    4492
                                                    4493          END
```

```
     SYMBOL        TYPE   VALUE       LENGTH    DEFN  REFERENCES

BEGIN               I     00000200         2     215   180   211   212   213
CC                  U     0000000A         1     705   313   335
CCCHECK             U     0000031C         1     301   283
CCFAILED            X     00000071         1     722   302   322   385
CCFOUND             X     00000070         1     721   309   313   342   380
CCMASK              U     0000000B         1     706
CCMSG               U     0000033E         1     321   314
CCPRTEXP            C     000010A6         1     651   339
CCPRTGOT            C     000010B6         1     654   346
CCPRTLINE           C     0000105A        13     646   656   349
CCPRTLNG            U     0000005E         1     656   348
CCPRTNAME           C     00001090         8     649   332
CCPRTNUM            C     00001067         3     647   330
CCPSW               F     00000068         4     720   306   881   929   977  1025  1073  1121  1169  1219  1267  1319  1367  1415
                                                      1463  1511  1559  1607  1657  1705  1757  1805  1853  1901  1949  1997  2045
                                                      2095  2143  2196  2244  2292  2340  2388  2436  2484  2534  2582  2636  2684
                                                      2732  2782  2830  2878  2926  2974  3022  3074  3122  3170  3220  3268  3316
                                                      3364  3412  3460  3512  3560  3610  3658  3706  3754  3802  3850  3898  3951
                                                      3999  4049  4097  4145  4193  4241  4289  4337
CCR15               F     000003B8         8     356   350   353
CTLR0               F     00000674         4     581   225   226   227   228
DECNUM              C     00001136        16     684   327   329   336   338   343   345   411   413   420   422   428   430   457
                                                       459   466   468   473   475
DONEXT              U     00000314         1     294
E6TEST              4     00000000       120     698   274
E6TESTS             F     0000478C         4    4353   267
EDIT                X     0000110A        18     679   328   337   344   412   421   429   458   467   474
ENDTEST             U     00000548         1     495   272
EOJ                 I     00000658         4     571   260   498
EOJPSW              D     00000648         8     569   571
FAILCONT            U     00000538         1     485
FAILED              F     00001000         4     619   440   487   496
FAILMSG             U     000004D0         1     455   292
FAILPSW             D     00000660         8     573   575
FAILTEST            I     00000670         4     575   499
FB0001              F     00000290         8     244   248   249   251
FPCINIT             X     0000067C         4     583
I3                  U     00000008         1     703   419   465
IMAGE               1     00000000     18620       0
K                   U     00000400         1     602   603   604   605
K64                 U     00010000         1     604
M4                  U     00000009         1     704   427   472
MB                  U     00100000         1     605
MSG                 I     00000590         4     531   259   514
MSGCMD              C     000005DE         9     561   544   545
MSGMSG              C     000005E7        95     562   538   559   536
MSGMVC              I     000005D8         6     559   542
MSGOK               I     000005A6         2     540   537
MSGRET              I     000005C6         4     555   548   551
MSGSAVE             F     000005CC         4     558   534   555
NEXTE6              U     000002E4         1     269   296   490
OPNAME              C     00000010         8     709   332   375   416   462
PAGE                U     00001000         1     603
PRT3                C     00001120        18     682   328   329   330   337   338   339   344   345   346   412   413   414   421
                                                       422   423   429   430   431   458   459   460   467   468   469   474   475
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     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

PRTI3               C    0000104F         3    633   469
PRTLINE             C    00001008        13    628   638   479
PRTLNG              U    00000052         1    638   478
PRTM4               C    00001057         2    636   476
PRTNAME             C    0000103E         8    631   462
PRTNUM              C    00001015         3    629   460
R0                  U    00000000         1   4439   174   225   228   248   250   251   252   257   276   277   348   434   439
                                                     440   478   486   487   513   515   531   534   536   538   540   555   880
                                                     928   976  1024  1072  1120  1168  1218  1266  1318  1366  1414  1462  1510
                                                    1558  1606  1656  1704  1756  1804  1852  1900  1948  1996  2044  2094  2142
                                                    2195  2243  2291  2339  2387  2435  2483  2533  2581  2635  2683  2731  2781
                                                    2829  2877  2925  2973  3021  3073  3121  3169  3219  3267  3315  3363  3411
                                                    3459  3511  3559  3609  3657  3705  3753  3801  3849  3897  3950  3998  4048
                                                    4096  4144  4192  4240  4288  4336
R1                  U    00000001         1   4440   258   290   291   306   307   308   309   349   402   403   435   479   496
                                                     497   545   559
R10                 U    0000000A         1   4449   213   222   223
R11                 U    0000000B         1   4450   279   280   883   931   979  1027  1075  1123  1171  1221  1269  1321  1369
                                                    1417  1465  1513  1561  1609  1659  1707  1759  1807  1855  1903  1951  1999
                                                    2047  2097  2145  2198  2246  2294  2342  2390  2438  2486  2536  2584  2638
                                                    2686  2734  2784  2832  2880  2928  2976  3024  3076  3124  3172  3222  3270
                                                    3318  3366  3414  3462  3514  3562  3612  3660  3708  3756  3804  3852  3900
                                                    3953  4001  4051  4099  4147  4195  4243  4291  4339
R12                 U    0000000C         1   4451   267   270   295   489
R13                 U    0000000D         1   4452
R14                 U    0000000E         1   4453
R15                 U    0000000F         1   4454   283   286   316   350   351   353   354   368   370   376   381   386   405
                                                     433   436   437   441   480   508   518   519
R2                  U    00000002         1   4441   259   326   327   334   335   336   341   342   343   410   411   418   419
                                                     420   426   427   428   456   457   464   465   466   471   472   473   513
                                                     514   515   532   534   540   541   542   544   550   555   556   869   870
                                                     880   881   917   918   928   929   965   966   976   977  1013  1014  1024
                                                    1025  1061  1062  1072  1073  1109  1110  1120  1121  1157  1158  1168  1169
                                                    1207  1208  1218  1219  1255  1256  1266  1267  1307  1308  1318  1319  1355
                                                    1356  1366  1367  1403  1404  1414  1415  1451  1452  1462  1463  1499  1500
                                                    1510  1511  1547  1548  1558  1559  1595  1596  1606  1607  1645  1646  1656
                                                    1657  1693  1694  1704  1705  1745  1746  1756  1757  1793  1794  1804  1805
                                                    1841  1842  1852  1853  1889  1890  1900  1901  1937  1938  1948  1949  1985
                                                    1986  1996  1997  2033  2034  2044  2045  2083  2084  2094  2095  2131  2132
                                                    2142  2143  2184  2185  2195  2196  2232  2233  2243  2244  2280  2281  2291
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                                                    2472  2473  2483  2484  2522  2523  2533  2534  2570  2571  2581  2582  2624
                                                    2625  2635  2636  2672  2673  2683  2684  2720  2721  2731  2732  2770  2771
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                                                    2926  2962  2963  2973  2974  3010  3011  3021  3022  3062  3063  3073  3074
                                                    3110  3111  3121  3122  3158  3159  3169  3170  3208  3209  3219  3220  3256
                                                    3257  3267  3268  3304  3305  3315  3316  3352  3353  3363  3364  3400  3401
                                                    3411  3412  3448  3449  3459  3460  3500  3501  3511  3512  3548  3549  3559
                                                    3560  3598  3599  3609  3610  3646  3647  3657  3658  3694  3695  3705  3706
                                                    3742  3743  3753  3754  3790  3791  3801  3802  3838  3839  3849  3850  3886
                                                    3887  3897  3898  3939  3940  3950  3951  3987  3988  3998  3999  4037  4038
                                                    4048  4049  4085  4086  4096  4097  4133  4134  4144  4145  4181  4182  4192
                                                    4193  4229  4230  4240  4241  4277  4278  4288  4289  4325  4326  4336  4337
R3                  U    00000003         1   4442
R4                  U    00000004         1   4443
R5                  U    00000005         1   4444   270   271   274   509   517   843   885   891   933   939   981   987  1029
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     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

                                                     1035  1077  1083  1125  1131  1173  1181  1223  1229  1271  1281  1323  1329
                                                     1371  1377  1419  1425  1467  1473  1515  1521  1563  1569  1611  1619  1661
                                                     1667  1709  1719  1761  1767  1809  1815  1857  1863  1905  1911  1953  1959
                                                     2001  2007  2049  2057  2099  2105  2147  2158  2200  2206  2248  2254  2296
                                                     2302  2344  2350  2392  2398  2440  2446  2488  2496  2538  2544  2586  2598
                                                     2640  2646  2688  2694  2736  2744  2786  2792  2834  2840  2882  2888  2930
                                                     2936  2978  2984  3026  3036  3078  3084  3126  3132  3174  3182  3224  3230
                                                     3272  3278  3320  3326  3368  3374  3416  3422  3464  3474  3516  3522  3564
                                                     3572  3614  3620  3662  3668  3710  3716  3758  3764  3806  3812  3854  3860
                                                     3902  3913  3955  3961  4003  4011  4053  4059  4101  4107  4149  4155  4197
                                                     4203  4245  4251  4293  4299  4341
R6                  U    00000006           1  4445
R7                  U    00000007           1  4446
R8                  U    00000008           1  4447   211   215   216   217   219
R9                  U    00000009           1  4448   212   219   220   222
RE1                 F    00001224           4   884   852   855
RE10                F    000018E4           4  1322  1290  1293
RE11                F    000019A4           4  1370  1338  1341
RE12                F    00001A64           4  1418  1386  1389
RE13                F    00001B24           4  1466  1434  1437
RE14                F    00001BE4           4  1514  1482  1485
RE15                F    00001CA4           4  1562  1530  1533
RE16                F    00001D64           4  1610  1578  1581
RE17                F    00001E24           4  1660  1628  1631
RE18                F    00001EE4           4  1708  1676  1679
RE19                F    00001FA4           4  1760  1728  1731
RE2                 F    000012E4           4   932   900   903
RE20                F    00002064           4  1808  1776  1779
RE21                F    00002124           4  1856  1824  1827
RE22                F    000021E4           4  1904  1872  1875
RE23                F    000022A4           4  1952  1920  1923
RE24                F    00002364           4  2000  1968  1971
RE25                F    00002424           4  2048  2016  2019
RE26                F    000024E4           4  2098  2066  2069
RE27                F    000025A4           4  2146  2114  2117
RE28                F    00002664           4  2199  2167  2170
RE29                F    00002724           4  2247  2215  2218
RE3                 F    000013A4           4   980   948   951
RE30                F    000027E4           4  2295  2263  2266
RE31                F    000028A4           4  2343  2311  2314
RE32                F    00002964           4  2391  2359  2362
RE33                F    00002A24           4  2439  2407  2410
RE34                F    00002AE4           4  2487  2455  2458
RE35                F    00002BA4           4  2537  2505  2508
RE36                F    00002C64           4  2585  2553  2556
RE37                F    00002D24           4  2639  2607  2610
RE38                F    00002DE4           4  2687  2655  2658
RE39                F    00002EA4           4  2735  2703  2706
RE4                 F    00001464           4  1028   996   999
RE40                F    00002F64           4  2785  2753  2756
RE41                F    00003024           4  2833  2801  2804
RE42                F    000030E4           4  2881  2849  2852
RE43                F    000031A4           4  2929  2897  2900
RE44                F    00003264           4  2977  2945  2948
RE45                F    00003324           4  3025  2993  2996
RE46                F    000033E4           4  3077  3045  3048
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
RE47	F	000034A4	4	3125	3093	3096	
RE48	F	00003564	4	3173	3141	3144	
RE49	F	00003624	4	3223	3191	3194	
RE5	F	00001524	4	1076	1044	1047	
RE50	F	000036E4	4	3271	3239	3242	
RE51	F	000037A4	4	3319	3287	3290	
RE52	F	00003864	4	3367	3335	3338	
RE53	F	00003924	4	3415	3383	3386	
RE54	F	000039E4	4	3463	3431	3434	
RE55	F	00003AA4	4	3515	3483	3486	
RE56	F	00003B64	4	3563	3531	3534	
RE57	F	00003C24	4	3613	3581	3584	
RE58	F	00003CE4	4	3661	3629	3632	
RE59	F	00003DA4	4	3709	3677	3680	
RE6	F	000015E4	4	1124	1092	1095	
RE60	F	00003E64	4	3757	3725	3728	
RE61	F	00003F24	4	3805	3773	3776	
RE62	F	00003FE4	4	3853	3821	3824	
RE63	F	000040A4	4	3901	3869	3872	
RE64	F	00004164	4	3954	3922	3925	
RE65	F	00004224	4	4002	3970	3973	
RE66	F	000042E4	4	4052	4020	4023	
RE67	F	000043A4	4	4100	4068	4071	
RE68	F	00004464	4	4148	4116	4119	
RE69	F	00004524	4	4196	4164	4167	
RE7	F	000016A4	4	1172	1140	1143	
RE70	F	000045E4	4	4244	4212	4215	
RE71	F	000046A4	4	4292	4260	4263	
RE72	F	00004764	4	4340	4308	4311	
RE8	F	00001764	4	1222	1190	1193	
RE9	F	00001824	4	1270	1238	1241	
READDR	A	0000001C	4	711	290		
REG2LOW	U	000000DD	1	609			
REG2PATT	U	AABBCCDD	1	608			
RELEN	A	00000018	4	710			
RPTDWSAV	D	00000580	8	524	513	515	
RPTERROR	I	00000556	4	508	351	436	480
RPTSAVE	F	00000574	4	521	508	518	
RPTSVR5	F	00000578	4	522	509	517	
SKIPXC	C	00000040	8	715	366	374	
SKL0001	U	0000005E	1	241	257		
SKT0001	C	0000022A	22	238	241	258	
SVOLDPSW	U	00000140	0	176			
T1	A	00001188	4	844	4356		
T10	A	00001848	4	1282	4365		
T11	A	00001908	4	1330	4366		
T12	A	000019C8	4	1378	4367		
T13	A	00001A88	4	1426	4368		
T14	A	00001B48	4	1474	4369		
T15	A	00001C08	4	1522	4370		
T16	A	00001CC8	4	1570	4371		
T17	A	00001D88	4	1620	4372		
T18	A	00001E48	4	1668	4373		
T19	A	00001F08	4	1720	4374		
T2	A	00001248	4	892	4357		
T20	A	00001FC8	4	1768	4375		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T21	A	00002088	4	1816	4376
T22	A	00002148	4	1864	4377
T23	A	00002208	4	1912	4378
T24	A	000022C8	4	1960	4379
T25	A	00002388	4	2008	4380
T26	A	00002448	4	2058	4381
T27	A	00002508	4	2106	4382
T28	A	000025C8	4	2159	4383
T29	A	00002688	4	2207	4384
T3	A	00001308	4	940	4358
T30	A	00002748	4	2255	4385
T31	A	00002808	4	2303	4386
T32	A	000028C8	4	2351	4387
T33	A	00002988	4	2399	4388
T34	A	00002A48	4	2447	4389
T35	A	00002B08	4	2497	4390
T36	A	00002BC8	4	2545	4391
T37	A	00002C88	4	2599	4392
T38	A	00002D48	4	2647	4393
T39	A	00002E08	4	2695	4394
T4	A	000013C8	4	988	4359
T40	A	00002EC8	4	2745	4395
T41	A	00002F88	4	2793	4396
T42	A	00003048	4	2841	4397
T43	A	00003108	4	2889	4398
T44	A	000031C8	4	2937	4399
T45	A	00003288	4	2985	4400
T46	A	00003348	4	3037	4401
T47	A	00003408	4	3085	4402
T48	A	000034C8	4	3133	4403
T49	A	00003588	4	3183	4404
T5	A	00001488	4	1036	4360
T50	A	00003648	4	3231	4405
T51	A	00003708	4	3279	4406
T52	A	000037C8	4	3327	4407
T53	A	00003888	4	3375	4408
T54	A	00003948	4	3423	4409
T55	A	00003A08	4	3475	4410
T56	A	00003AC8	4	3523	4411
T57	A	00003B88	4	3573	4412
T58	A	00003C48	4	3621	4413
T59	A	00003D08	4	3669	4414
T6	A	00001548	4	1084	4361
T60	A	00003DC8	4	3717	4415
T61	A	00003E88	4	3765	4416
T62	A	00003F48	4	3813	4417
T63	A	00004008	4	3861	4418
T64	A	000040C8	4	3914	4419
T65	A	00004188	4	3962	4420
T66	A	00004248	4	4012	4421
T67	A	00004308	4	4060	4422
T68	A	000043C8	4	4108	4423
T69	A	00004488	4	4156	4424
T7	A	00001608	4	1132	4362
T70	A	00004548	4	4204	4425
T71	A	00004608	4	4252	4426

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T72	A	000046C8	4	4300	4427
T8	A	000016C8	4	1182	4363
T9	A	00001788	4	1230	4364
TESTING	F	00001004	4	620	277
TNUM	H	00000004	2	700	276 326 410 456
TSUB	A	00000000	4	699	279
TTABLE	F	0000478C	4	4355	
V0	U	00000000	1	4460	
V1	U	00000001	1	4461	
V10	U	0000000A	1	4470	
V11	U	0000000B	1	4471	
V12	U	0000000C	1	4472	
V13	U	0000000D	1	4473	
V14	U	0000000E	1	4474	
V15	U	0000000F	1	4475	
V16	U	00000010	1	4476	
V17	U	00000011	1	4477	
V18	U	00000012	1	4478	
V19	U	00000013	1	4479	
V1FUDGE	X	00001156	16	690	868 916 964 1012 1060 1108 1156 1206 1254 1306 1354 1402 1450 1498 1546 1594 1644 1692 1744 1792 1840 1888 1936 1984 2032 2082 2130 2183 2231 2279 2327 2375 2423 2471 2521 2569 2623 2671 2719 2769 2817 2865 2913 2961 3009 3061 3109 3157 3207 3255 3303 3351 3399 3447 3499 3547 3597 3645 3693 3741 3789 3837 3885 3938 3986 4036 4084 4132 4180 4228 4276 4324
V1INPUT	X	00001166	16	691	
V1O1	X	000011B0	16	857	882
V1O10	X	00001870	16	1295	1320
V1O11	X	00001930	16	1343	1368
V1O12	X	000019F0	16	1391	1416
V1O13	X	00001AB0	16	1439	1464
V1O14	X	00001B70	16	1487	1512
V1O15	X	00001C30	16	1535	1560
V1O16	X	00001CF0	16	1583	1608
V1O17	X	00001DB0	16	1633	1658
V1O18	X	00001E70	16	1681	1706
V1O19	X	00001F30	16	1733	1758
V1O2	X	00001270	16	905	930
V1O20	X	00001FF0	16	1781	1806
V1O21	X	000020B0	16	1829	1854
V1O22	X	00002170	16	1877	1902
V1O23	X	00002230	16	1925	1950
V1O24	X	000022F0	16	1973	1998
V1O25	X	000023B0	16	2021	2046
V1O26	X	00002470	16	2071	2096
V1O27	X	00002530	16	2119	2144
V1O28	X	000025F0	16	2172	2197
V1O29	X	000026B0	16	2220	2245
V1O3	X	00001330	16	953	978
V1O30	X	00002770	16	2268	2293
V1O31	X	00002830	16	2316	2341
V1O32	X	000028F0	16	2364	2389
V1O33	X	000029B0	16	2412	2437
V1O34	X	00002A70	16	2460	2485
V1O35	X	00002B30	16	2510	2535
V1O36	X	00002BF0	16	2558	2583

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     SYMBOL        TYPE   VALUE      LENGTH    DEFN  REFERENCES

V1037               X    00002CB0        16  2612  2637
V1038               X    00002D70        16  2660  2685
V1039               X    00002E30        16  2708  2733
V104                X    000013F0        16  1001  1026
V1040               X    00002EF0        16  2758  2783
V1041               X    00002FB0        16  2806  2831
V1042               X    00003070        16  2854  2879
V1043               X    00003130        16  2902  2927
V1044               X    000031F0        16  2950  2975
V1045               X    000032B0        16  2998  3023
V1046               X    00003370        16  3050  3075
V1047               X    00003430        16  3098  3123
V1048               X    000034F0        16  3146  3171
V1049               X    000035B0        16  3196  3221
V105                X    000014B0        16  1049  1074
V1050               X    00003670        16  3244  3269
V1051               X    00003730        16  3292  3317
V1052               X    000037F0        16  3340  3365
V1053               X    000038B0        16  3388  3413
V1054               X    00003970        16  3436  3461
V1055               X    00003A30        16  3488  3513
V1056               X    00003AF0        16  3536  3561
V1057               X    00003BB0        16  3586  3611
V1058               X    00003C70        16  3634  3659
V1059               X    00003D30        16  3682  3707
V106                X    00001570        16  1097  1122
V1060               X    00003DF0        16  3730  3755
V1061               X    00003EB0        16  3778  3803
V1062               X    00003F70        16  3826  3851
V1063               X    00004030        16  3874  3899
V1064               X    000040F0        16  3927  3952
V1065               X    000041B0        16  3975  4000
V1066               X    00004270        16  4025  4050
V1067               X    00004330        16  4073  4098
V1068               X    000043F0        16  4121  4146
V1069               X    000044B0        16  4169  4194
V107                X    00001630        16  1145  1170
V1070               X    00004570        16  4217  4242
V1071               X    00004630        16  4265  4290
V1072               X    000046F0        16  4313  4338
V108                X    000016F0        16  1195  1220
V109                X    000017B0        16  1243  1268
V1OUTPUT            X    00000028        16   713   291
V2                  U    00000002         1  4462   868   870   882   916   918   930   964   966   978  1012  1014  1026  1060
                                                     1062  1074  1108  1110  1122  1156  1158  1170  1206  1208  1220  1254  1256
                                                     1268  1306  1308  1320  1354  1356  1368  1402  1404  1416  1450  1452  1464
                                                     1498  1500  1512  1546  1548  1560  1594  1596  1608  1644  1646  1658  1692
                                                     1694  1706  1744  1746  1758  1792  1794  1806  1840  1842  1854  1888  1890
                                                     1902  1936  1938  1950  1984  1986  1998  2032  2034  2046  2082  2084  2096
                                                     2130  2132  2144  2183  2185  2197  2231  2233  2245  2279  2281  2293  2327
                                                     2329  2341  2375  2377  2389  2423  2425  2437  2471  2473  2485  2521  2523
                                                     2535  2569  2571  2583  2623  2625  2637  2671  2673  2685  2719  2721  2733
                                                     2769  2771  2783  2817  2819  2831  2865  2867  2879  2913  2915  2927  2961
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                                                     3171  3207  3209  3221  3255  3257  3269  3303  3305  3317  3351  3353  3365
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
					3599 3611 3645 3647 3659 3693 3695 3707 3741 3743 3755 3789 3791
					3803 3837 3839 3851 3885 3887 3899 3938 3940 3952 3986 3988 4000
					4036 4038 4050 4084 4086 4098 4132 4134 4146 4180 4182 4194 4228
					4230 4242 4276 4278 4290 4324 4326 4338
V20	U	00000014	1	4480	
V21	U	00000015	1	4481	
V22	U	00000016	1	4482	
V23	U	00000017	1	4483	
V24	U	00000018	1	4484	
V25	U	00000019	1	4485	
V26	U	0000001A	1	4486	
V27	U	0000001B	1	4487	
V28	U	0000001C	1	4488	
V29	U	0000001D	1	4489	
V2ADDR	A	0000000C	4	708	402
V2_1	A	00001194	4	852	869
V2_10	A	00001854	4	1290	1307
V2_11	A	00001914	4	1338	1355
V2_12	A	000019D4	4	1386	1403
V2_13	A	00001A94	4	1434	1451
V2_14	A	00001B54	4	1482	1499
V2_15	A	00001C14	4	1530	1547
V2_16	A	00001CD4	4	1578	1595
V2_17	A	00001D94	4	1628	1645
V2_18	A	00001E54	4	1676	1693
V2_19	A	00001F14	4	1728	1745
V2_2	A	00001254	4	900	917
V2_20	A	00001FD4	4	1776	1793
V2_21	A	00002094	4	1824	1841
V2_22	A	00002154	4	1872	1889
V2_23	A	00002214	4	1920	1937
V2_24	A	000022D4	4	1968	1985
V2_25	A	00002394	4	2016	2033
V2_26	A	00002454	4	2066	2083
V2_27	A	00002514	4	2114	2131
V2_28	A	000025D4	4	2167	2184
V2_29	A	00002694	4	2215	2232
V2_3	A	00001314	4	948	965
V2_30	A	00002754	4	2263	2280
V2_31	A	00002814	4	2311	2328
V2_32	A	000028D4	4	2359	2376
V2_33	A	00002994	4	2407	2424
V2_34	A	00002A54	4	2455	2472
V2_35	A	00002B14	4	2505	2522
V2_36	A	00002BD4	4	2553	2570
V2_37	A	00002C94	4	2607	2624
V2_38	A	00002D54	4	2655	2672
V2_39	A	00002E14	4	2703	2720
V2_4	A	000013D4	4	996	1013
V2_40	A	00002ED4	4	2753	2770
V2_41	A	00002F94	4	2801	2818
V2_42	A	00003054	4	2849	2866
V2_43	A	00003114	4	2897	2914
V2_44	A	000031D4	4	2945	2962
V2_45	A	00003294	4	2993	3010
V2_46	A	00003354	4	3045	3062

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V2_47	A	00003414	4	3093	3110
V2_48	A	000034D4	4	3141	3158
V2_49	A	00003594	4	3191	3208
V2_5	A	00001494	4	1044	1061
V2_50	A	00003654	4	3239	3256
V2_51	A	00003714	4	3287	3304
V2_52	A	000037D4	4	3335	3352
V2_53	A	00003894	4	3383	3400
V2_54	A	00003954	4	3431	3448
V2_55	A	00003A14	4	3483	3500
V2_56	A	00003AD4	4	3531	3548
V2_57	A	00003B94	4	3581	3598
V2_58	A	00003C54	4	3629	3646
V2_59	A	00003D14	4	3677	3694
V2_6	A	00001554	4	1092	1109
V2_60	A	00003DD4	4	3725	3742
V2_61	A	00003E94	4	3773	3790
V2_62	A	00003F54	4	3821	3838
V2_63	A	00004014	4	3869	3886
V2_64	A	000040D4	4	3922	3939
V2_65	A	00004194	4	3970	3987
V2_66	A	00004254	4	4020	4037
V2_67	A	00004314	4	4068	4085
V2_68	A	000043D4	4	4116	4133
V2_69	A	00004494	4	4164	4181
V2_7	A	00001614	4	1140	1157
V2_70	A	00004554	4	4212	4229
V2_71	A	00004614	4	4260	4277
V2_72	A	000046D4	4	4308	4325
V2_8	A	000016D4	4	1190	1207
V2_9	A	00001794	4	1238	1255
V3	U	00000003	1	4463	400
V30	U	0000001E	1	4490	
V31	U	0000001F	1	4491	
V4	U	00000004	1	4464	
V5	U	00000005	1	4465	
V6	U	00000006	1	4466	
V7	U	00000007	1	4467	
V8	U	00000008	1	4468	
V9	U	00000009	1	4469	
WK1	F	00000074	4	724	
X0001	U	000002B8	1	247	235 248
X1	F	000011FC	4	867	844
X10	F	000018BC	4	1305	1282
X11	F	0000197C	4	1353	1330
X12	F	00001A3C	4	1401	1378
X13	F	00001AFC	4	1449	1426
X14	F	00001BBC	4	1497	1474
X15	F	00001C7C	4	1545	1522
X16	F	00001D3C	4	1593	1570
X17	F	00001DFC	4	1643	1620
X18	F	00001EBC	4	1691	1668
X19	F	00001F7C	4	1743	1720
X2	F	000012BC	4	915	892
X20	F	0000203C	4	1791	1768
X21	F	000020FC	4	1839	1816

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X22	F	000021BC	4	1887	1864
X23	F	0000227C	4	1935	1912
X24	F	0000233C	4	1983	1960
X25	F	000023FC	4	2031	2008
X26	F	000024BC	4	2081	2058
X27	F	0000257C	4	2129	2106
X28	F	0000263C	4	2182	2159
X29	F	000026FC	4	2230	2207
X3	F	0000137C	4	963	940
X30	F	000027BC	4	2278	2255
X31	F	0000287C	4	2326	2303
X32	F	0000293C	4	2374	2351
X33	F	000029FC	4	2422	2399
X34	F	00002ABC	4	2470	2447
X35	F	00002B7C	4	2520	2497
X36	F	00002C3C	4	2568	2545
X37	F	00002CFC	4	2622	2599
X38	F	00002DBC	4	2670	2647
X39	F	00002E7C	4	2718	2695
X4	F	0000143C	4	1011	988
X40	F	00002F3C	4	2768	2745
X41	F	00002FFC	4	2816	2793
X42	F	000030BC	4	2864	2841
X43	F	0000317C	4	2912	2889
X44	F	0000323C	4	2960	2937
X45	F	000032FC	4	3008	2985
X46	F	000033BC	4	3060	3037
X47	F	0000347C	4	3108	3085
X48	F	0000353C	4	3156	3133
X49	F	000035FC	4	3206	3183
X5	F	000014FC	4	1059	1036
X50	F	000036BC	4	3254	3231
X51	F	0000377C	4	3302	3279
X52	F	0000383C	4	3350	3327
X53	F	000038FC	4	3398	3375
X54	F	000039BC	4	3446	3423
X55	F	00003A7C	4	3498	3475
X56	F	00003B3C	4	3546	3523
X57	F	00003BFC	4	3596	3573
X58	F	00003CBC	4	3644	3621
X59	F	00003D7C	4	3692	3669
X6	F	000015BC	4	1107	1084
X60	F	00003E3C	4	3740	3717
X61	F	00003EFC	4	3788	3765
X62	F	00003FBC	4	3836	3813
X63	F	0000407C	4	3884	3861
X64	F	0000413C	4	3937	3914
X65	F	000041FC	4	3985	3962
X66	F	000042BC	4	4035	4012
X67	F	0000437C	4	4083	4060
X68	F	0000443C	4	4131	4108
X69	F	000044FC	4	4179	4156
X7	F	0000167C	4	1155	1132
X70	F	000045BC	4	4227	4204
X71	F	0000467C	4	4275	4252
X72	F	0000473C	4	4323	4300

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
X8	F	0000173C	4	1205	1182
X9	F	000017FC	4	1253	1230
XC0001	U	000002E0	1	261	253
XCFAILMSG	H	0000041C	2	409	404
XCHECK	U	000003C0	1	365	286
XC01	X	000011D8	16	861	
XC010	X	00001898	16	1299	
XC011	X	00001958	16	1347	
XC012	X	00001A18	16	1395	
XC013	X	00001AD8	16	1443	
XC014	X	00001B98	16	1491	
XC015	X	00001C58	16	1539	
XC016	X	00001D18	16	1587	
XC017	X	00001DD8	16	1637	
XC018	X	00001E98	16	1685	
XC019	X	00001F58	16	1737	
XC02	X	00001298	16	909	
XC020	X	00002018	16	1785	
XC021	X	000020D8	16	1833	
XC022	X	00002198	16	1881	
XC023	X	00002258	16	1929	
XC024	X	00002318	16	1977	
XC025	X	000023D8	16	2025	
XC026	X	00002498	16	2075	
XC027	X	00002558	16	2123	
XC028	X	00002618	16	2176	
XC029	X	000026D8	16	2224	
XC03	X	00001358	16	957	
XC030	X	00002798	16	2272	
XC031	X	00002858	16	2320	
XC032	X	00002918	16	2368	
XC033	X	000029D8	16	2416	
XC034	X	00002A98	16	2464	
XC035	X	00002B58	16	2514	
XC036	X	00002C18	16	2562	
XC037	X	00002CD8	16	2616	
XC038	X	00002D98	16	2664	
XC039	X	00002E58	16	2712	
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XC040	X	00002F18	16	2762	
XC041	X	00002FD8	16	2810	
XC042	X	00003098	16	2858	
XC043	X	00003158	16	2906	
XC044	X	00003218	16	2954	
XC045	X	000032D8	16	3002	
XC046	X	00003398	16	3054	
XC047	X	00003458	16	3102	
XC048	X	00003518	16	3150	
XC049	X	000035D8	16	3200	
XC05	X	000014D8	16	1053	
XC050	X	00003698	16	3248	
XC051	X	00003758	16	3296	
XC052	X	00003818	16	3344	
XC053	X	000038D8	16	3392	
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XC055	X	00003A58	16	3492	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
XC056	X	00003B18	16	3540	
XC057	X	00003BD8	16	3590	
XC058	X	00003C98	16	3638	
XC059	X	00003D58	16	3686	
XC06	X	00001598	16	1101	
XC060	X	00003E18	16	3734	
XC061	X	00003ED8	16	3782	
XC062	X	00003F98	16	3830	
XC063	X	00004058	16	3878	
XC064	X	00004118	16	3931	
XC065	X	000041D8	16	3979	
XC066	X	00004298	16	4029	
XC067	X	00004358	16	4077	
XC068	X	00004418	16	4125	
XC069	X	000044D8	16	4173	
XC07	X	00001658	16	1149	
XC070	X	00004598	16	4221	
XC071	X	00004658	16	4269	
XC072	X	00004718	16	4317	
XC08	X	00001718	16	1199	
XC09	X	000017D8	16	1247	
XCOUTPUT	X	00000050	16	717	400 403
XCPI3	C	000010FF	3	669	423
XCPLINE	C	000010B8	13	664	674 435
XCPLNG	U	00000052	1	674	434
XCPM4	C	00001107	2	672	431
XCPNAME	C	000010EE	8	667	416
XCPTNUM	C	000010C5	3	665	414
XCR15	F	000004C8	8	447	433 437
XCRESULT	X	00000498	16	444	
XCSKIP	C	00000007	1	702	367 369
XCV1	X	000004A8	16	445	
XCV2	X	000004B8	16	446	
ZVE6TST	J	00000000	18620	173	176 178 182 186 618 174
=A(E6TESTS)	A	0000069C	4	591	267
=AL2(L'MSGMSG)	R	000006A6	2	594	536
=CL1'N'	C	000006A8	1	595	302
=CL1'S'	C	000006AA	1	597	367
=CL1'Y'	C	000006A9	1	596	322 369 385
=CL8' '	C	00000688	8	588	374
=CL8'SKIP XC '	C	00000680	8	587	366
=CL8'VCVDQ '	C	00000690	8	589	375
=F'1'	F	00000698	4	590	252 439 486
=H'0'	H	000006A4	2	593	531
=XL1'3'	X	000006AB	1	598	380
=XL4'3'	X	000006A0	4	592	308

MACRO	DEFN	REFERENCES															
FCHECK	69	234															
PTTABLE	805	4354															
VCVBQ	127	390															
VCVDQ	154	871	919	967	1015	1063	1111	1159	1209	1257	1309	1357	1405	1453	1501	1549	1597	1647
		1695	1747	1795	1843	1891	1939	1987	2035	2085	2133	2186	2234	2282	2330	2378	2426	2474
		2524	2572	2626	2674	2722	2772	2820	2868	2916	2964	3012	3064	3112	3160	3210	3258	3306
		3354	3402	3450	3502	3550	3600	3648	3696	3744	3792	3840	3888	3941	3989	4039	4087	4135
		4183	4231	4279	4327													
VRI_J	737	841	889	937	985	1033	1081	1129	1179	1227	1279	1327	1375	1423	1471	1519	1567	1617
		1665	1717	1765	1813	1861	1909	1957	2005	2055	2103	2156	2204	2252	2300	2348	2396	2444
		2494	2542	2596	2644	2692	2742	2790	2838	2886	2934	2982	3034	3082	3130	3180	3228	3276
		3324	3372	3420	3472	3520	3570	3618	3666	3714	3762	3810	3858	3911	3959	4009	4057	4105
		4153	4201	4249	4297													

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   DESC     SYMBOL   SIZE     POS        ADDR

Entry: 0

Image      IMAGE    18620  0000-48BB  0000-48BB
  Region            18620  0000-48BB  0000-48BB
    CSECT  ZVE6TST  18620  0000-48BB  0000-48BB
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STMT	FILE NAME
1	/devstor/sharedvfp/tests/zvector-e6-24-VCVDQ.asm

** NO ERRORS FOUND **